

Practical No: 1

Aim: Implement the following:

A. Create a simple sequence-based project.

Steps:

1. Open Project.
2. Select Sequence activity, double click it.
3. Select from activities panel Input dialog and configure it
4. Create a variable Name (string) to hold the output.
5. Select from activities panel Message Box & configure by tying "Your Name is'+ Name.
6. Run the project.

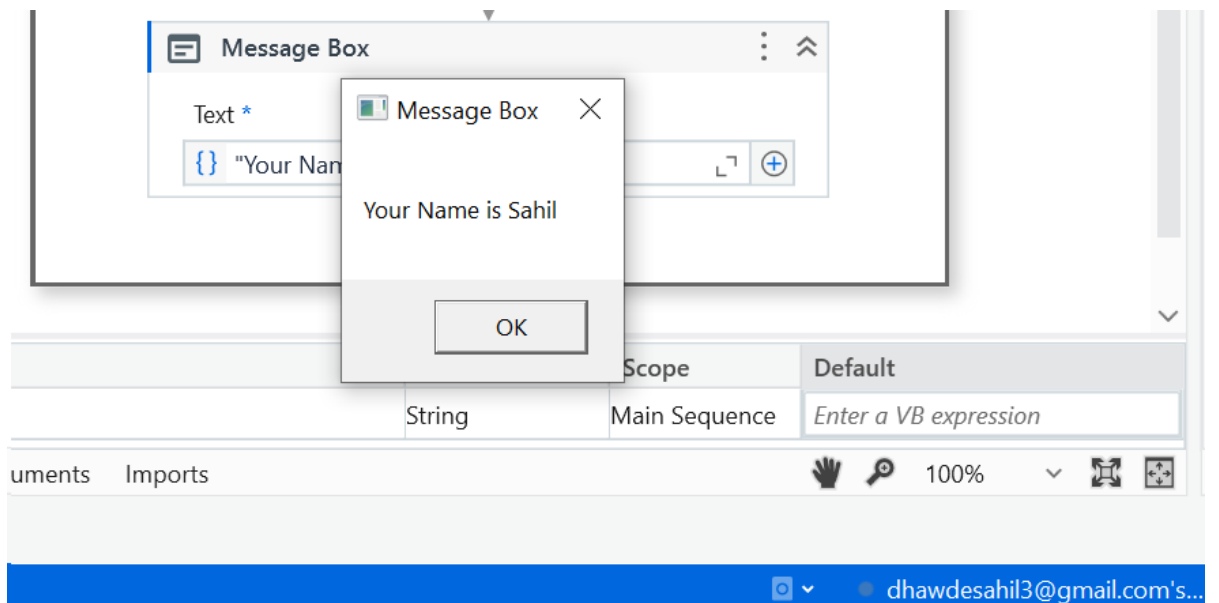
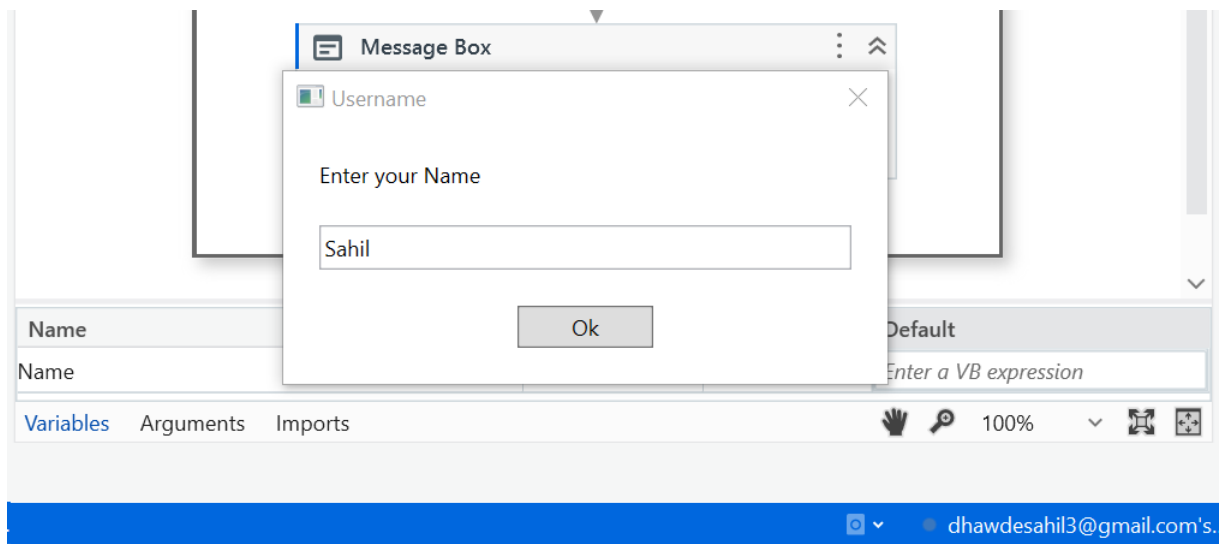
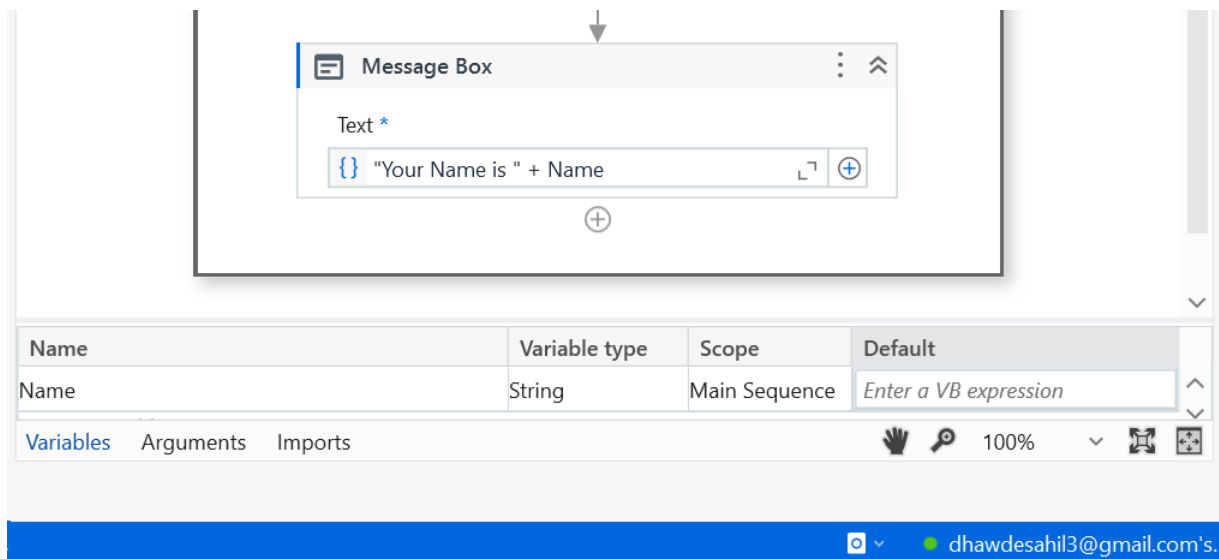
Process & Output:

The screenshot displays the Visual Studio Code IDE interface for a sequence-based project. The 'Main Sequence' activity is expanded, showing an 'Input Dialog' activity followed by a 'Message Box' activity. The 'Input Dialog' is configured with the following settings:

- Dialog Title: Username
- Input Label: Enter your Name
- Input Type: Text Box
- Value entered: Name

The 'Message Box' activity is connected to the 'Input Dialog' and is configured to display the value of the 'Name' variable. The 'Variables' panel at the bottom shows a variable named 'Name' of type 'String' with scope 'Main Sequence'.

Name	Variable type	Scope	Default
Name	String	Main Sequence	Enter a VB expression

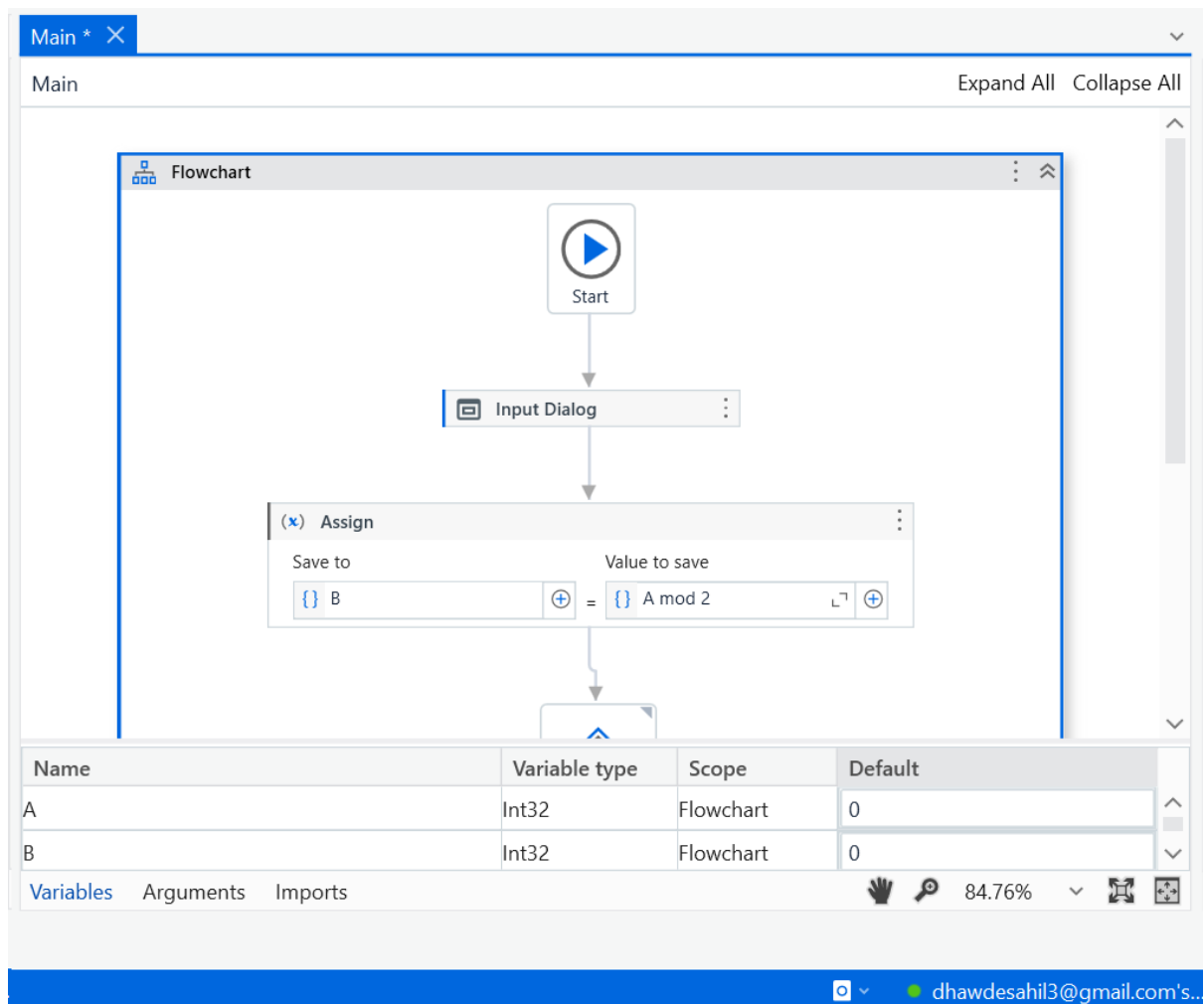


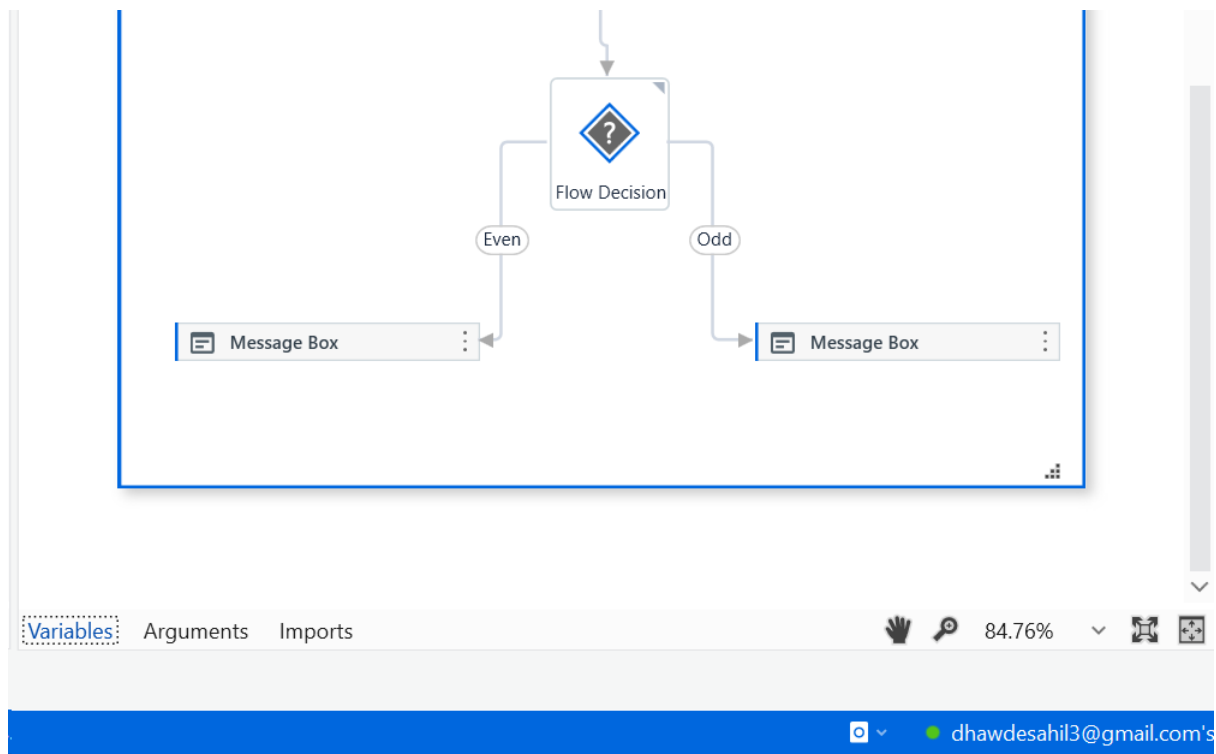
B. Create a flowchart-based project.

Steps:

1. Open Project.
2. Select Flowchart activity, double click it.
3. Create a two variables A and B (int32).
4. Select from activities panel Input dialog and enter A in Value entered.
5. Select Assign activity and enter “B = A mod 2”.
6. Select from activities panel Flow Decision & type in Condition “B.Equals(0).
7. And change False and True labels to Odd and Even.
8. Select two Message box for Odd and Even.
9. And type “Number” + A.ToString “is even.” for even label and “Number” + A.ToString “is odd.” for odd label.
10. Run the project.

Process & Output:





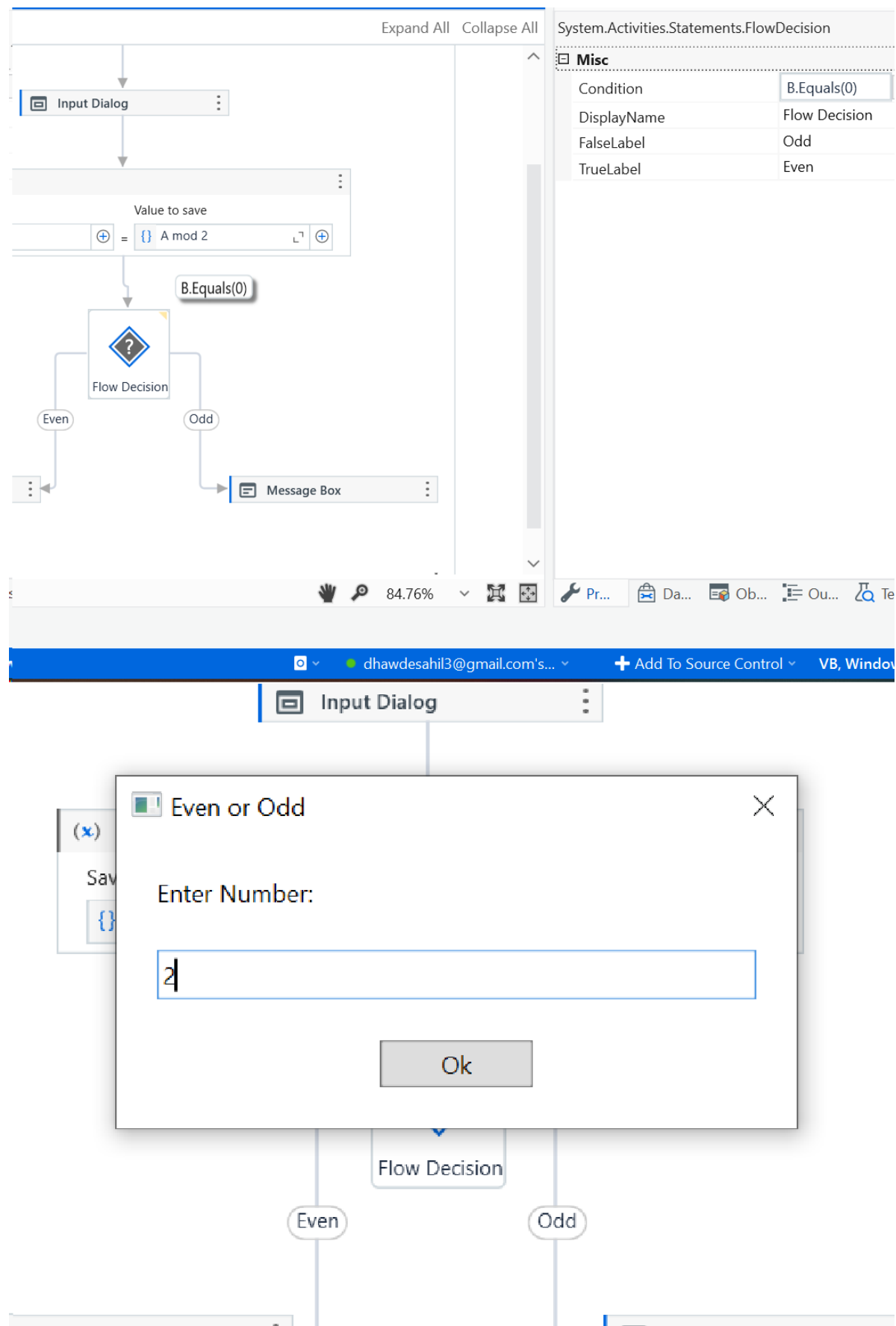
Input Dialog

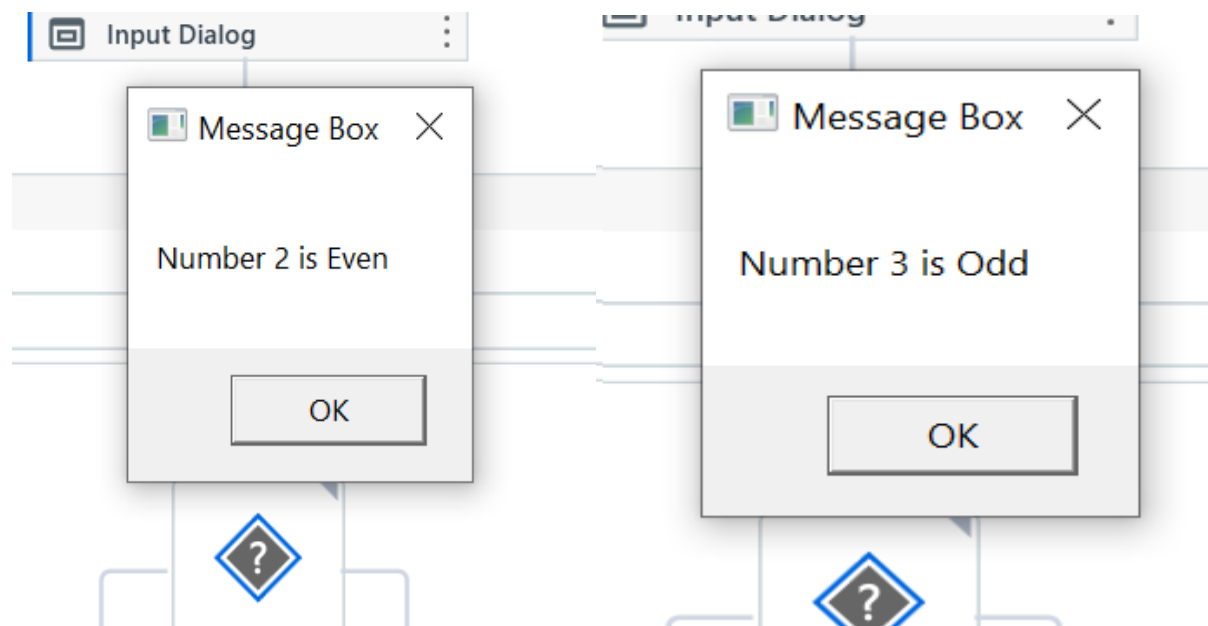
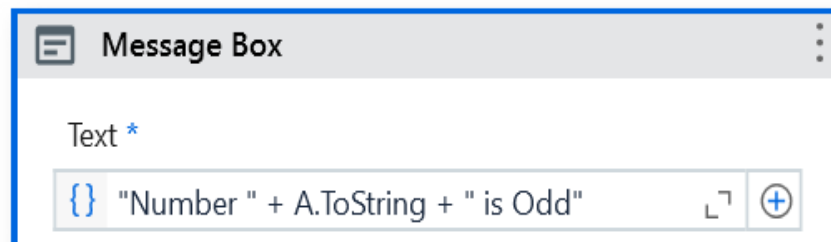
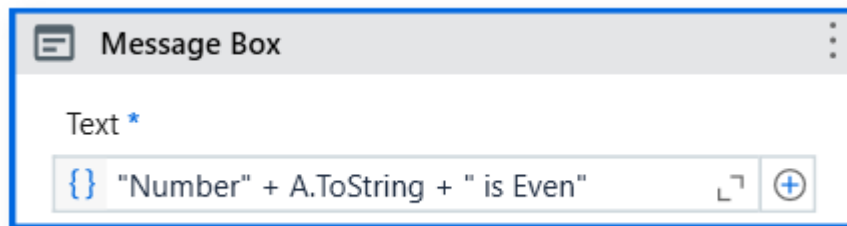
Dialog Title
{} "Even or Odd" [L] [R] [X] [Y] [Z]

Input Label
{} "Enter Number:" [L] [R] [X] [Y] [Z]

Input Type
Text Box [v]

Value entered
{} A [X] [Y] [Z]



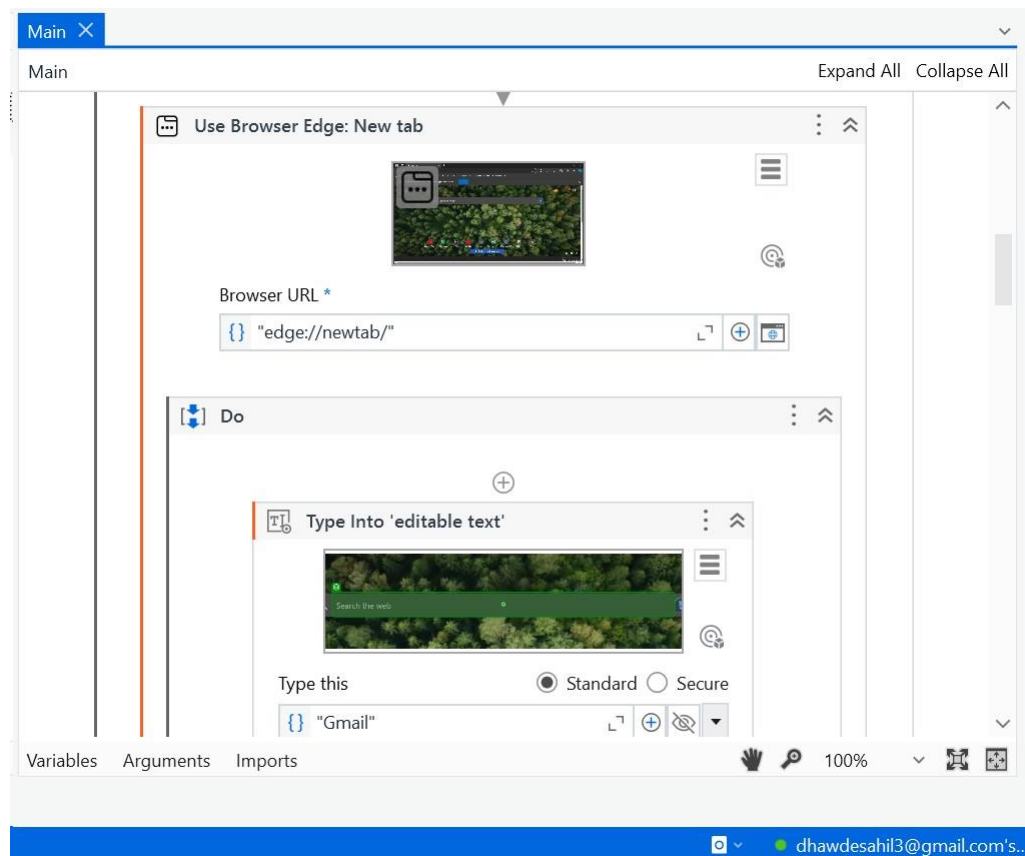


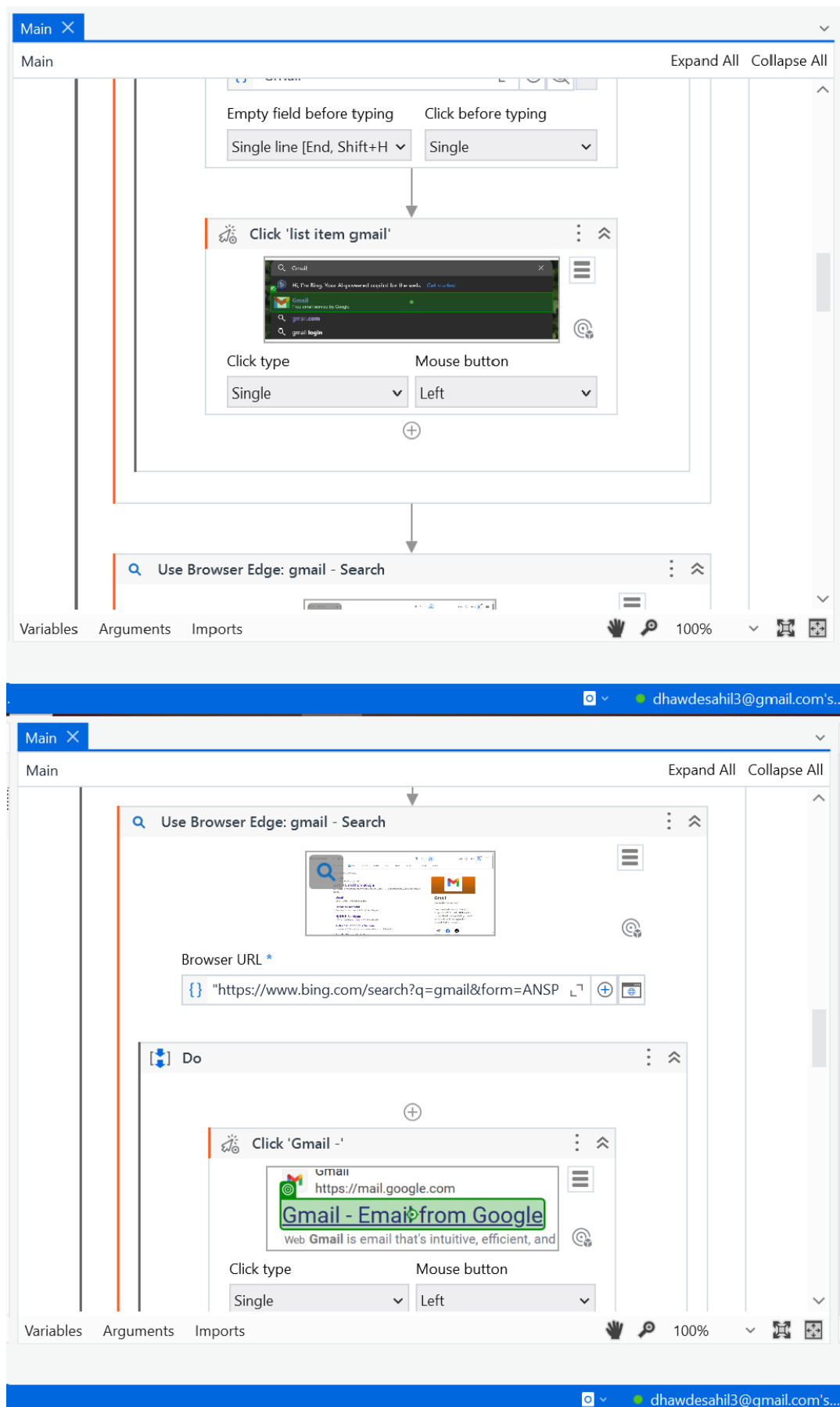
C. Create an UiPath Robot which can empty a folder in Gmail solely on basis of recording.

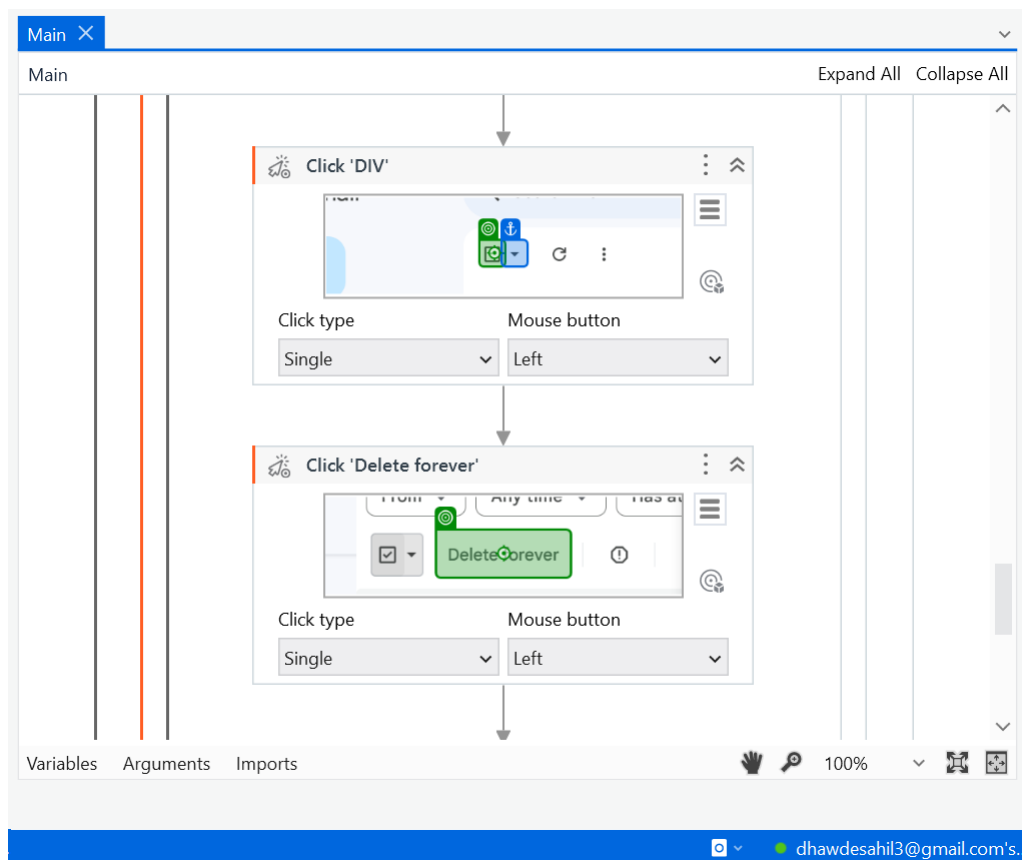
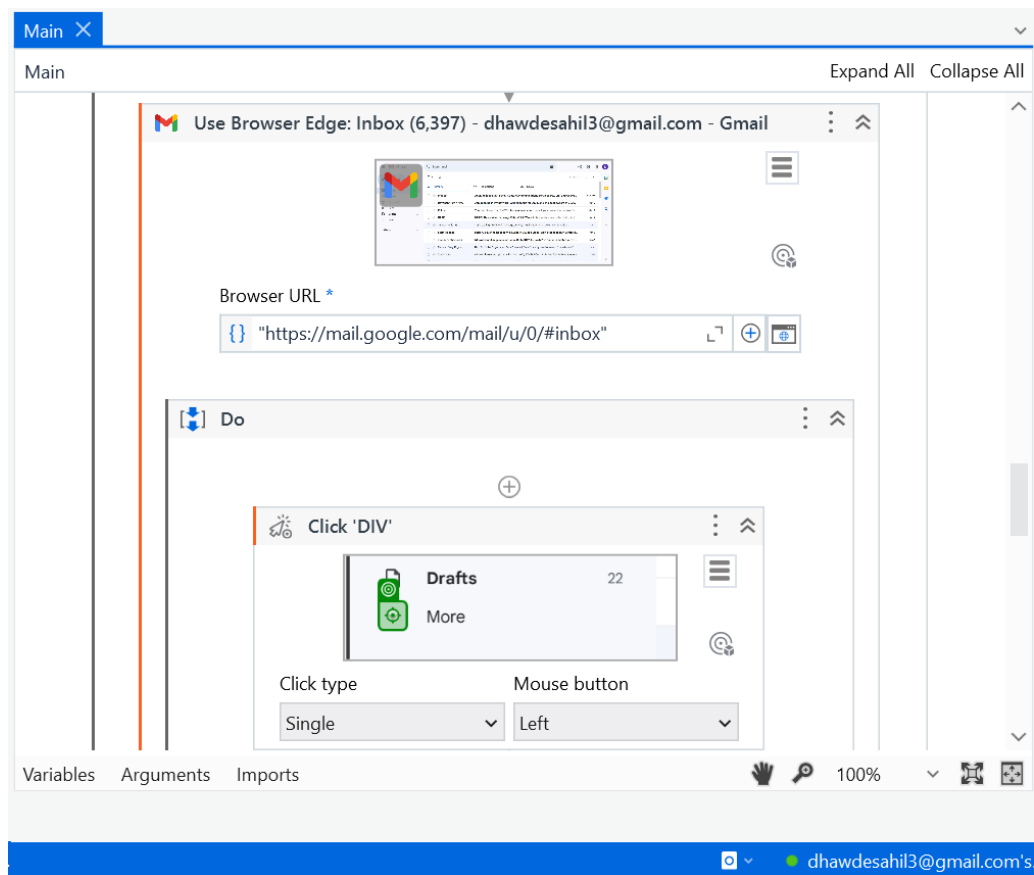
Steps:

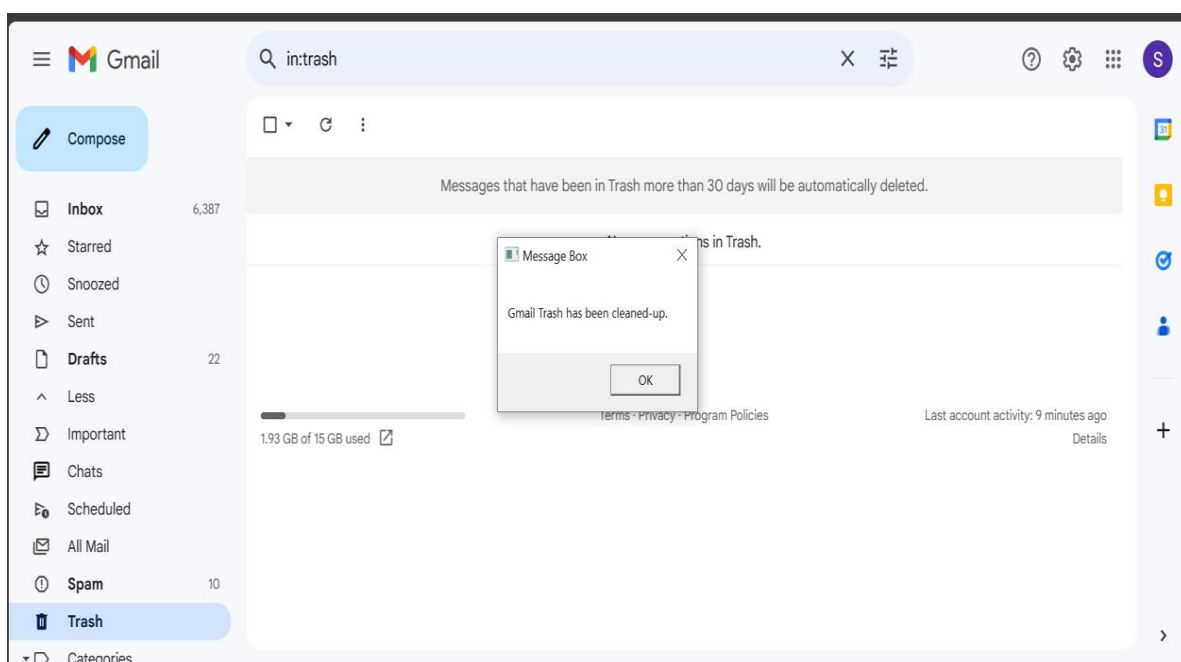
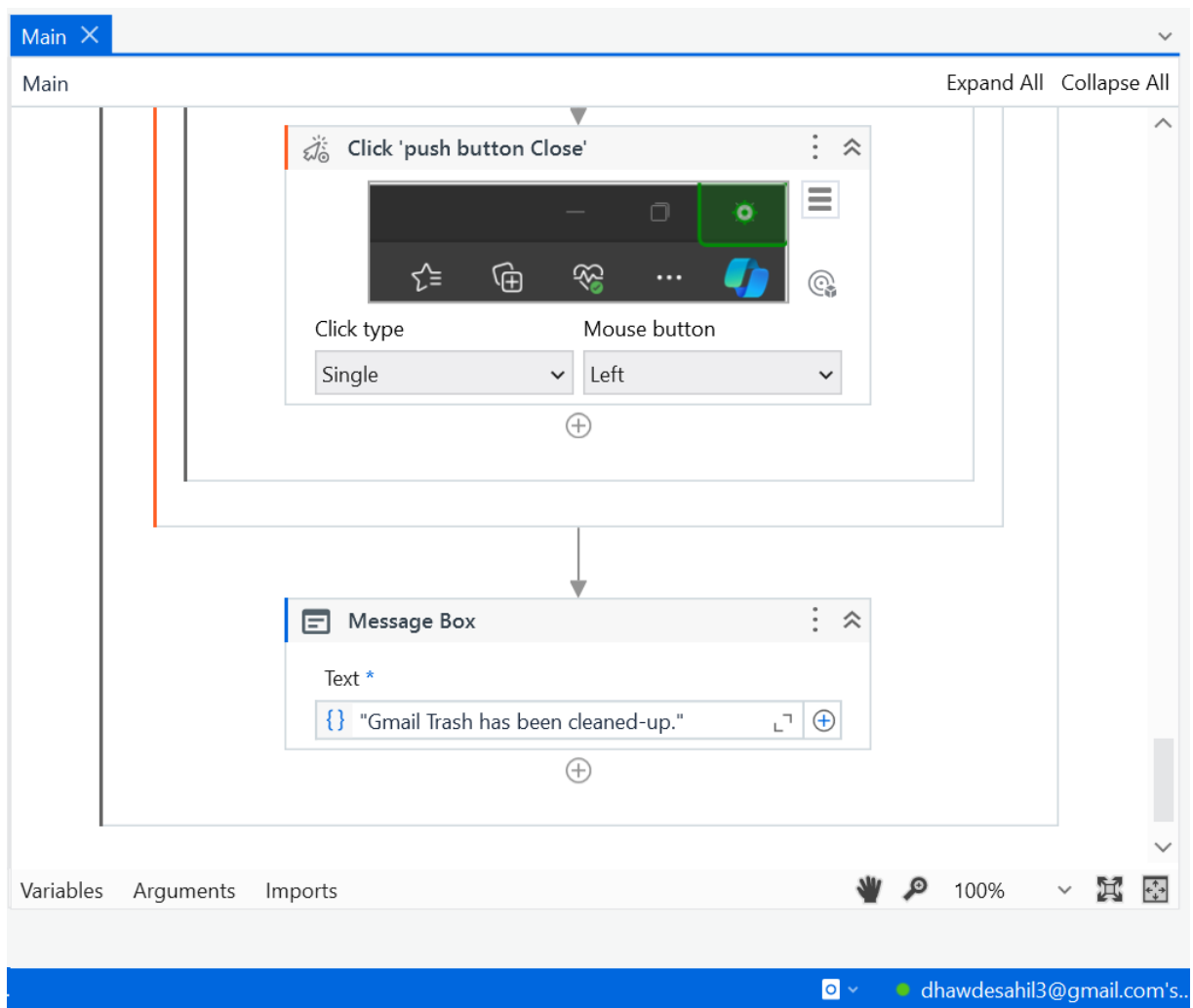
1. Open project.
2. Select Sequence activity.
3. Select App/Web Recorder icon on Ribbon panel of UiPath studio.
4. Select Microsoft Bing icon from task bar and select "Single Click" option.
5. In browser, select search bar with "Single Click" option and select Type Into activity to Type this as "Gmail".
6. Click "Gmail" on list item.
7. And select first link on browser with "Single Click" options.
8. In Gmail, select Trash that is on left side with "Single Click" options.
9. In Trash, check select all button and select "Delete Forever" button with "Single Click" options.
10. Select close button with "Single Click" options to close the browser.
11. Select from activities panel Message box activity, type "Gmail Trash has been cleaned-up."
12. Run the project.

Process & Output:









Practical No: 2

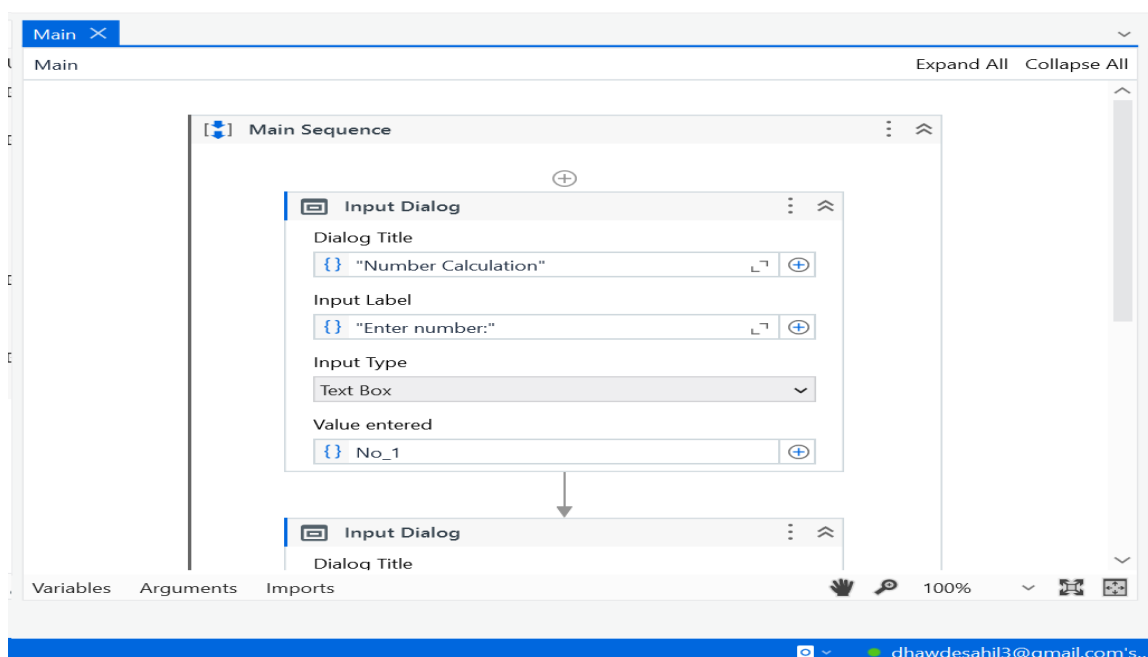
Aim: Implement the following:

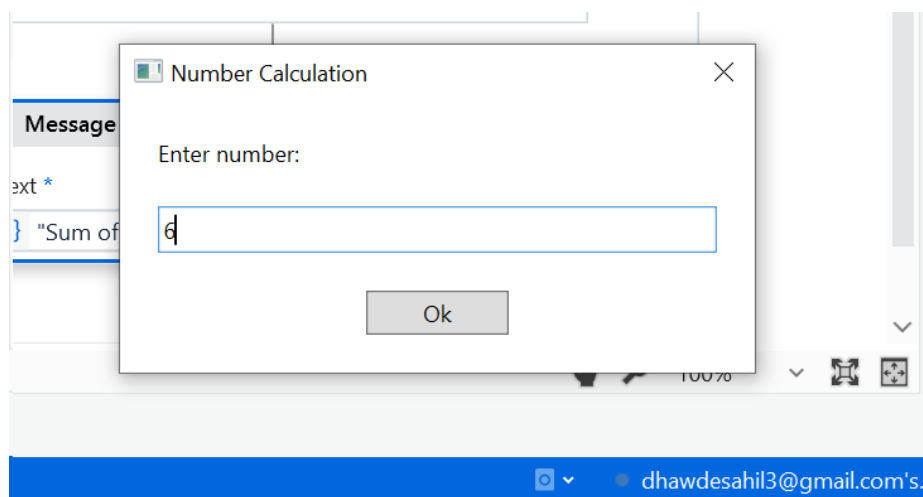
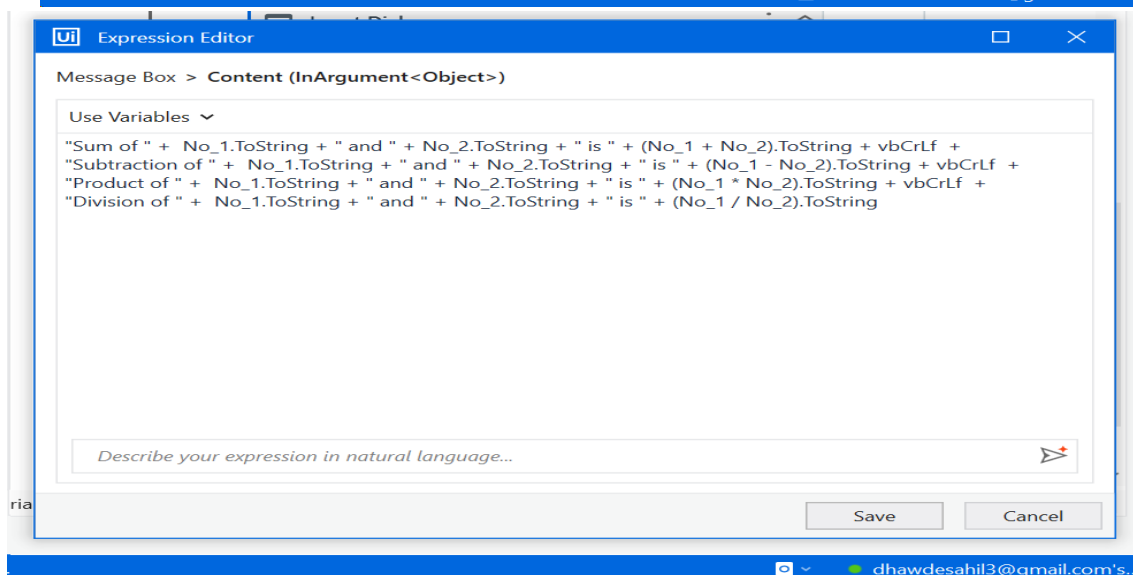
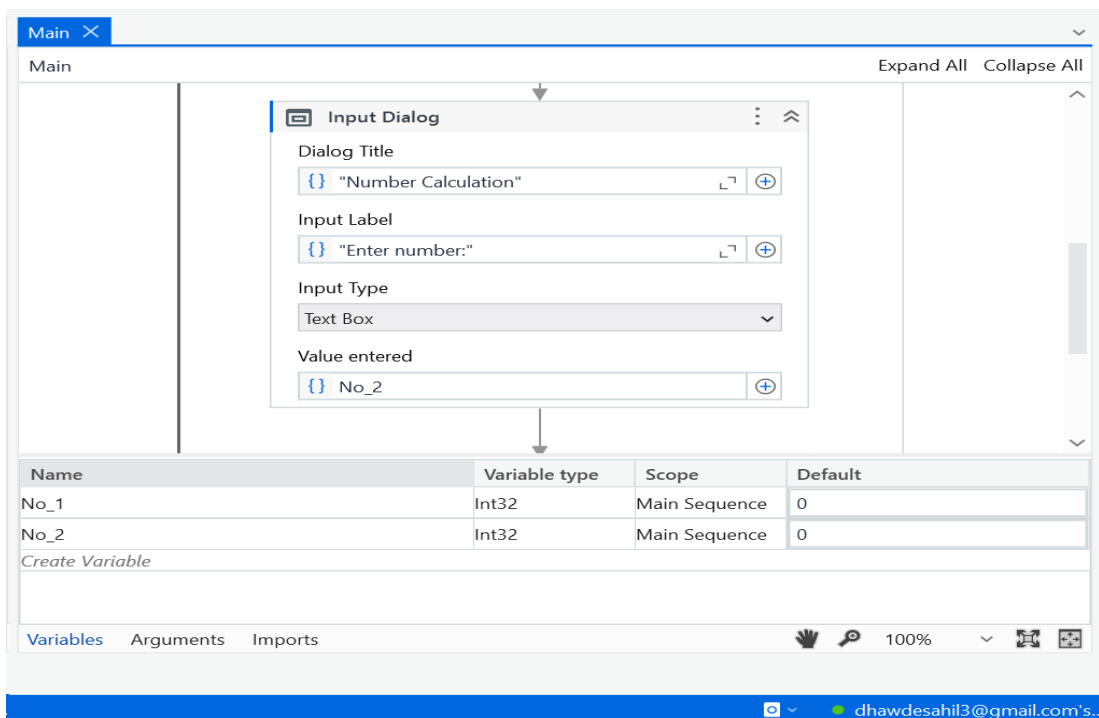
A. Automate UiPath Number Calculation (Subtraction, Multiplication, Division of numbers).

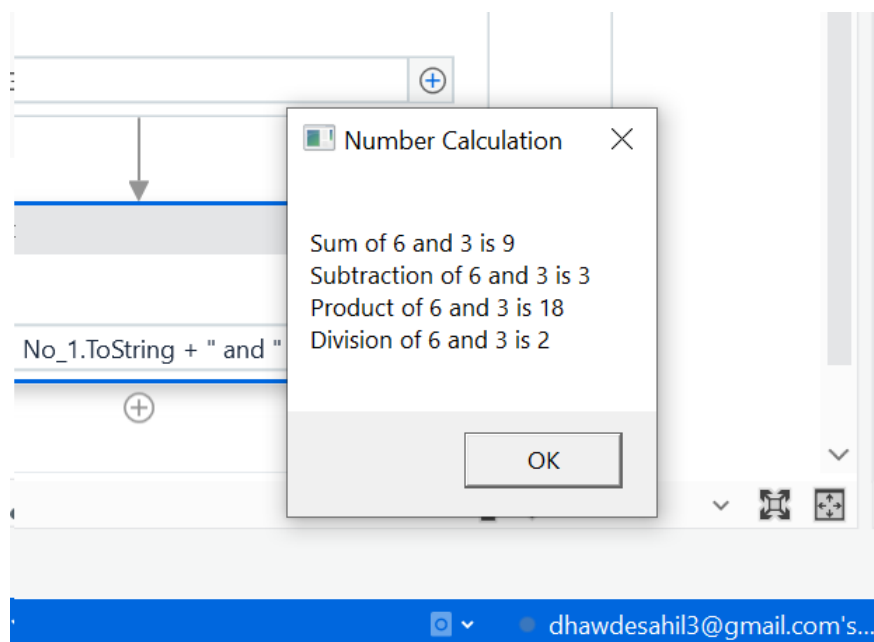
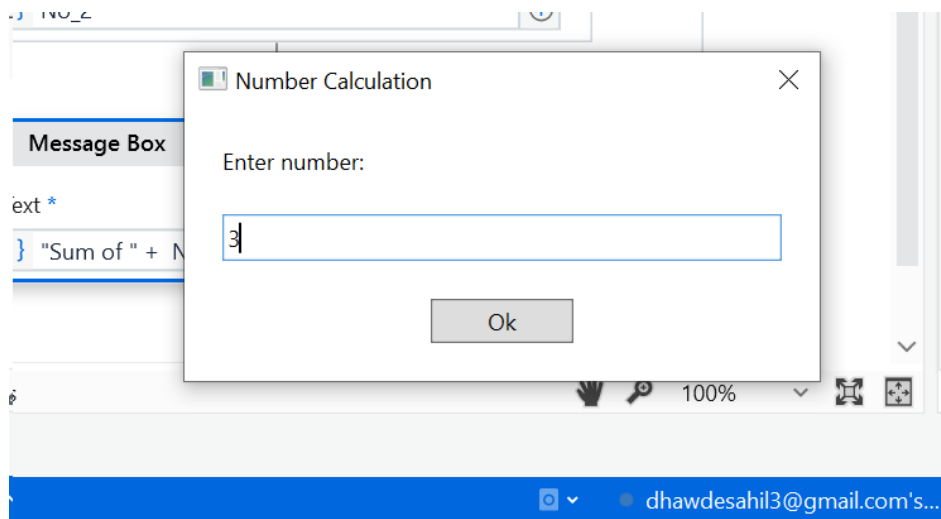
Steps:

1. Open Project.
2. Select Sequence activity.
3. Create two integer variables No_1, No_2 to hold the output.
4. Select from activities panel two Input dialog and configure it.
5. Select from activities panel Message box.
6. And type
"Sum of " + No_1.ToString + " and " + No_2.ToString + " is " + (No_1 + No_2).ToString + vbCrLf +
"Subtraction of " + No_1.ToString + " and " + No_2.ToString + " is " + (No_1 - No_2).ToString + vbCrLf +
"Product of " + No_1.ToString + " and " + No_2.ToString + " is " + (No_1 * No_2).ToString + vbCrLf +
"Division of " + No_1.ToString + " and " + No_2.ToString + " is " + (No_1 / No_2).ToString".
7. Run the project.

Process & Output:







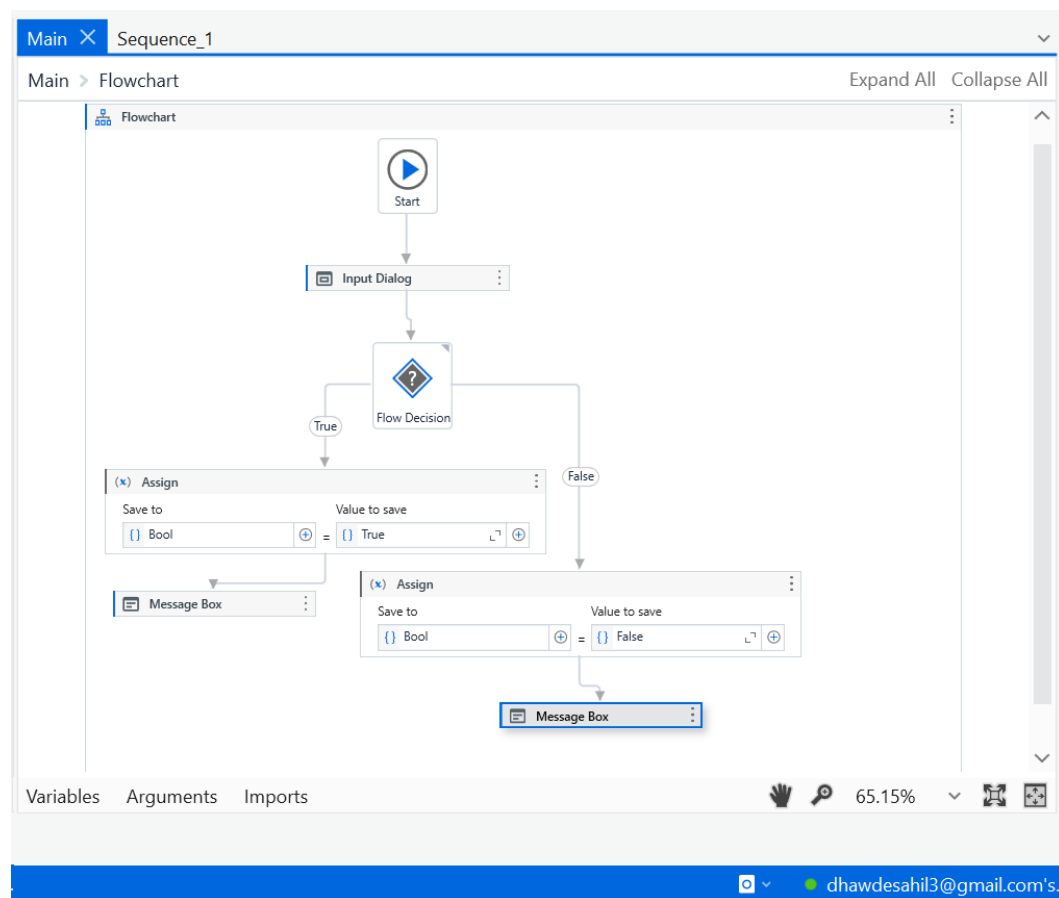
B. Create an automation UiPath project using different types of variables (number, datetime, Boolean)

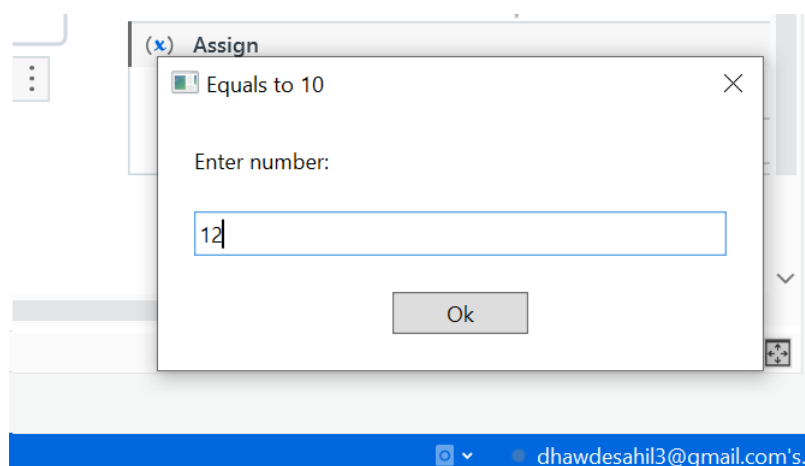
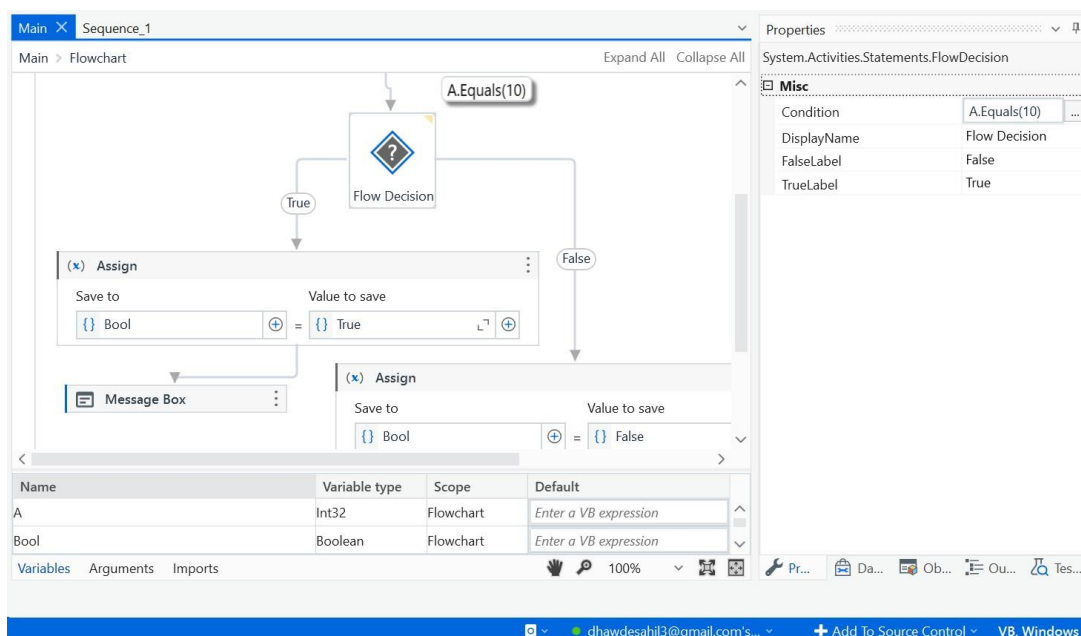
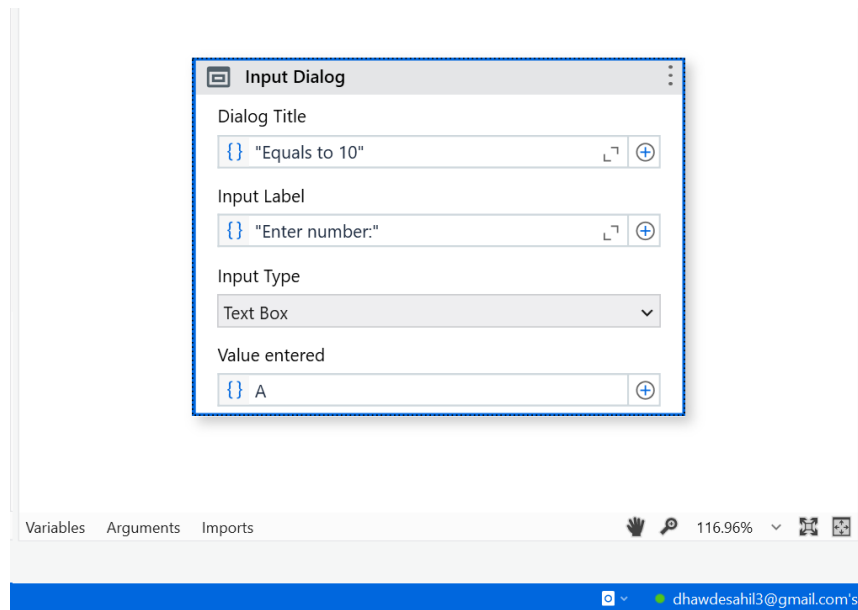
i. Boolean and Number

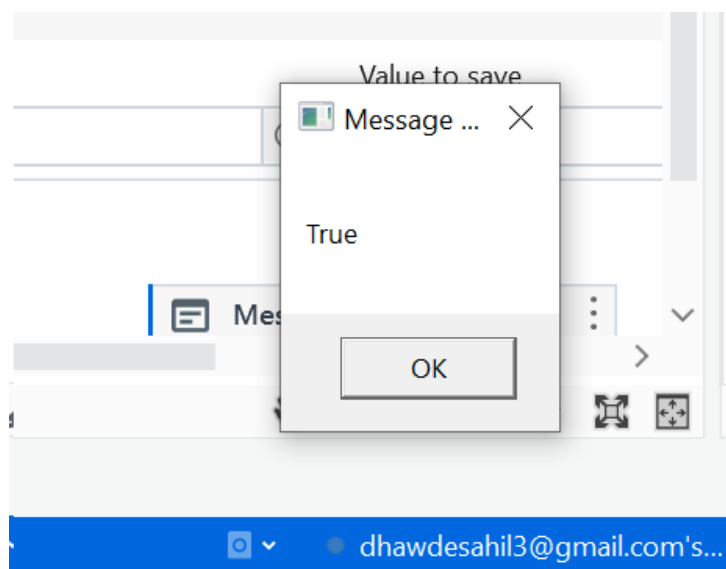
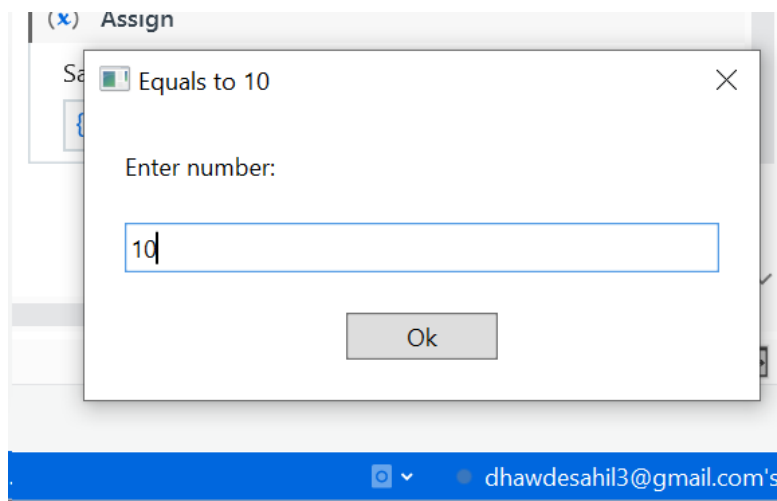
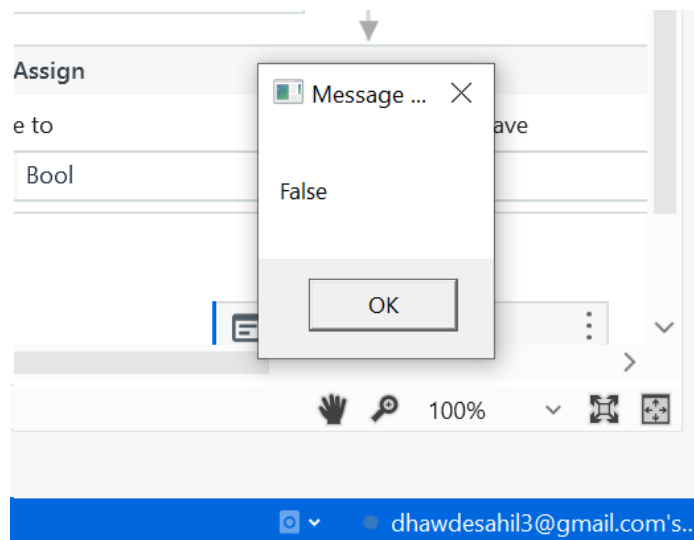
Steps:

1. Open Project.
2. Select Flowchart activity, double click it.
3. Create a two variables A (int32) and Bool (Boolean).
4. Select from activities panel Input dialog and enter A in Value entered.
5. Select from activities panel Flow Decision & type in Condition “A.Equals(10).
6. Select two Assign activity, one for True and other for False.
7. Assign bool = True for True side and Assign bool = False for False side
8. Select two Message box activity for True and False.
9. And type “bool” for both True label and False label.
10. Run the project.

Process & Output:





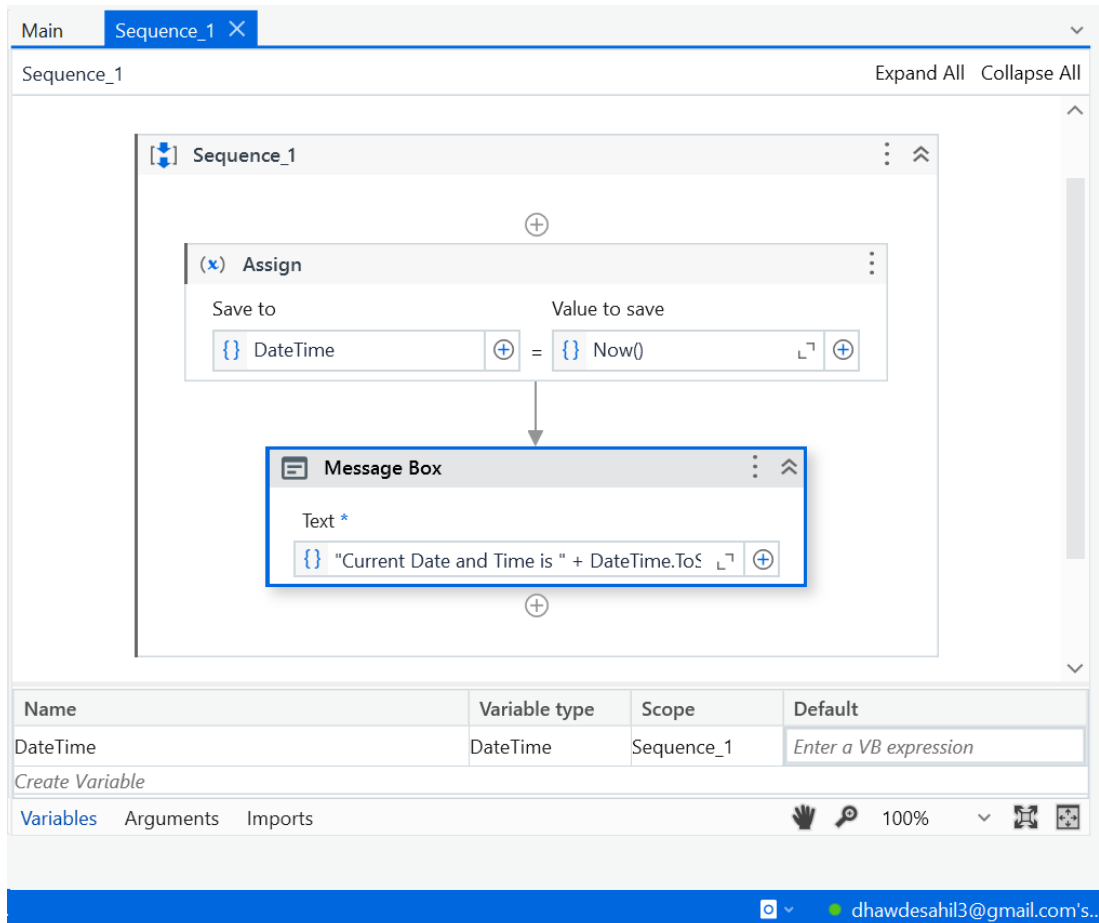


ii. Datetime

Steps:

1. Open Project.
2. Select Sequence activity.
3. Create a variables DateTime of DateTime type.
4. Select Assign activity, assign DateTime = Now().
5. Select from activities panel Message box activity and type "Current Date and Time is " + DateTime.ToString".
6. Run the project.

Process & Output:



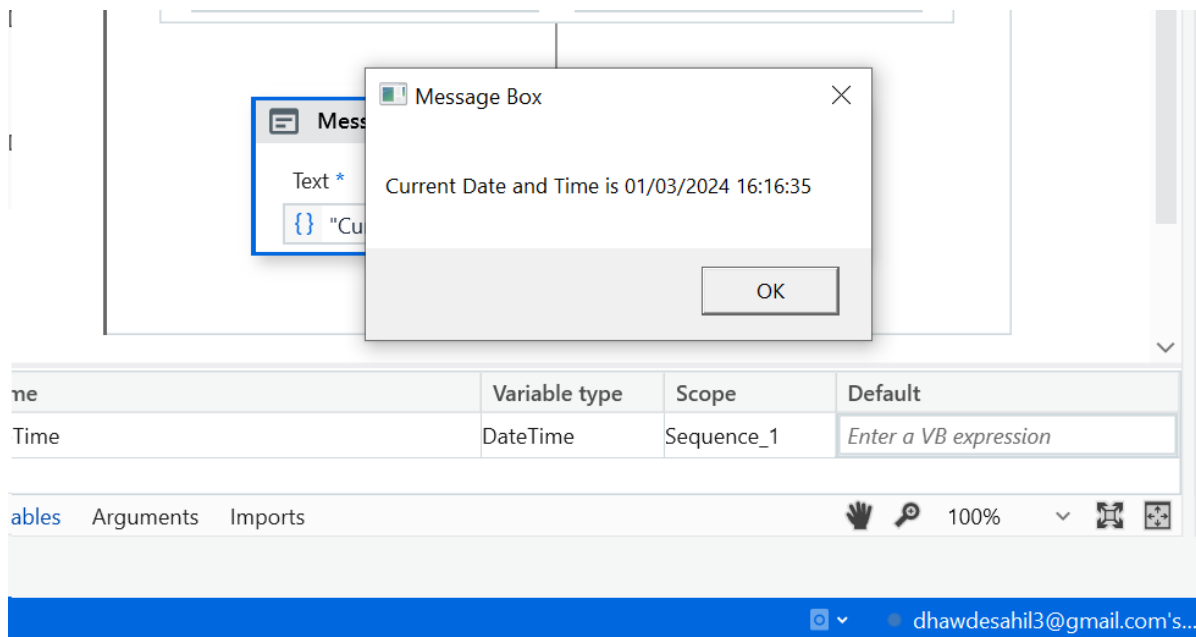
The screenshot displays the Visual Studio IDE with a project named 'Sequence_1'. The 'Sequence_1' activity is expanded, showing two sub-activities: 'Assign' and 'Message Box'. The 'Assign' activity is configured to save the value of 'Now()' to a variable named 'DateTime'. The 'Message Box' activity is configured to display the text 'Current Date and Time is ' + DateTime.ToString(). Below the activities, a table shows the variable 'DateTime' with type 'DateTime' and scope 'Sequence_1'. The bottom status bar shows the user 'dhawdesahil3@gmail.com's...'.

Name	Variable type	Scope	Default
DateTime	DateTime	Sequence_1	Enter a VB expression

Create Variable

Variables Arguments Imports

100%



Practical No: 3

Aim: Implement the following:

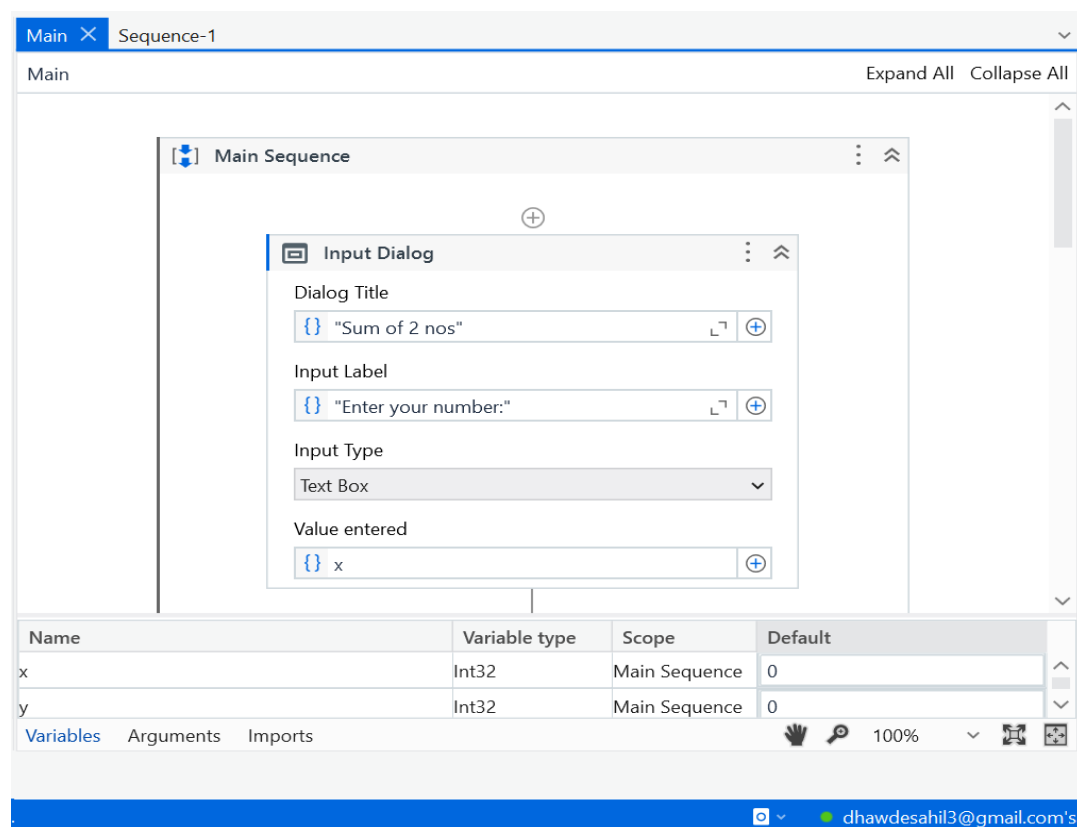
A. Create an automation UiPath Project using decision statements.

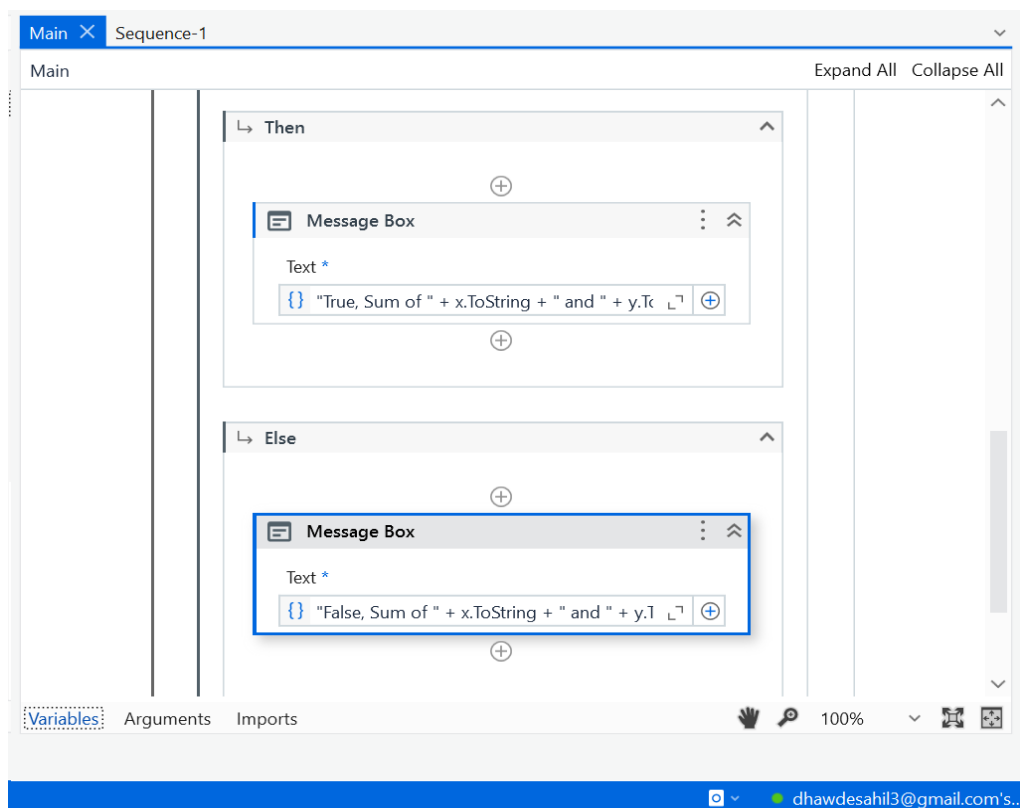
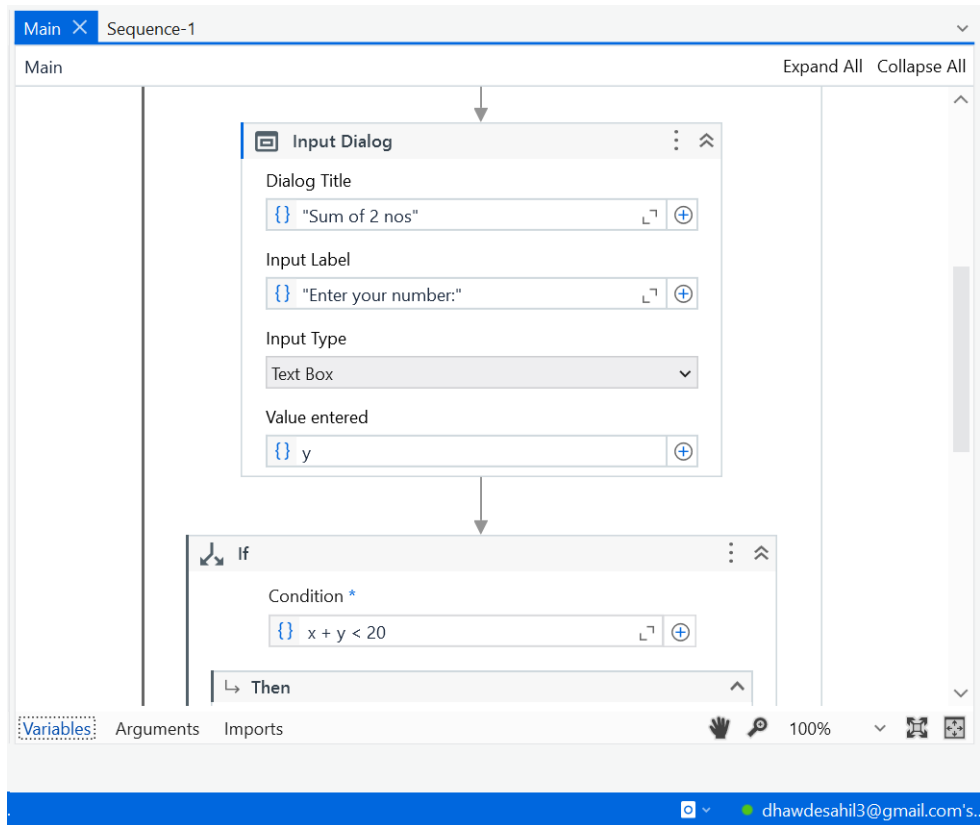
i. If statement

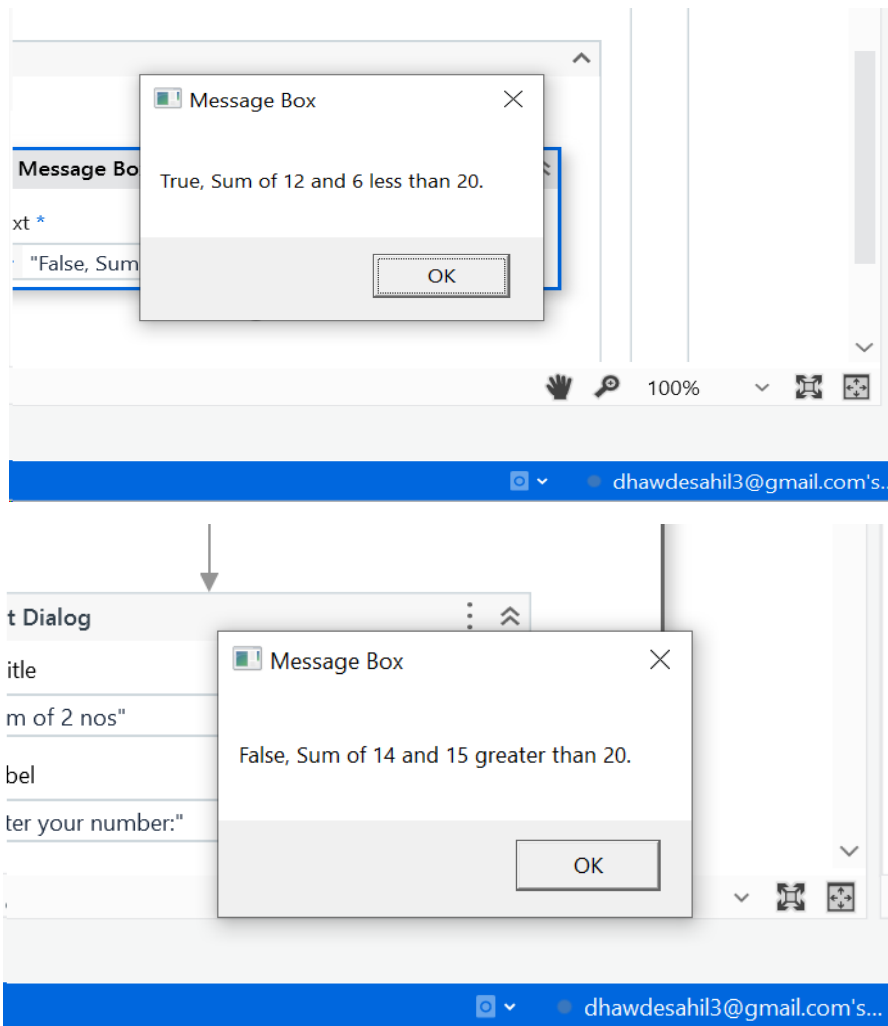
Steps:

1. Open Project.
2. Select Sequence activity.
3. Select from activities panel two Input dialog and configure it.
4. Create two integer variables x, y to hold the output.
5. Select from activities panel If activity & in condition type $x + y < 20$.
6. Drag n Drop a Write line activity to "Then" part & enter text "True, Sum " + x.ToString
" and " + y.ToString + "less than 20" .
7. Drag n Drop a Write line activity to "Else" part & enter text "False, Sum " + x.ToString
" and " + y.ToString + " greater than 20" .
8. Run the project.

Process & Output:





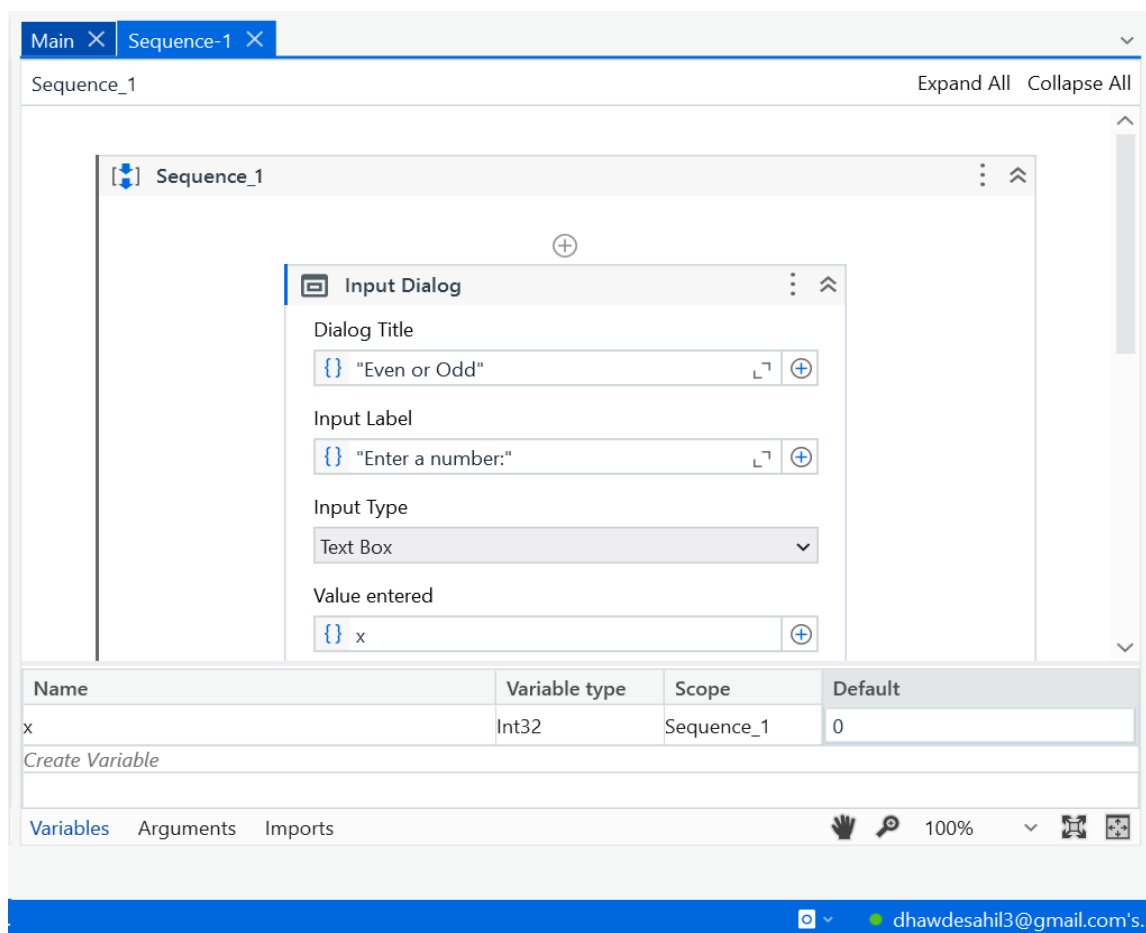


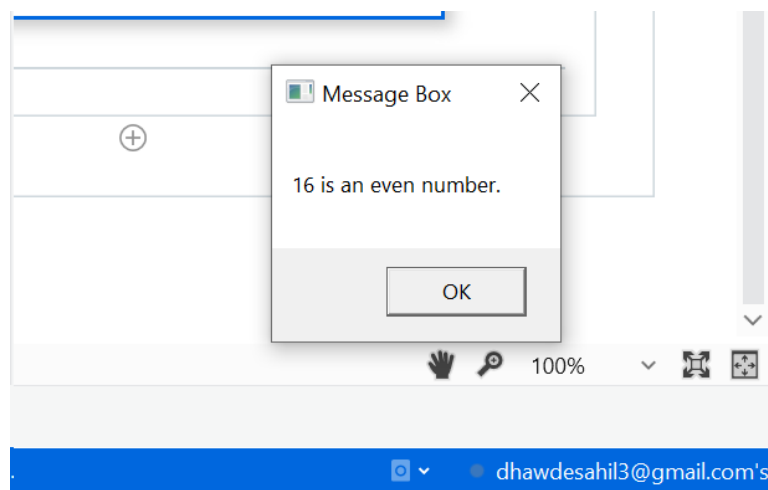
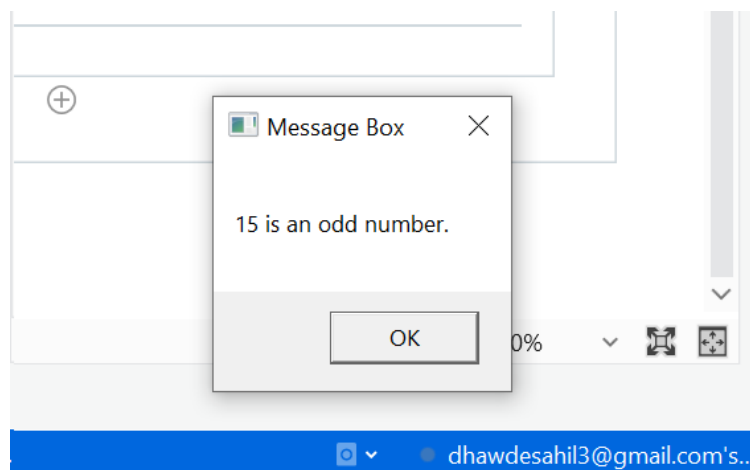
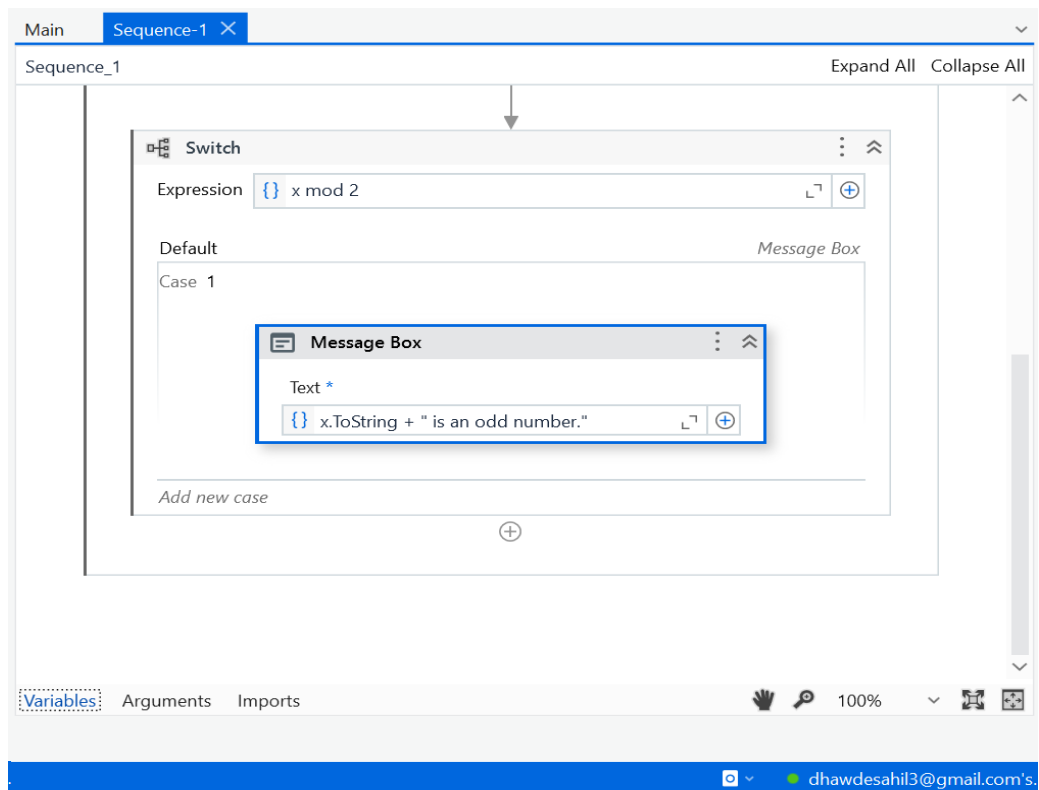
ii. Switch statement

Steps:

1. Open Project.
2. Select Sequence activity.
3. Select from activities panel, input dialog, and configure it.
4. Create one integer variables x, to hold the output.
5. Select from activities panel Switch activity.
6. In Expression type "x Mod 2".
7. In Case 0 (default case) Drag n Drop a message Box activity & enter text "x.toString + is an even number".
8. Add Case, In Case 1 Drag n Drop a message Box activity & enter text "x.toString + is an Odd number".
9. Run the project.

Process & Output:





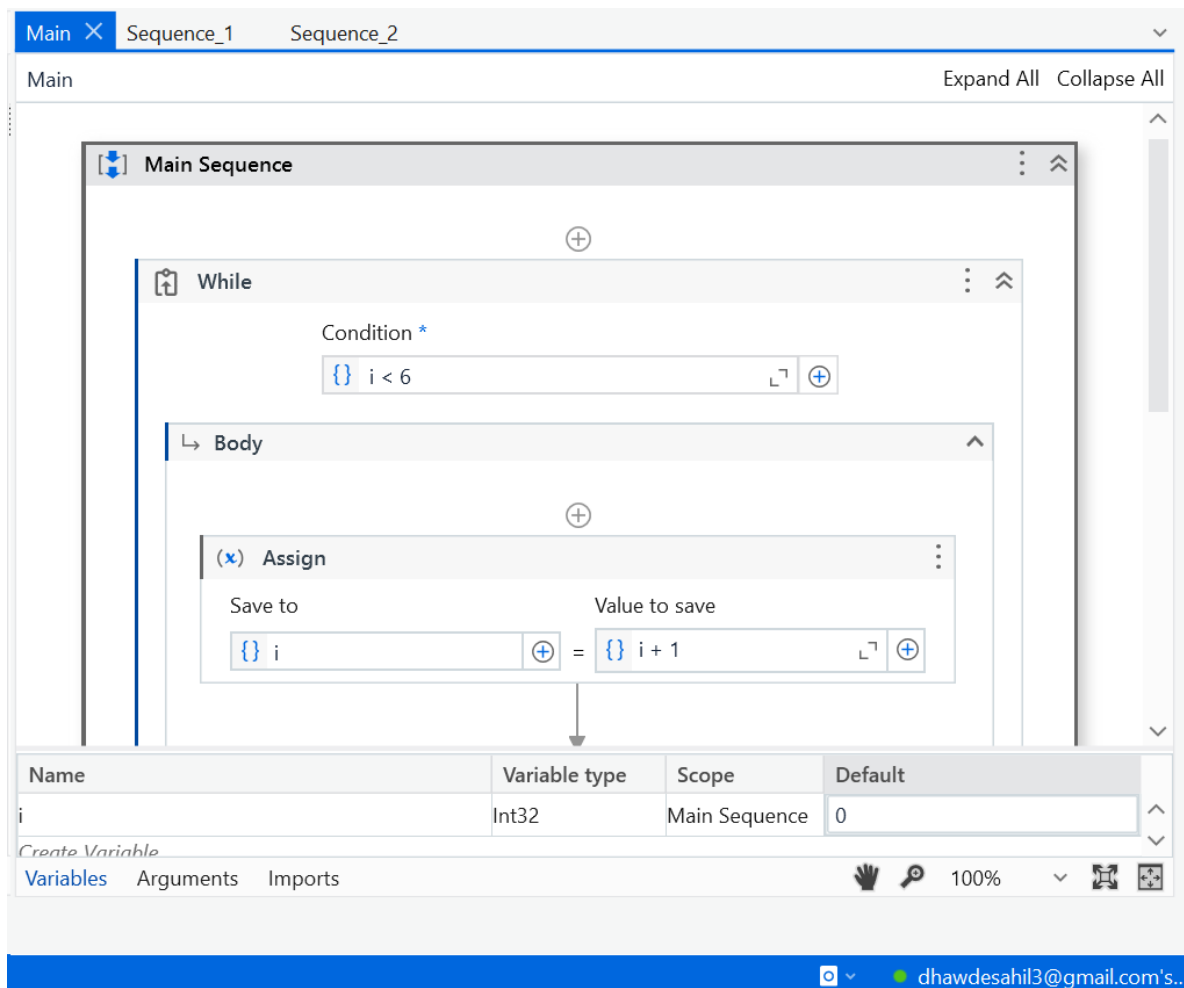
B. Create an automation UiPath Project using looping statements.

i. While loop

Steps:

1. Open Project.
2. Select the Sequence activity.
3. Create a variable i (int32).
4. Create and assign variable i = 0.
5. Select While activity.
6. In condition type "i < 6"
7. Inside the body of While, select the Assign activity & enter text "i = i+1"
8. Then select Delay activity to delay the output of write line.
9. Now select Write line activity and enter text "i.ToString".
10. Run the project.

Process & Output:



The screenshot displays the Visual Studio Code interface for a sequence editor. The top bar shows tabs for 'Main', 'Sequence_1', and 'Sequence_2'. The 'Main' tab is active, showing a sequence diagram with two blocks: 'Delay' and 'Write Line'. The 'Delay' block has a duration of '00h 00m 05.000s'. The 'Write Line' block has a text field containing '{ } i.ToString'. Below the sequence diagram, there is a table with the following columns: Name, Variable type, Scope, and Default. The table contains one row with the variable 'i' of type 'Int32' and scope 'Main Sequence', with a default value of '0'. Below the table, there are tabs for 'Variables', 'Arguments', and 'Imports'. The 'Variables' tab is active, showing a search bar and a list of variables. The 'Output' panel is visible at the bottom, showing a search bar and a list of output messages. The output messages include 'Looping execution started', '1', '2', '3', '4', '5', '6', and 'Looping execution ended in: 00:00:30'. The bottom status bar shows the file name 'dhawdesahil3@gmail.com's...'.

Sequence Editor Interface:

- Sequence Diagram:
 - Delay (Duration: 00h 00m 05.000s)
 - Write Line (Text: { } i.ToString)
- Variable Table:

Name	Variable type	Scope	Default
i	Int32	Main Sequence	0
- Output Console:

Search

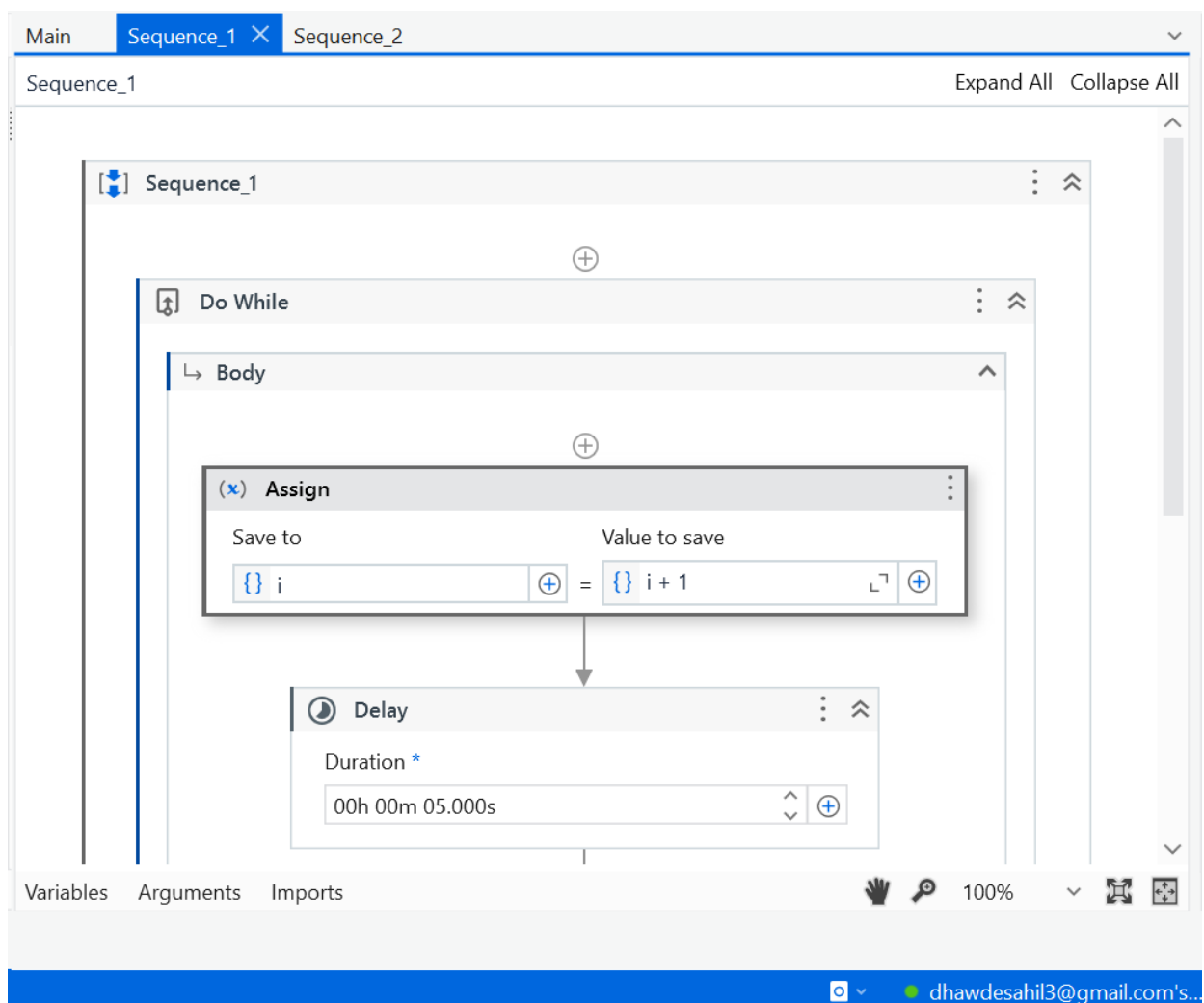
 - Looping execution started
 - 1
 - 2
 - 3
 - 4
 - 5
 - 6
 - Looping execution ended in: 00:00:30

ii. Do while loop

Steps:

1. Open Project.
2. Select the Sequence activity.
3. Create a variable i (int32).
4. Create and assign variable i = 0.
5. Select Do While activity.
6. Inside the body of Do while, select the Assign activity & enter text "i = i+1"
7. Then select Delay activity to delay the output of write line.
8. Now select Write line activity and enter text "i.ToString".
9. In condition type "i < 1"
10. Run the project.

Process & Output:



The screenshot displays a sequence editor interface with a tab labeled 'Sequence_1'. The sequence contains three steps: a 'Duration' step set to '00h 00m 05.000s', a 'Write Line' step with the text 'i.ToString', and a 'Condition' step with the expression 'i < 6'. Below the sequence, a table lists the variable 'i' with type 'Int32' and scope 'Sequence_1', having a default value of '0'. The bottom status bar shows the email 'dhawdesahil3@gmail.com's'.

Name	Variable type	Scope	Default
i	Int32	Sequence_1	0

Variables Arguments Imports

Output

Looping execution started

1

2

3

4

5

6

Looping execution ended in: 00:00:30

Output Breakpoints Error List

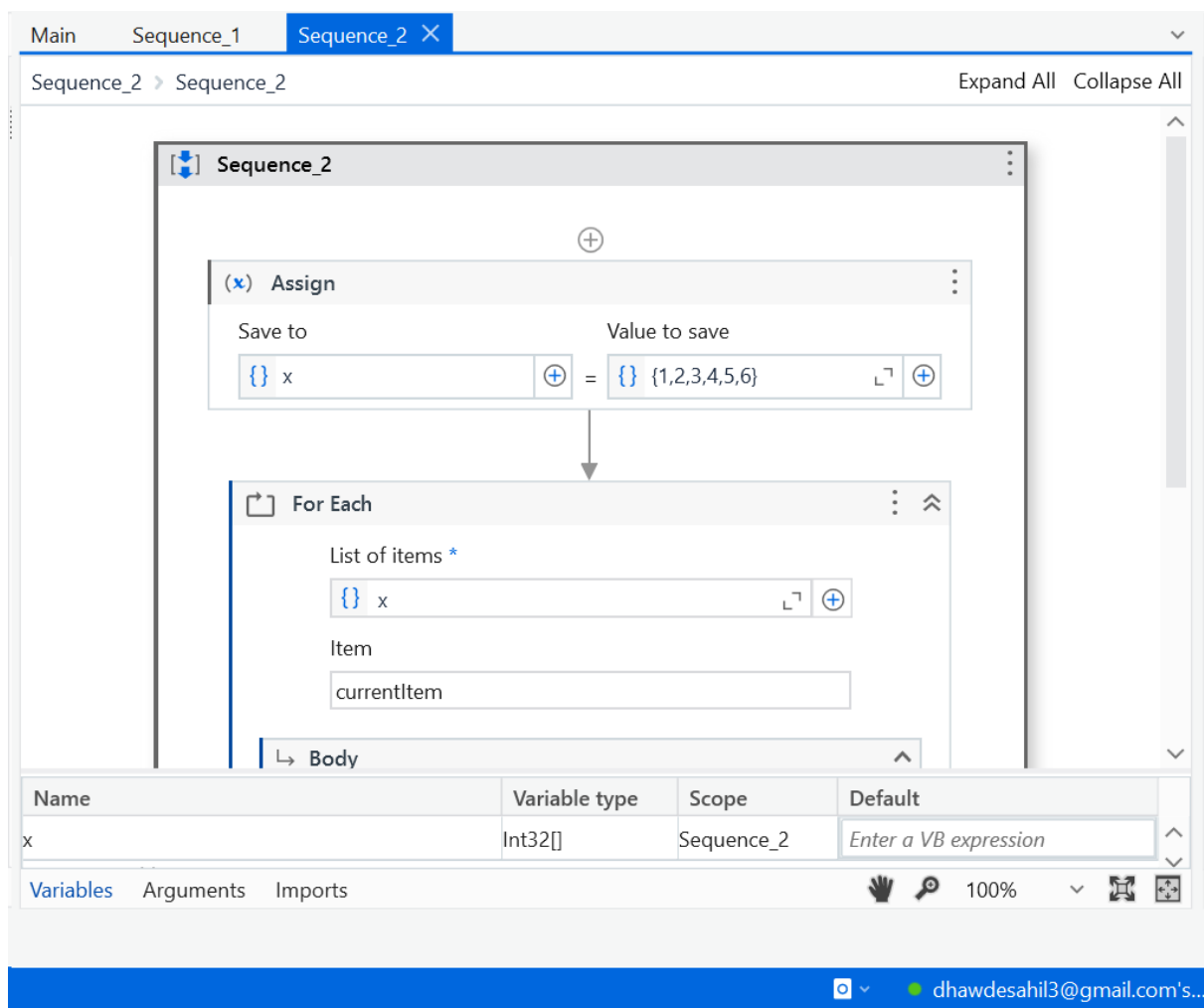
dhawdesahil3@gmail.com's

iii. For each loop

Steps:

1. Open Project.
2. Select the Sequence activity.
3. Create two variables x (array int32).
4. Select Assign activity & set x = {1,2,3,4,5,6}.
5. Select For Each activity.
6. Set values of List of items as "x".
7. Set values of Item as "currentItem".
8. Inside the Body of For Each.
9. Select the Write line activity & enter text "currentItem.ToString".
10. Run the project.

Process & Output:



Body

Write Line

Text

{ } currentItem.ToString

Name	Variable type	Scope	Default
x	Int32[]	Sequence_2	Enter a VB expression

Variables Arguments Imports

Output

Search

- Looping execution started
- 1
- 2
- 3
- 4
- 5
- 6

Breakpoints Error List

dhawdesahil3

Practical No: 4

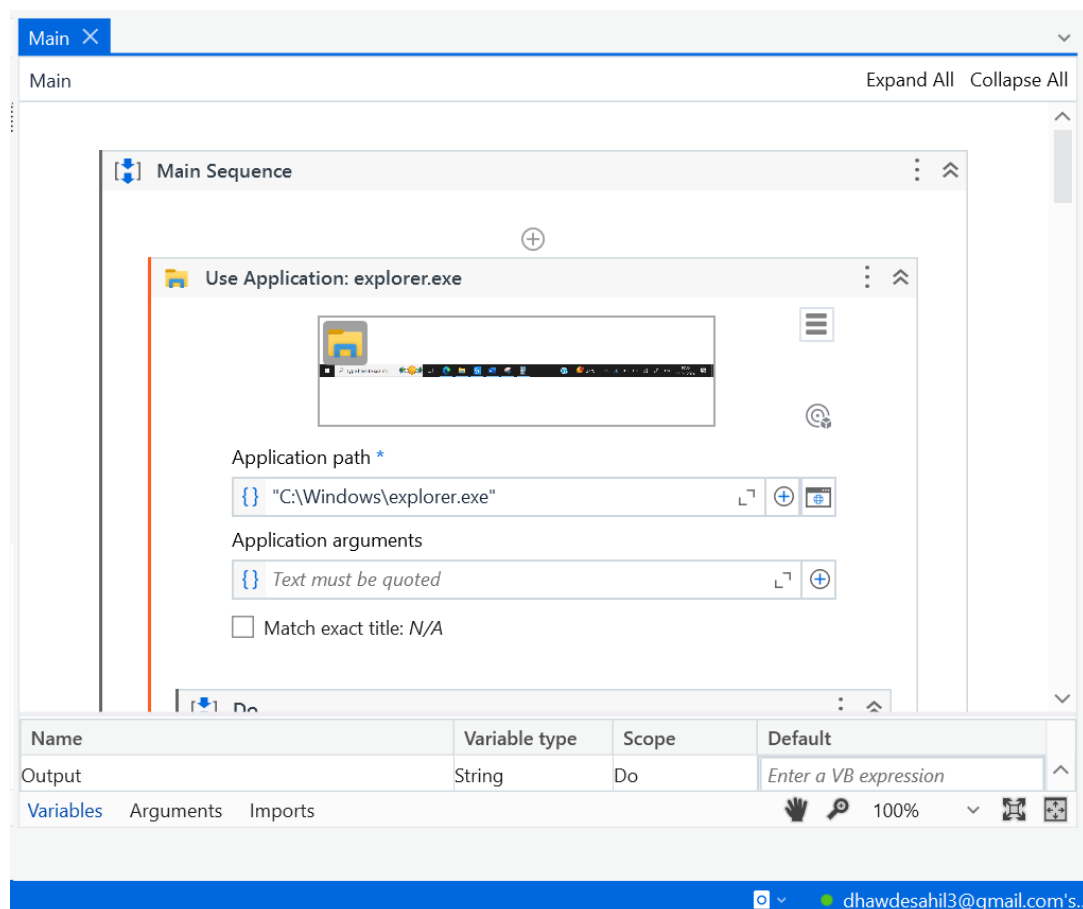
Aim: Implement the following:

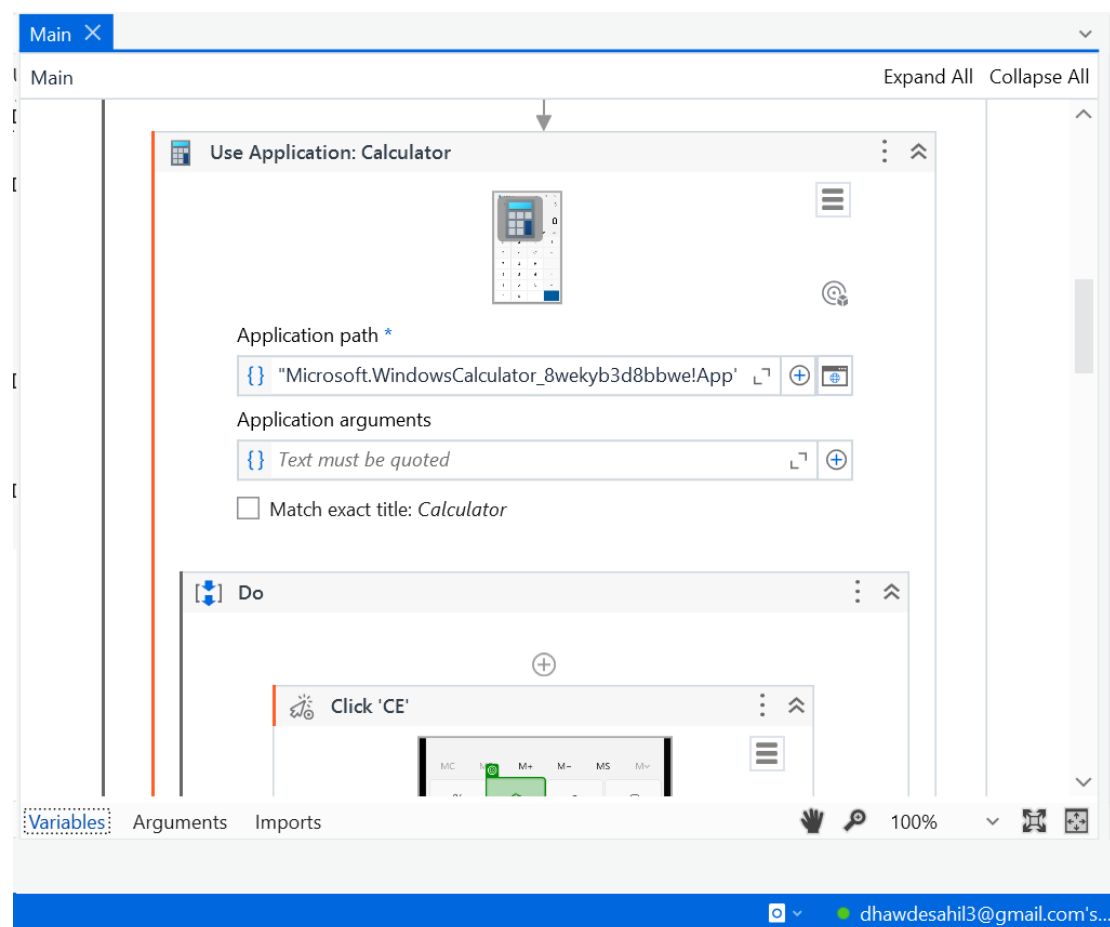
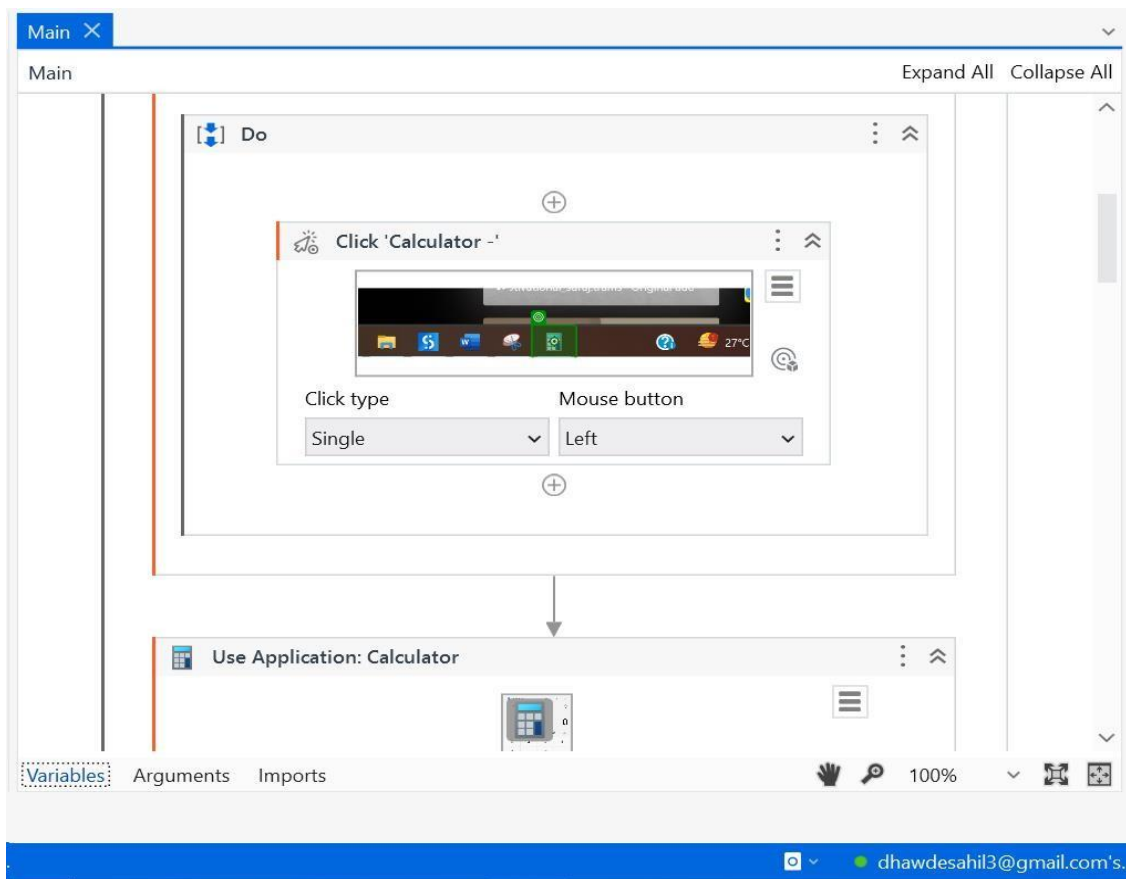
A. Automate any process using basic recording.

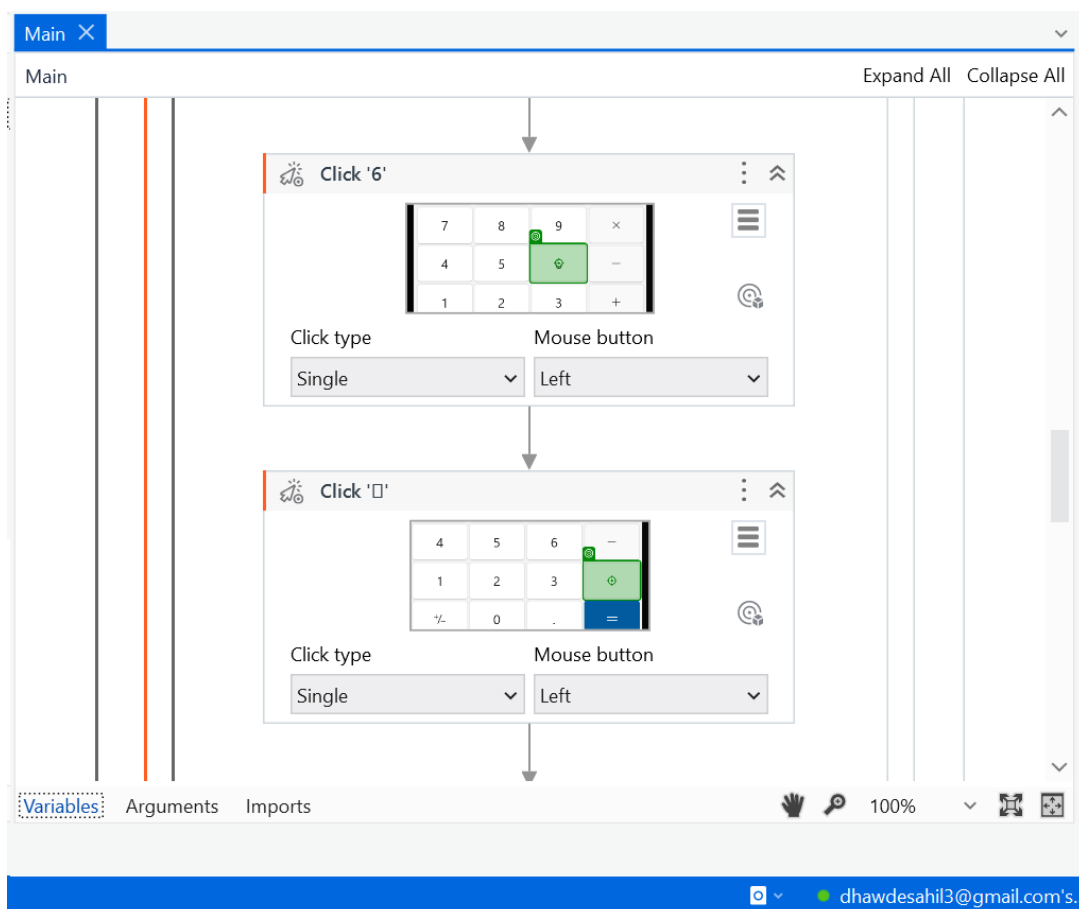
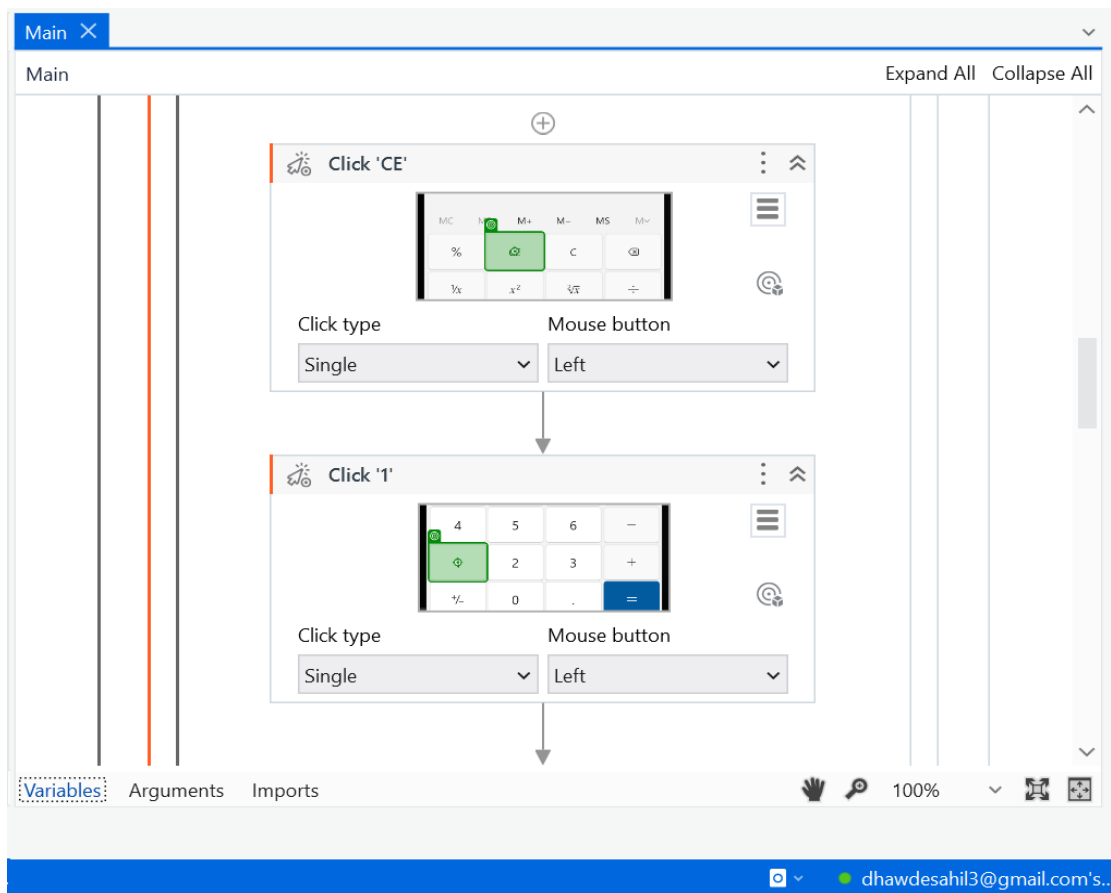
Steps:

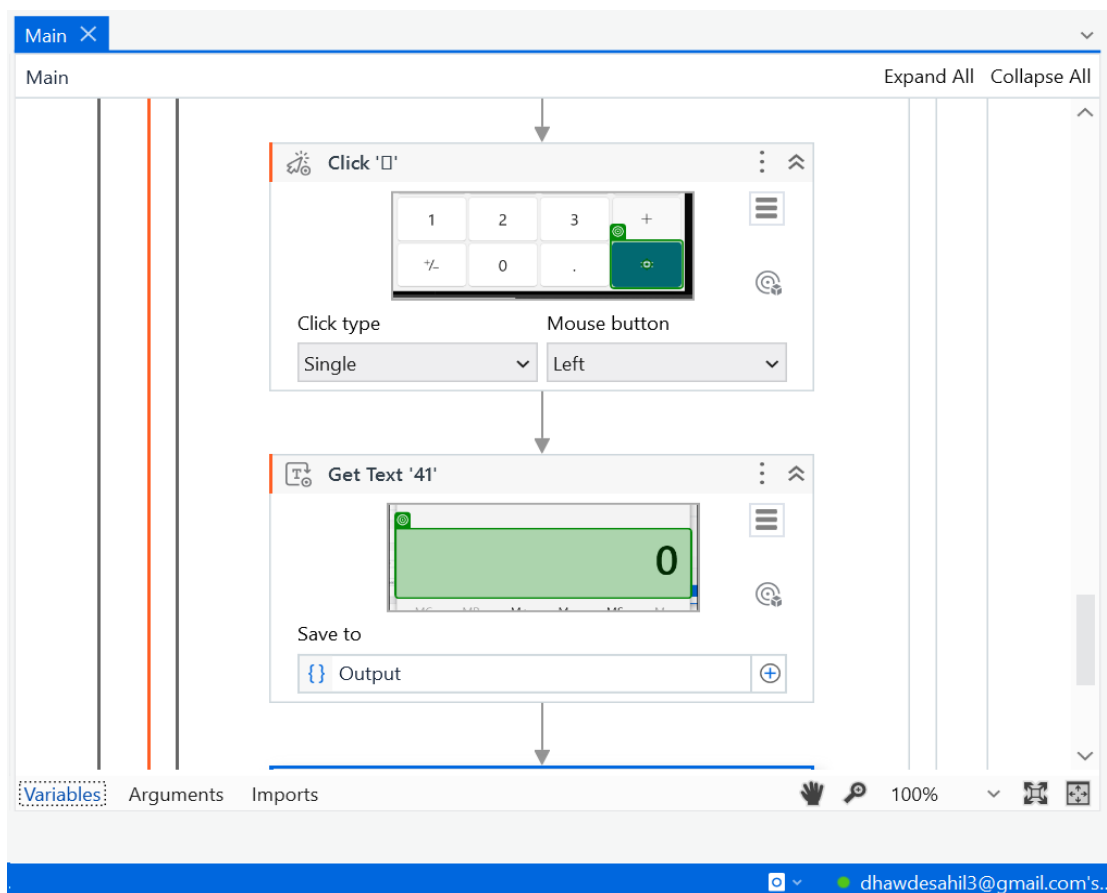
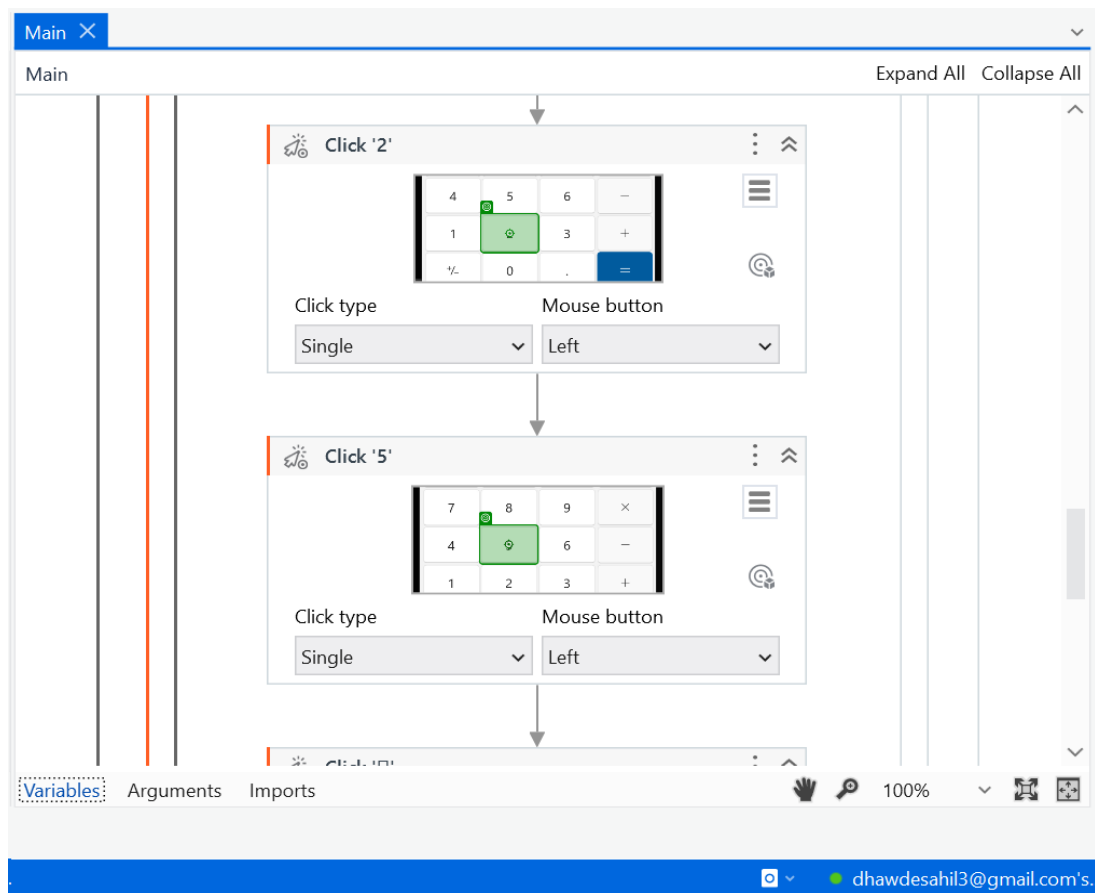
1. Open project.
2. Select Sequence activity.
3. Select App/Web Recorder icon on Ribbon panel of UiPath studio.
4. Select Calculator icon from task bar and select “Single Click” option.
5. In calculator, select number to calculate with “Single Click” option.
6. Select output display window of calculator and select “Get Text” option.
7. In Get Text activity, save to as “Output.”
8. Select from activities panel Message box activity, type “Calculation has been completed and sum is " + Output.”
9. Run the project.

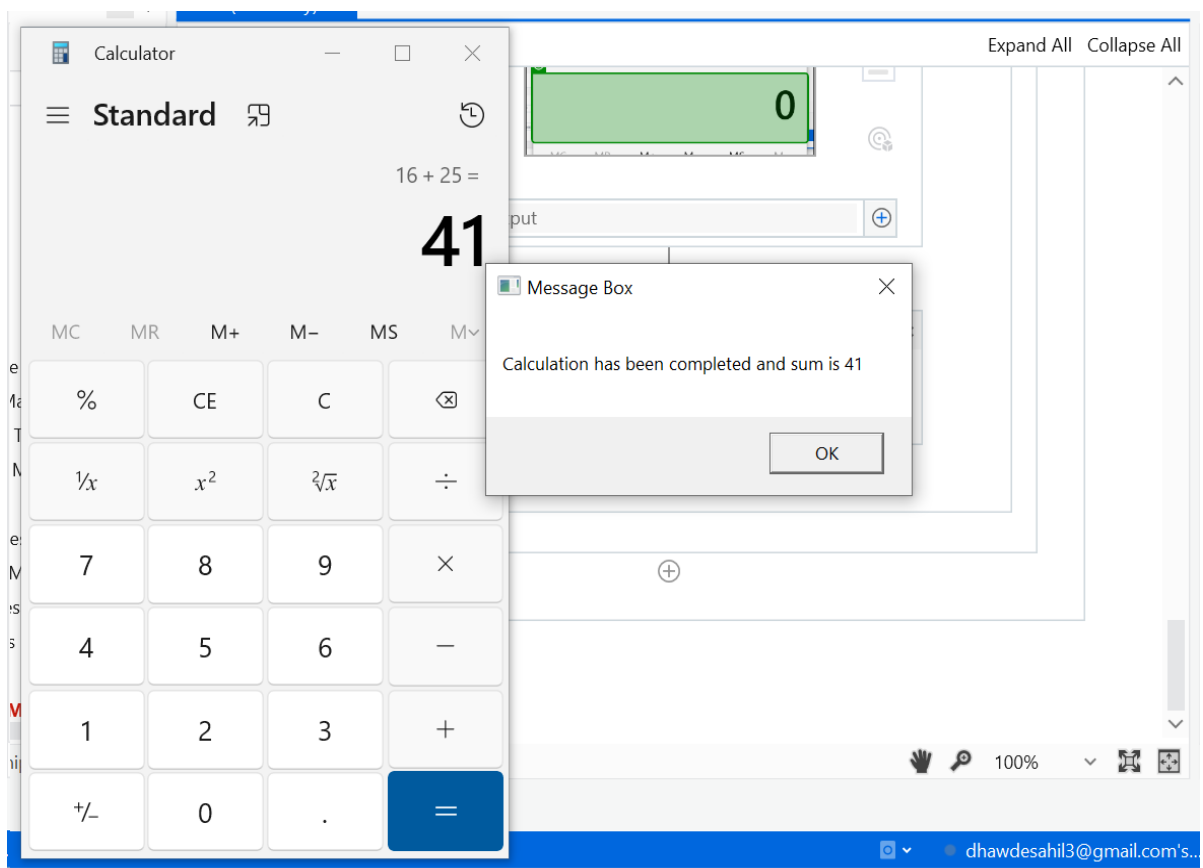
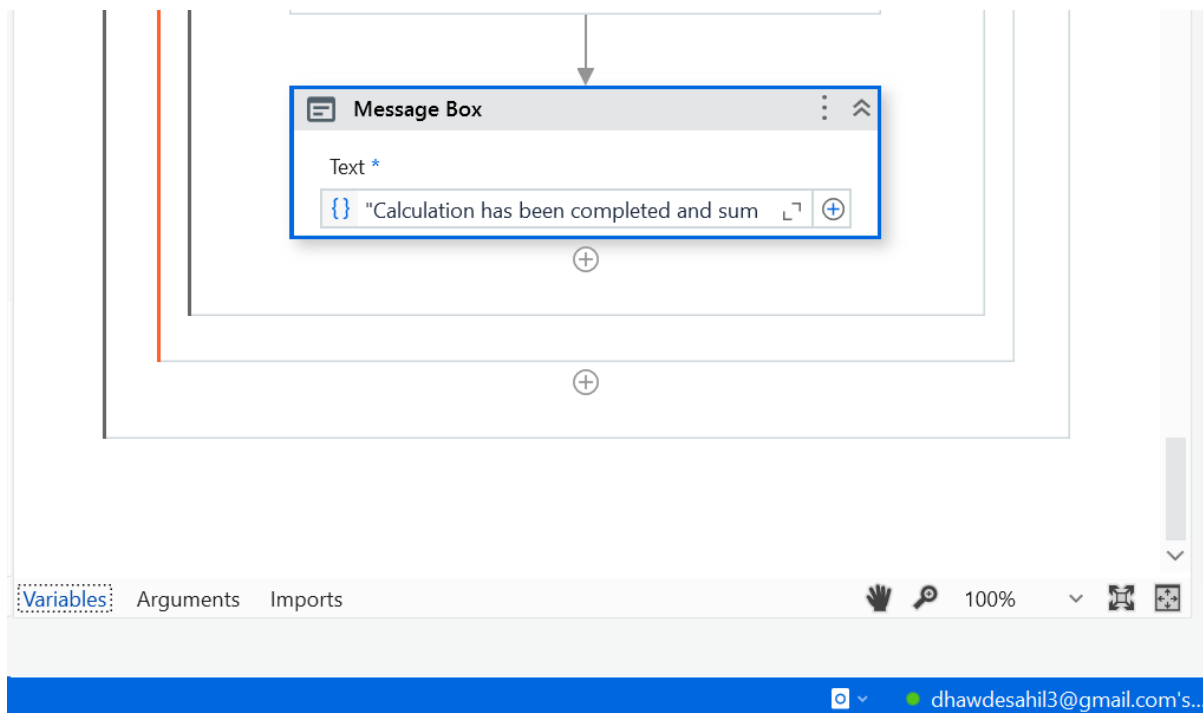
Process & Output:









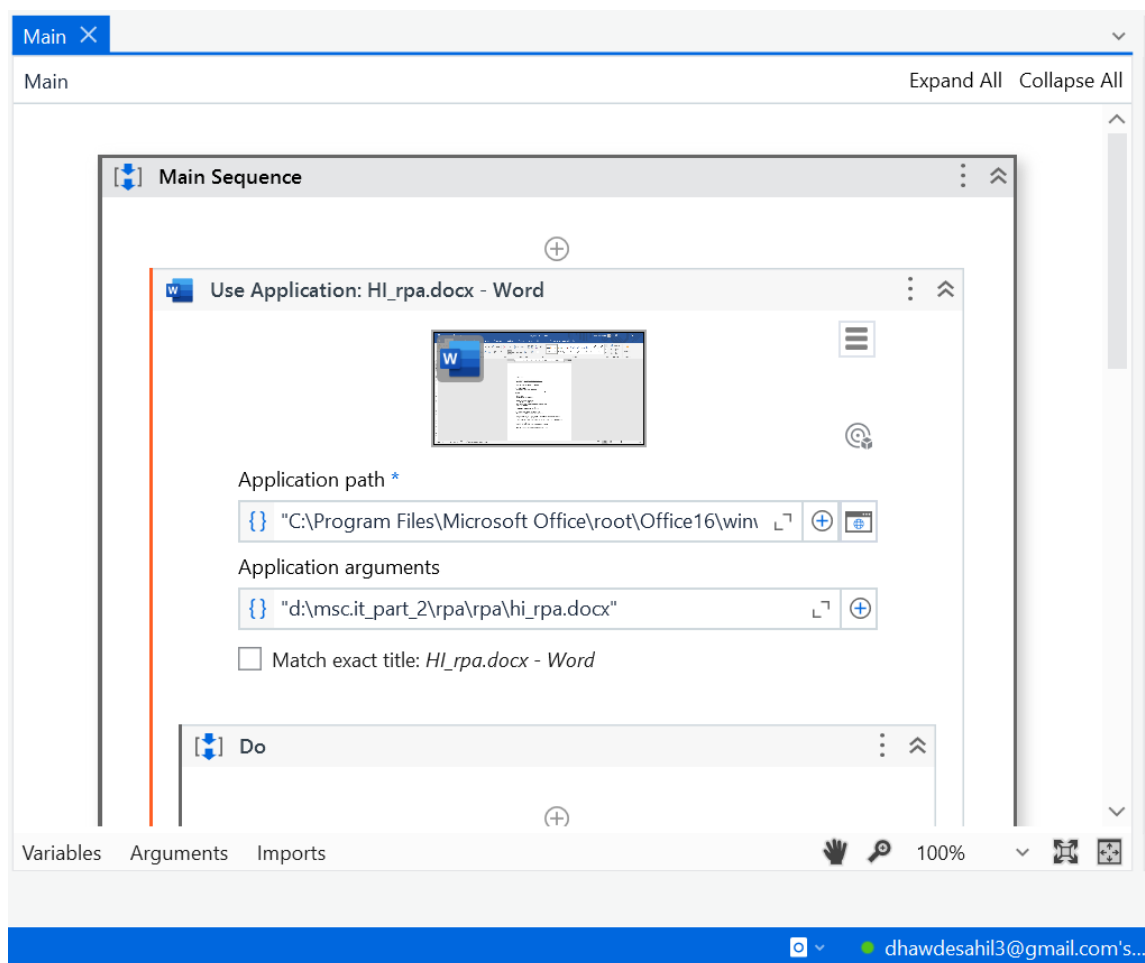


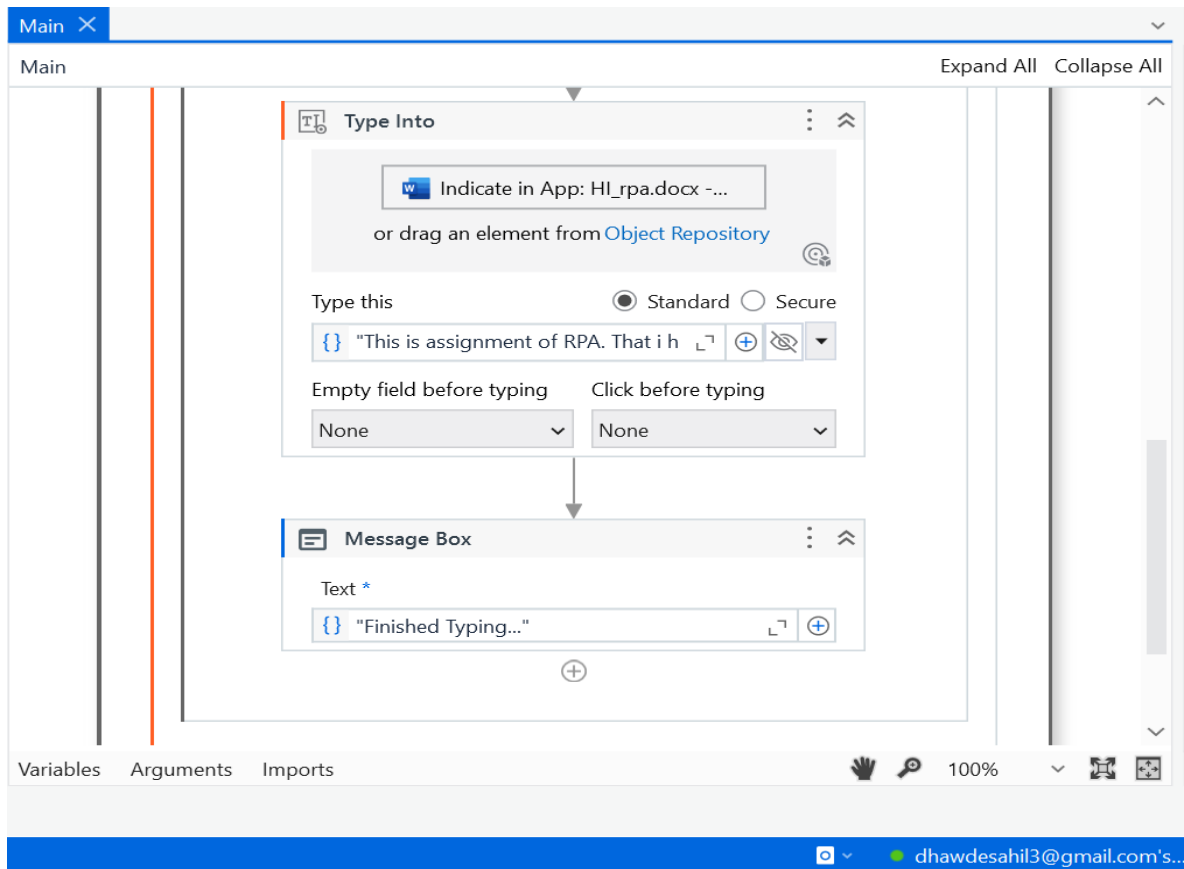
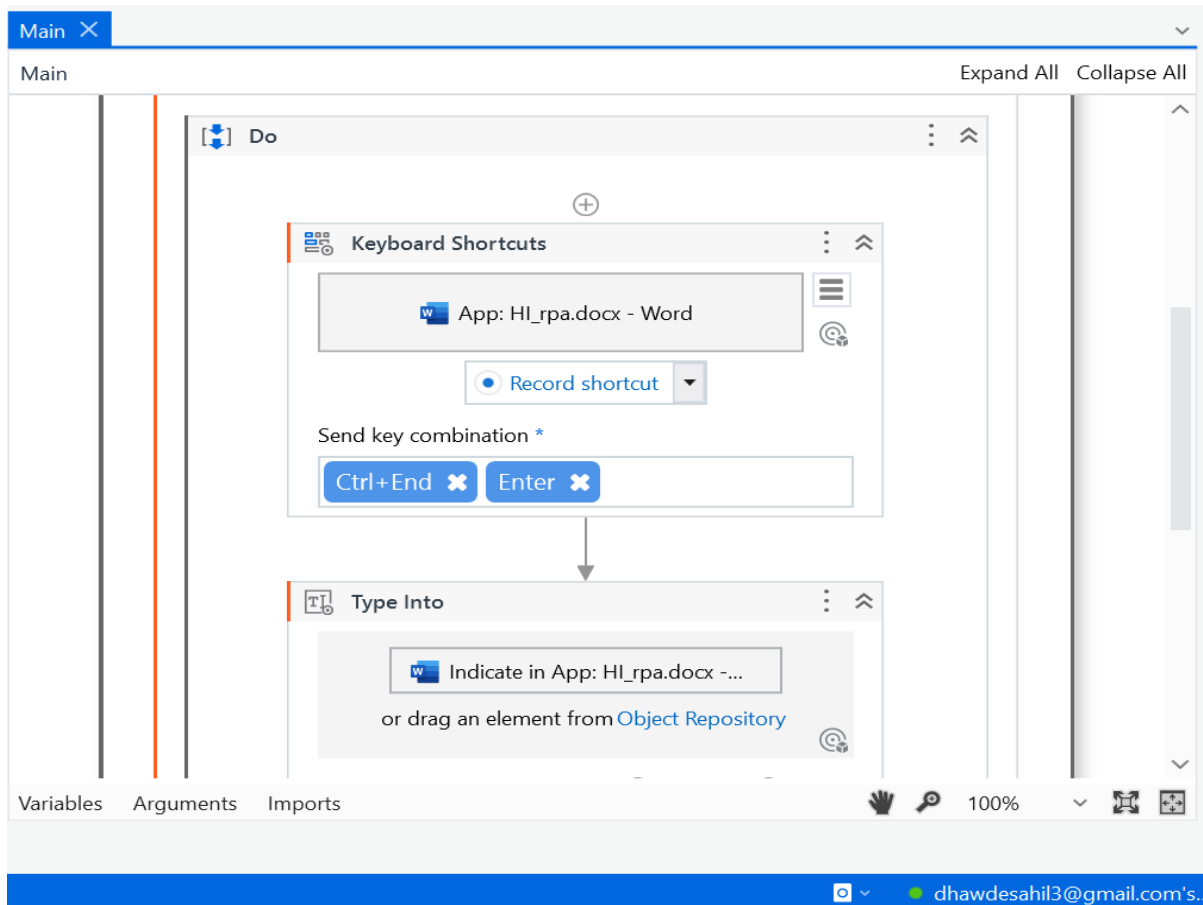
B. Automate any process using desktop recording.

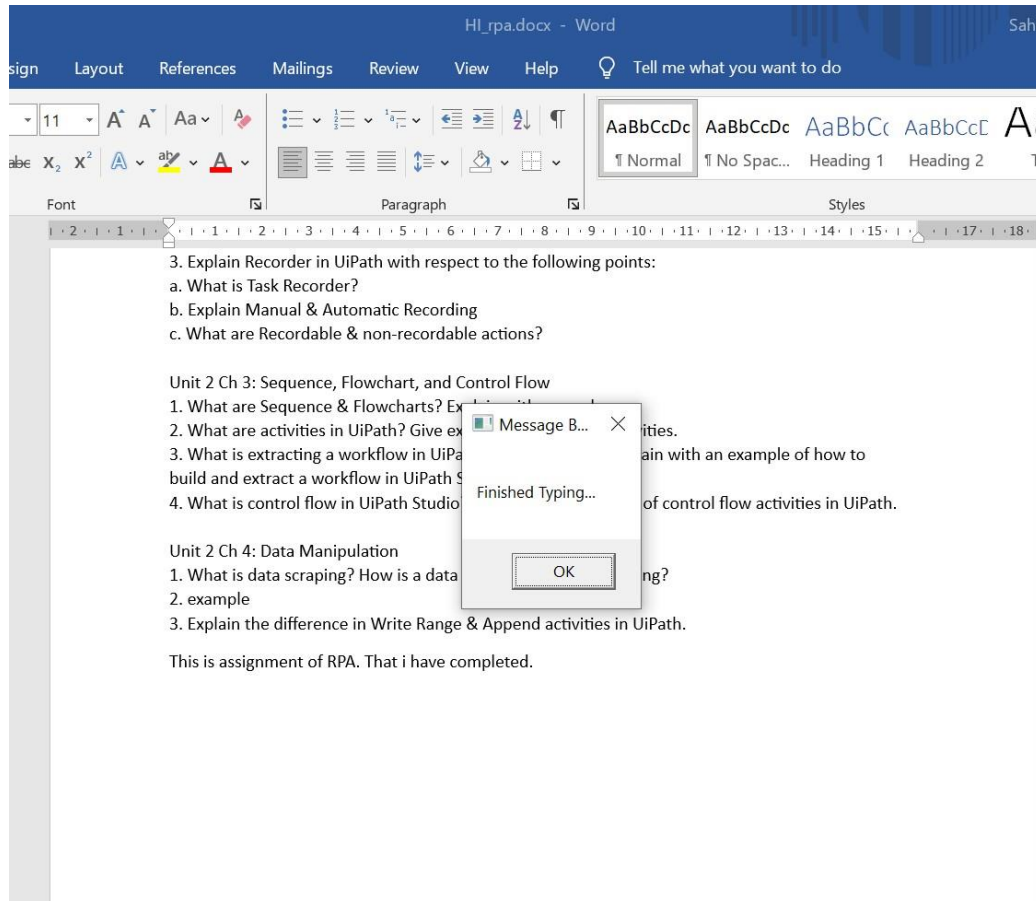
Steps:

1. Open project.
2. Select Sequence activity.
3. Select from activities panel Use Application/Browser activity, and indicate the word file.
4. In Do block, select from activities panel Key Shortcut activity and record shortcut as “Clrt + end and Enter” from Keyboard.
5. Select from activities panel Type into activity.
6. In Type this of Type into enter “This is assignment of RPA. That i have completed. ”.
7. Select from activities panel Message box activity, type "Finished Typing..."
8. Run the project.

Process & Output:







C. Automate any process using web recording.

Steps:

1. Open project.
2. Select Sequence activity.
3. Select from activities panel Use Application/Browser activity, and indicate the website of collage.
4. Select Table extraction icon on Ribbon panel of UiPath studio.
5. Select elements on the website to be added as columns in Data Table.
6. Preview data table, make change as per your choice and save it.
7. Select from activities panel Write Range Workbook activity, and give the excel file name, sheet name and cell value as "A1" to write Data Table.
8. Give the content as "ExtractDataTable".
9. Select Message box and type "Extraction of Data Table has been completed.".
10. Run the project.

Process & Output:

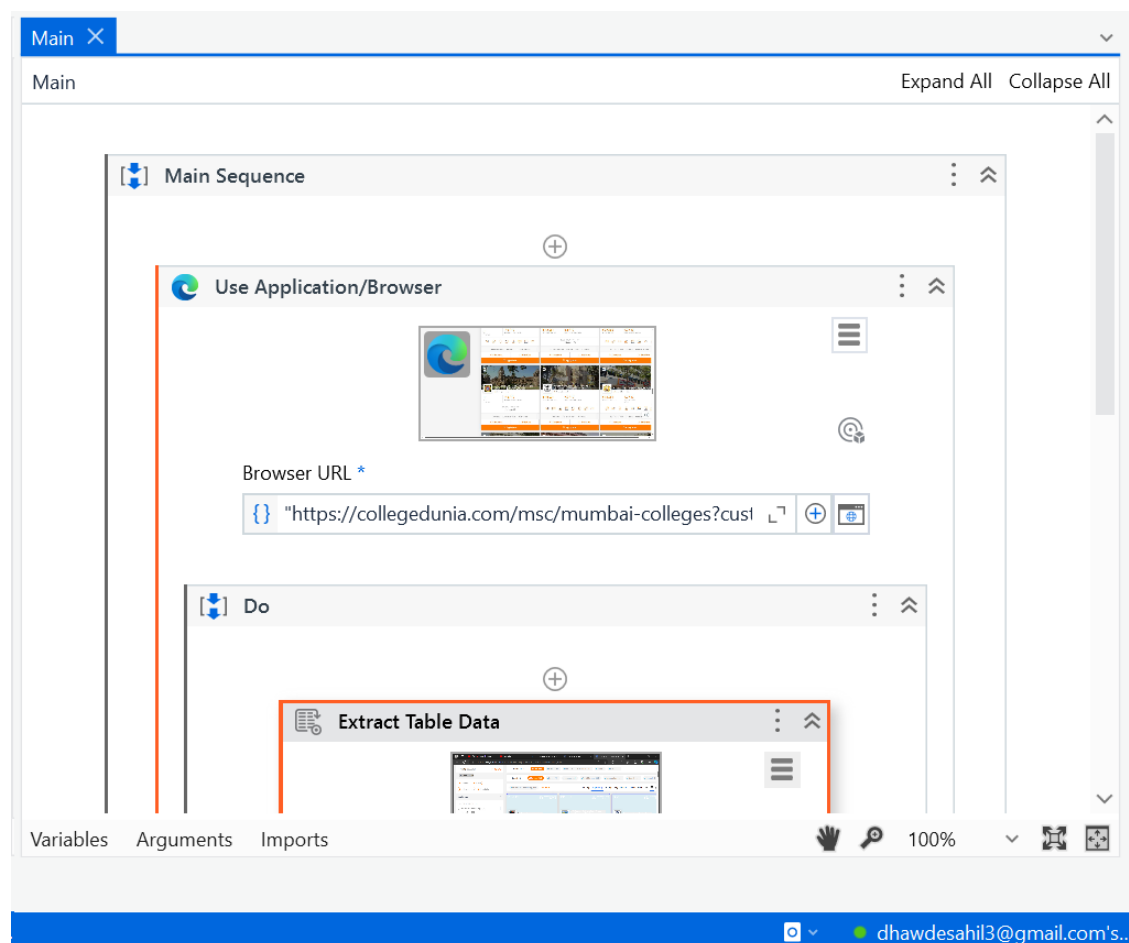


Table Extraction

Selection stopped

If the data is on multiple pages, enable Extract data from multiple pages.

6 Columns 60 Rows

Column	Value
College Name	[St. Xavier's College, ...]
Fees	[₹ 19,600, --, ₹ 50,00, ...]
Location	[Mumbai, Maharash...]
University	[UGC, AICTE, UGC, ...]
Rating	[8.1 / 10, 7.9 / 10, 8.2, ...]
College Logo Src	[https://images.colle...]

Select Data to Extract

Main

Main

Expand All Collapse All

Do

Extract Table Data

Extract to

{ } ExtractDataTable

Write Range Workbook

{ } "D:\MSc.IT_Part_2\RPA\RPA_Excel.xlsx"

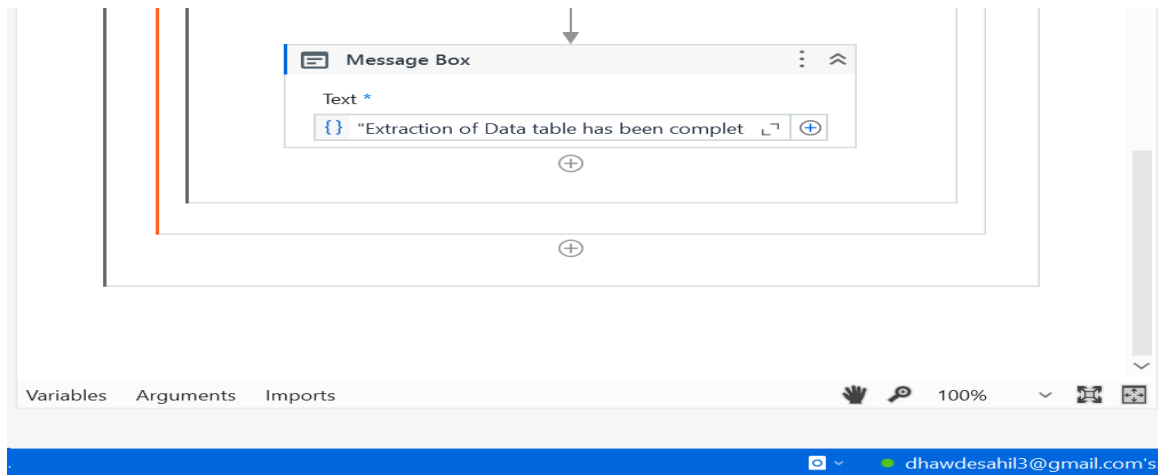
{ } "MSc.IT_College" "A1"

{ } ExtractDataTable

Variables Arguments Imports

100%

dhawdesahil3@gmail.com's...



College Name	Fees	Location	University	Rating	College Logo Src
1 College Name					
2 St. Xavier's College, Mumbai	19,600	Mumbai, Maharashtra	UGC	8.1 / 10	https://images.collegedunia.com/public/college_data/images/logos/150469574
3 D.G. Ruparel College of Arts Science and Commerce	50,000	Mumbai, Maharashtra	AICTE, UGC	7.9 / 10	https://images.collegedunia.com/public/college_data/images/logos/141681314
4 Mithibai College of Arts, Mumbai	37,295	Mumbai, Maharashtra	UGC	8.2 / 10	https://images.collegedunia.com/public/college_data/images/logos/156385675
5 Ramnarain Ruia Autonomous College, Mumbai	13,602	Mumbai, Maharashtra	UGC	7.9 / 10	https://images.collegedunia.com/public/college_data/images/logos/147141201
6 Jai Hind College - [JHC], Mumbai	50,920	Mumbai, Maharashtra	AICTE	8.5 / 10	https://images.collegedunia.com/public/college_data/images/logos/150512760
7 Kishinchand Chellaram College - [KC College], Mumbai	44,300	Mumbai, Maharashtra	AICTE	8.2 / 10	https://images.collegedunia.com/public/college_data/images/logos/147936750
8 Sophia College for Women, Mumbai	3,00,000	Mumbai, Maharashtra	UGC	8.6 / 10	https://images.collegedunia.com/public/college_data/images/logos/139512246
9 IIT Bombay - Indian Institute of Technology - Bombay	3,50,000	Mumbai, Maharashtra	UGC	8.1 / 10	https://images.collegedunia.com/public/college_data/images/logos/143195316
10 JIBIMS, Mumbai	57,250	Mumbai, Maharashtra	UGC	7.7 / 10	https://images.collegedunia.com/public/college_data/images/logos/159301934
11 TISS Tata Institute of Social Sciences, Mumbai	1,00,220	Mumbai, Maharashtra	UGC	7.5 / 10	https://images.collegedunia.com/public/college_data/images/logos/145095259
12 Institute of Chemical Technology - [ICT], Mumbai	67,230	Mumbai, Maharashtra	UGC	8.3 / 10	https://images.collegedunia.com/public/college_data/images/logos/150513139
13 Aikesh Dinesh Mody Institute for Financial and Management Studies	33,895	Mumbai, Maharashtra	UGC	7.7 / 10	https://images.collegedunia.com/public/college_data/images/logos/150044638
14 Seth GS Medical College - [GSMC], Mumbai	21,267	Mumbai, Maharashtra	UGC	7.5 / 10	https://images.collegedunia.com/public/college_data/images/logos/143858885
15 Mumbai University - [MU], Mumbai	40,215	Mumbai, Maharashtra	UGC	7.7 / 10	https://images.collegedunia.com/public/college_data/images/logos/148151820
16 Kelkar Education Trust's V.G. Vaze College of Arts, Science and Commerce	10,985	Mumbai, Maharashtra	UGC	8.3 / 10	https://images.collegedunia.com/public/college_data/images/logos/147074690
17 Mulund College of Commerce, Mumbai	53,000	Mumbai, Maharashtra	UGC	7.7 / 10	https://images.collegedunia.com/public/college_data/images/logos/col15945.jpg
18 Wilson College, Mumbai	11,575	Mumbai, Maharashtra	UGC	7.5 / 10	https://images.collegedunia.com/public/college_data/images/logos/146011045
19 Dr. B.N.M. College of Home Science, Mumbai	14,865	Mumbai, Maharashtra	UGC	7.8 / 10	https://images.collegedunia.com/public/college_data/images/logos/139885576
20 SIES College of Arts Science and Commerce - [SIES], Mumbai	27,245	Mumbai, Maharashtra	UGC	7.7 / 10	https://images.collegedunia.com/public/college_data/images/logos/col3860.jpg
21 Ramniranjan Jhunjhunwala College - [R.J.C.], Mumbai	27,245	Mumbai, Maharashtra	UGC	8.3 / 10	https://images.collegedunia.com/public/college_data/images/logos/151524149
22 Chhatrapati Natikhi Damodar Thackersey Mahavidyalaya - [CNDT], Mumbai	27,245	Mumbai, Maharashtra	UGC	7.5 / 10	https://images.collegedunia.com/public/college_data/images/logos/142701030

College Name	Fees	Location	University	Rating	College Logo Src
26 Marine Engineering And Research Institute - [MERI], Mumbai	1,85,000	Mumbai, Maharashtra	UGC	8.7 / 10	https://images.collegedunia.com/public/college_data/images/logos/147374989
27 The Second Wind - [TSW], Mumbai	1,85,000	Mumbai, Maharashtra	AICTE	6.8 / 10	https://images.collegedunia.com/public/college_data/images/logos/158226958
28 LN Group of Institutes, Mumbai	12,025	Mumbai, Maharashtra	UGC	6.8 / 10	https://images.collegedunia.com/public/college_data/images/logos/147824987
29 Maharshi Dayanand College of Arts Science and Commerce - [MDCSC], Mumbai	15,500	Mumbai, Maharashtra	UGC	6.2 / 10	https://images.collegedunia.com/public/college_data/images/logos/147537784
30 NG Acharya and DK Marathe College of Arts & Science - [NGAC], Mumbai	12,651	Mumbai, Maharashtra	UGC	7.6 / 10	https://images.collegedunia.com/public/college_data/images/logos/139589904
31 Rizvi College of Arts Science and Commerce, Mumbai	27,760	Mumbai, Maharashtra	UGC	7.5 / 10	https://images.collegedunia.com/public/college_data/images/logos/142769682
32 PTVA's Sathaye College, Mumbai	12,651	Mumbai, Maharashtra	UGC	7.2 / 10	https://images.collegedunia.com/public/college_data/images/logos/147929608
33 Bhavan's College, Mumbai	10,621	Mumbai, Maharashtra	UGC	8.1 / 10	https://images.collegedunia.com/public/college_data/images/logos/140868639
34 Guru Nanak Khalsa College of Arts, Science & Commerce - [GNKCC], Mumbai	41,601	Mumbai, Maharashtra	AICTE	7.2 / 10	https://images.collegedunia.com/public/college_data/images/logos/147143686
35 Guru Nanak College of Arts Science and Commerce - [GNC], Mumbai	51,730	Mumbai, Maharashtra	AICTE	7.5 / 10	https://images.collegedunia.com/public/college_data/images/logos/col1784.jpg
36 Kirti M Doongursee College of Arts Science and Commerce - [KMDC], Mumbai	51,730	Mumbai, Maharashtra	AICTE	7.2 / 10	https://images.collegedunia.com/public/college_data/images/logos/147928993
37 Nagindas Khandwala College, Mumbai	1,01,000	Mumbai, Maharashtra	AICTE	7.7 / 10	https://images.collegedunia.com/public/college_data/images/logos/147929558
38 Sir Sitaram and Lady Shantabai Patkar College - [SSLP], Mumbai	1,80,000	Mumbai, Maharashtra	AICTE	7.3 / 10	https://images.collegedunia.com/public/college_data/images/logos/col4538.jpg
39 Vivekanand Education Society's College of Arts, Science and Commerce - [VESC], Mumbai	65,500	Mumbai, Maharashtra	AICTE	8.1 / 10	https://images.collegedunia.com/public/college_data/images/logos/col4538.jpg
40 Vikas Night College of Arts Science and Commerce - [VNC], Mumbai	1,01,000	Mumbai, Maharashtra	AICTE	6.3 / 10	https://images.collegedunia.com/public/college_data/images/logos/168553120
41 Rachna Sansad School of Design, Mumbai	1,80,000	Mumbai, Maharashtra	AICTE, NBA	6 / 10	https://images.collegedunia.com/public/college_data/images/logos/139762566
42 National Institute for Research in Reproductive Health - [NIRRH], Mumbai	50,000	Mumbai, Maharashtra	UGC	8.7 / 10	https://images.collegedunia.com/public/college_data/images/logos/145769761
43 JD Institute of Fashion Technology, Mumbai	65,500	Mumbai, Maharashtra	AICTE	8.3 / 10	https://images.collegedunia.com/public/college_data/images/logos/142622396
44 Bunts Sangha's S.M. Shetty College of Science - [BSS], Mumbai	50,000	Mumbai, Maharashtra	AICTE	8.2 / 10	https://images.collegedunia.com/public/college_data/images/logos/161572266
45 Tata Memorial Centre - [TMC], Mumbai	50,000	Mumbai, Maharashtra	MCI	9.8 / 10	https://images.collegedunia.com/public/college_data/images/logos/142890834
46 Topiwala National Medical College - [TNMC], Mumbai	50,000	Mumbai, Maharashtra	MCI	9.8 / 10	https://images.collegedunia.com/public/college_data/images/logos/170124145
47 Anil Surdas Modi School of Commerce - [ASMC], Mumbai	50,000	Mumbai, Maharashtra	AICTE	8.1 / 10	https://images.collegedunia.com/public/college_data/images/logos/147920628

Practical No: 5

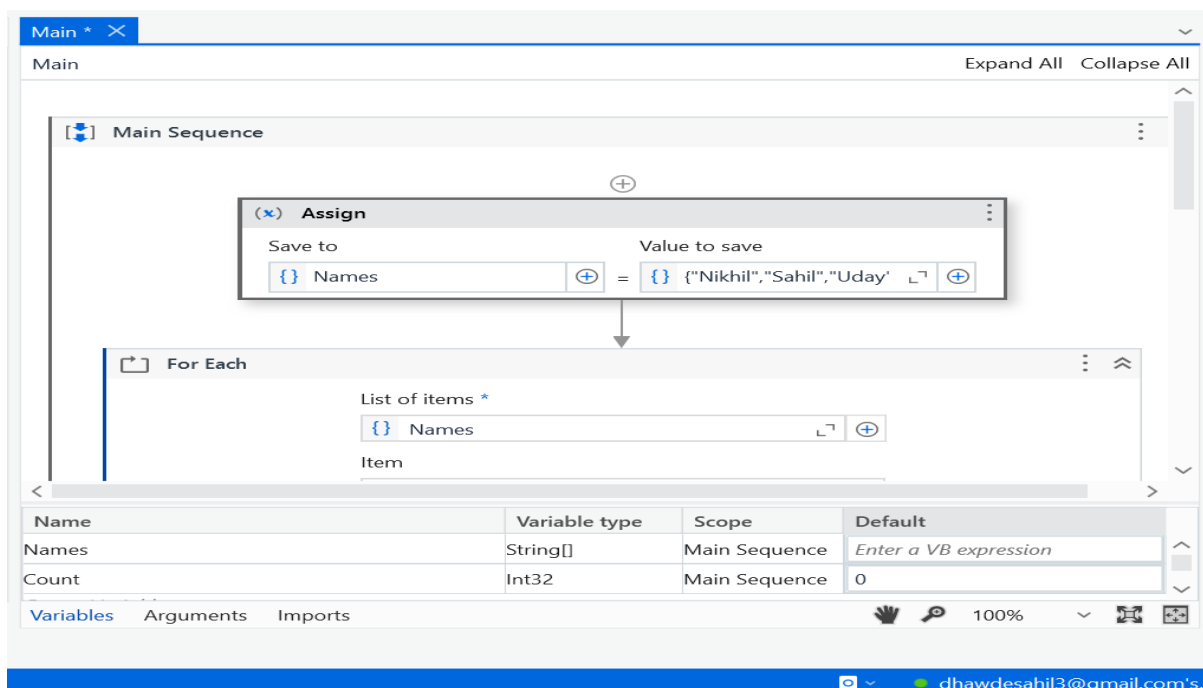
Aim: Implement the following:

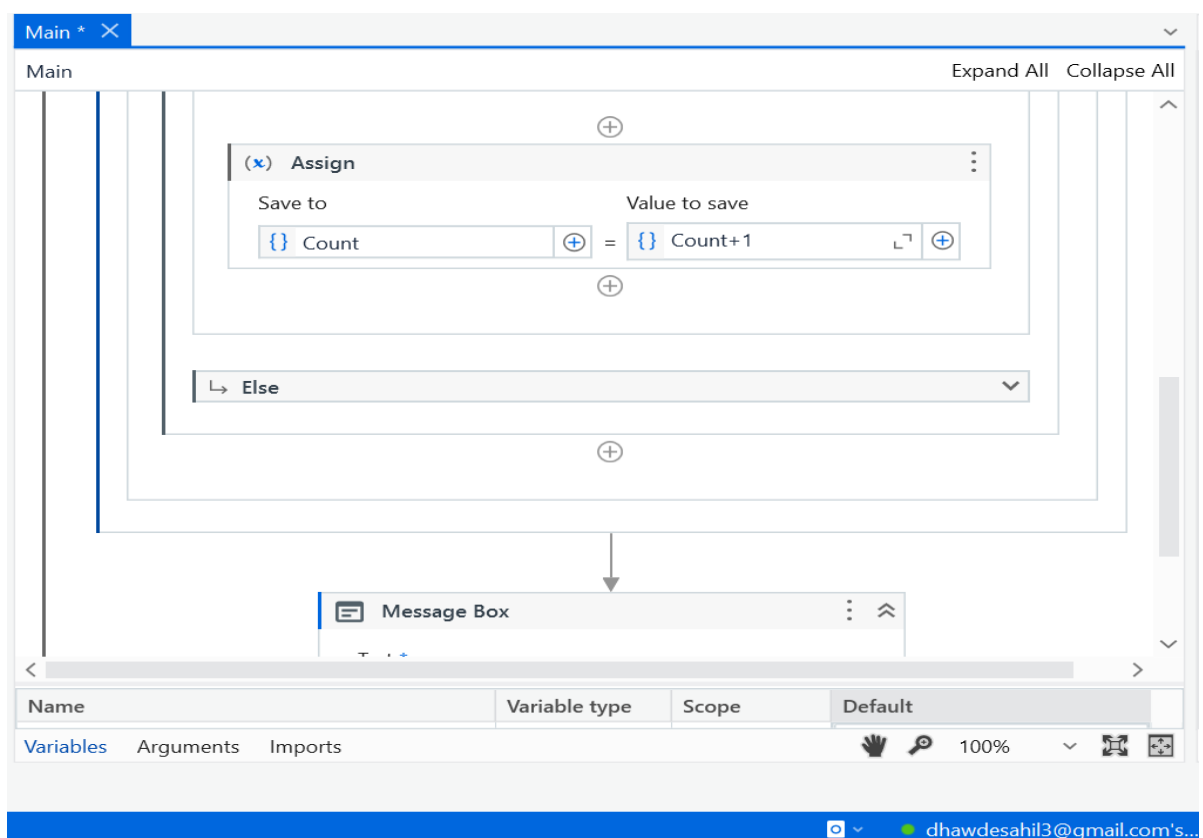
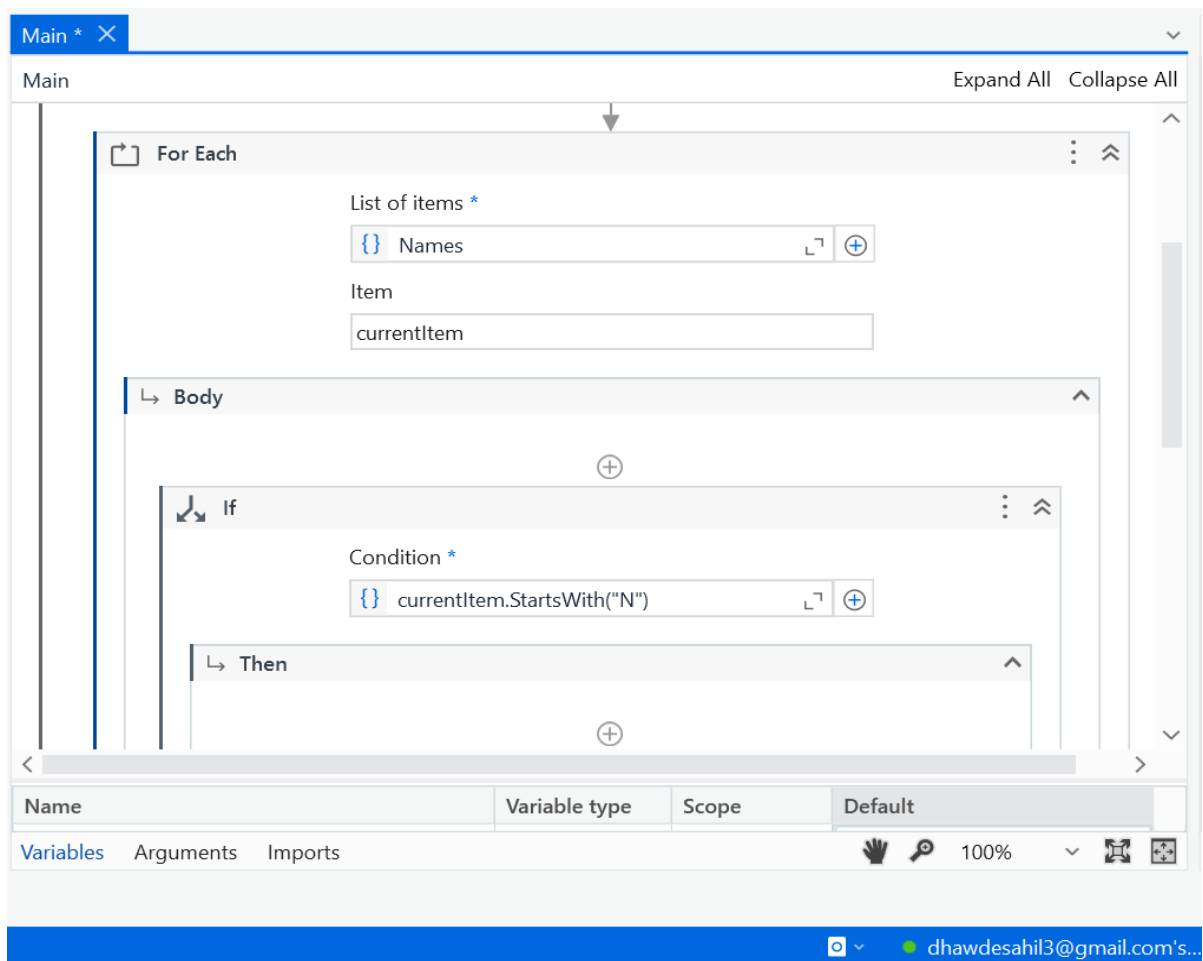
A. Consider an array of names. We have to find out how many of them start with the letter "N". Create an automation where the number of names starting with "N" is counted and the result is displayed.

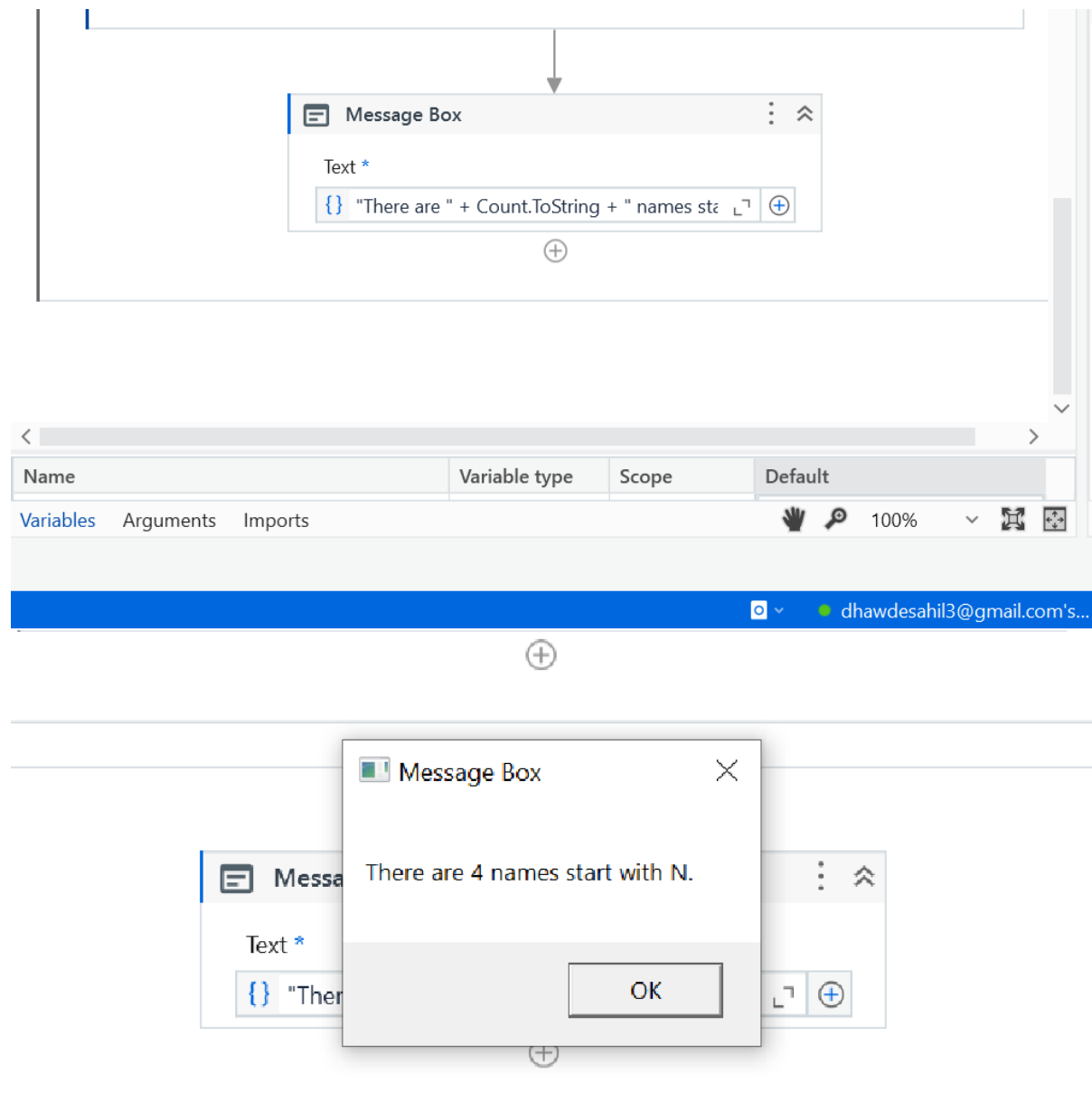
Steps:

1. Open Project.
2. Select Sequence activity.
3. Create two variables' Names (array of string) and count (int32).
4. Select from activities panel Assign activity, and assign Names = { "Nikhil", "Sahil", "Uday", "Nihal", "Nicks", "Nikita", "Anuj"}.
5. Select from activities panel For Each activity.
6. In For Each, set values of List of items as "Names".
7. Set values of Item as "currentItem"
8. Inside the Body of For Each
9. Select the If activity.
10. In If activity, type condition as "currentItem.StartsWith("N")"
11. In Then block, select Assign activity to assign "count = count + 1".
12. Run the project.

Process & Output:







Practical No: 6

Aim: Implement the following:

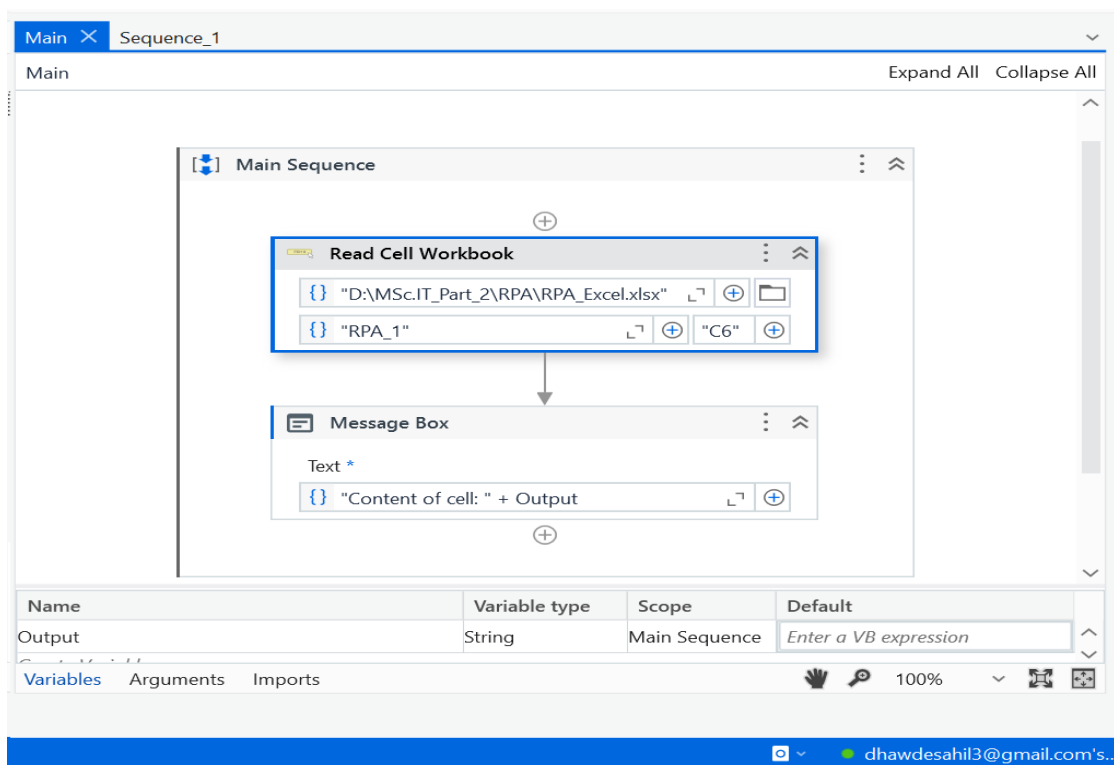
A. Create an application automating the read, write and append operation on excel file.

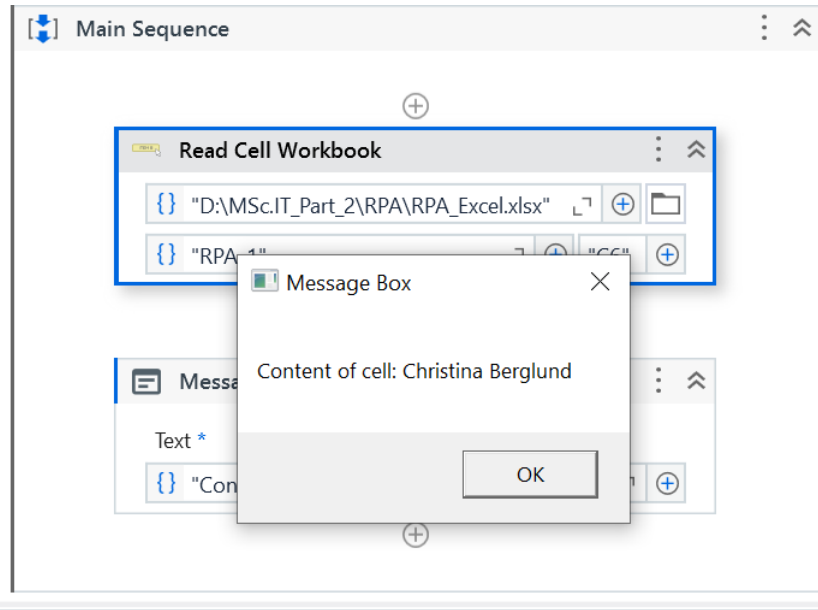
i. Read Cell

Steps:

1. Open Project.
2. Select Sequence activity.
3. Create variable Output (string).
4. Select from activities panel Read Cell Workbook activity, and give the excel file name, sheet name and cell value that you have to read.
5. Select from activities panel Message box activity, type "Content of cell: + Output".
6. Run the project.

Process & Output:





RPA_Excel.xlsx - Excel

	A	B	C	D	E	F	G	H
1	CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country	
2	1	Alfreds Futterkiste	Maria Anders	Obere Str. 57	Berlin	12209	Germany	
3	2	Ana Trujillo Empa	Ana Trujillo	Avda. de la Cor	México D.F.	05021	Mexico	
4	3	Antonio Moreno	Antonio Moreno	Mataderos 231	México D.F.	05023	Mexico	
5	4	Around the Horn	Thomas Hardy	120 Hanover Sc	London	WA1 1DP	UK	
6	5	Berglunds snabbk	Christina Berglund	Berguvsvägen 8	Luleå	S-958 22	Sweden	
7								
8								
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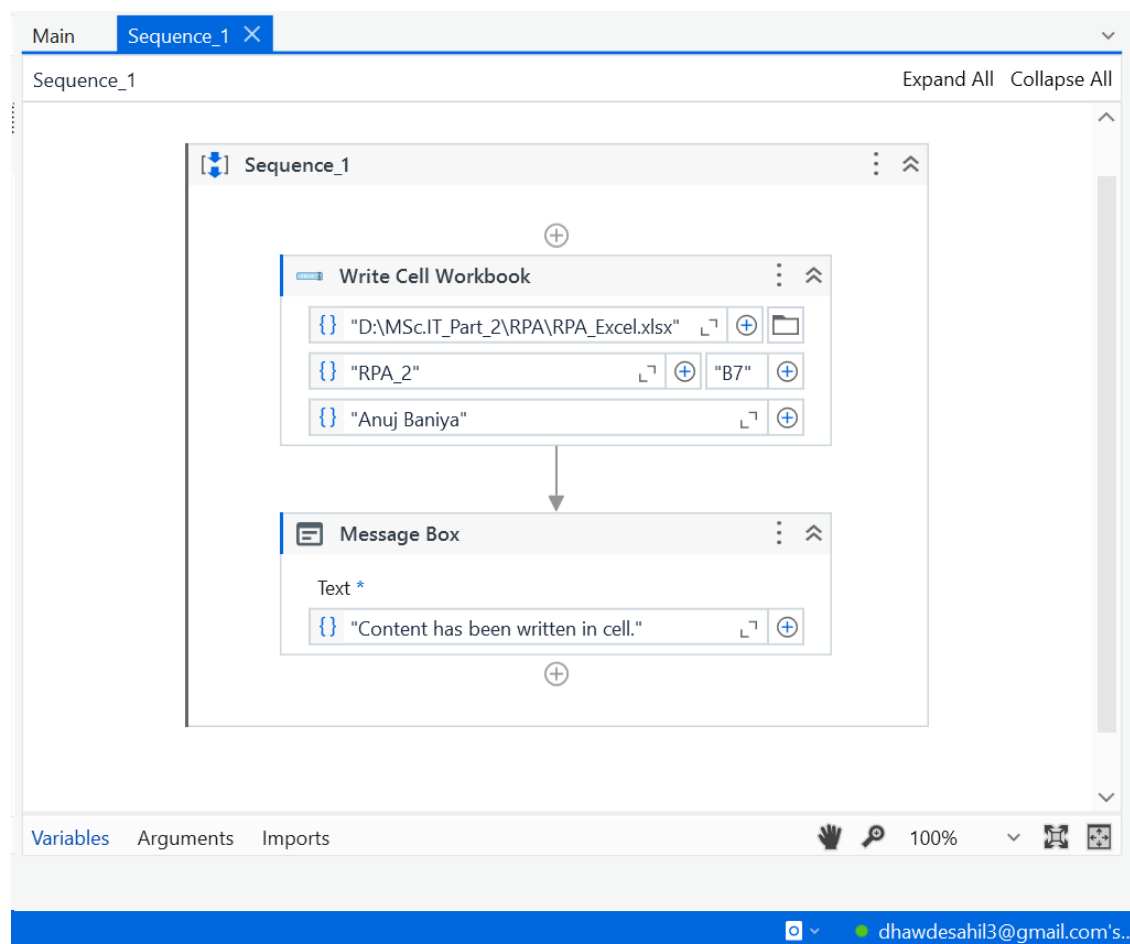
RPA_1 RPA_2

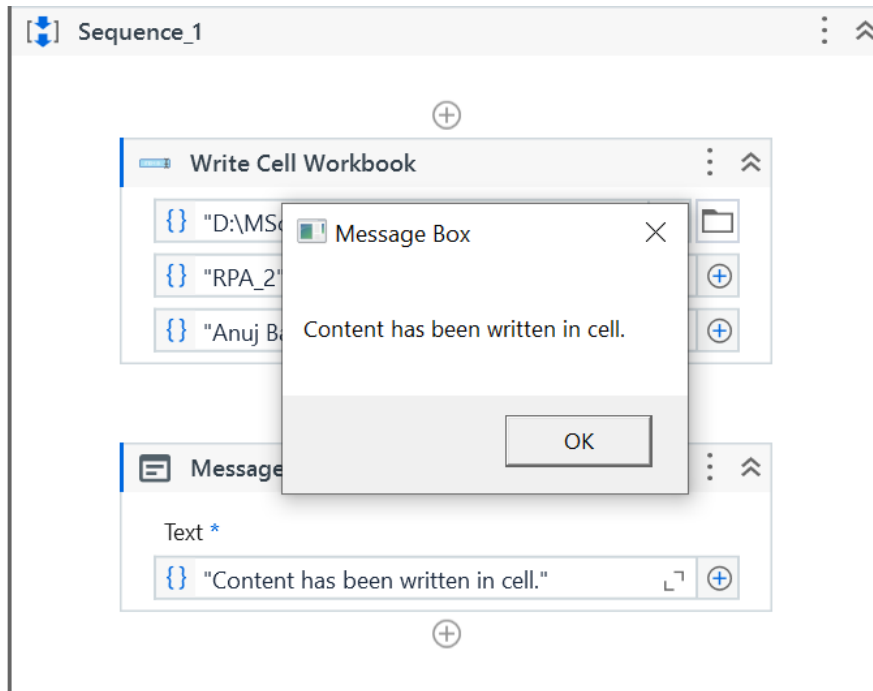
ii. Write Cell

Steps:

1. Open Project.
2. Select Sequence activity.
3. Select from activities panel Write Cell Workbook activity, and give the excel file name, sheet name and cell value on which you have to write the content.
4. Write the content in Content box.
5. Select from activities panel Message box activity, type “Content has been written in cell”.
6. Run the project.

Process & Output:





RPA_Excel.xlsx - Excel

	A	B	C	D	E	F	G
1	CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
2	1	Alfreds Futterkiste	Maria Anders	Oberer Str. 57	Berlin	12209	Germany
3	2	Ana Trujillo Emparedada	Ana Trujillo	Avda. de la Constitución	México D.F.	05021	Mexico
4	3	Antonio Moreno Taquería	Antonio Moreno	Mataderos 2312	México D.F.	05023	Mexico
5	4	Around the Horn	Thomas Hardy	120 Hanover Sq.	London	WA1 1DP	UK
6	5	Berglunds snabbköp	Christina Berglund	Berguvsvägen 8	Luleå	S-958 22	Sweden
7		Anuj Baniya					
8							
9							
10							
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16							
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22							

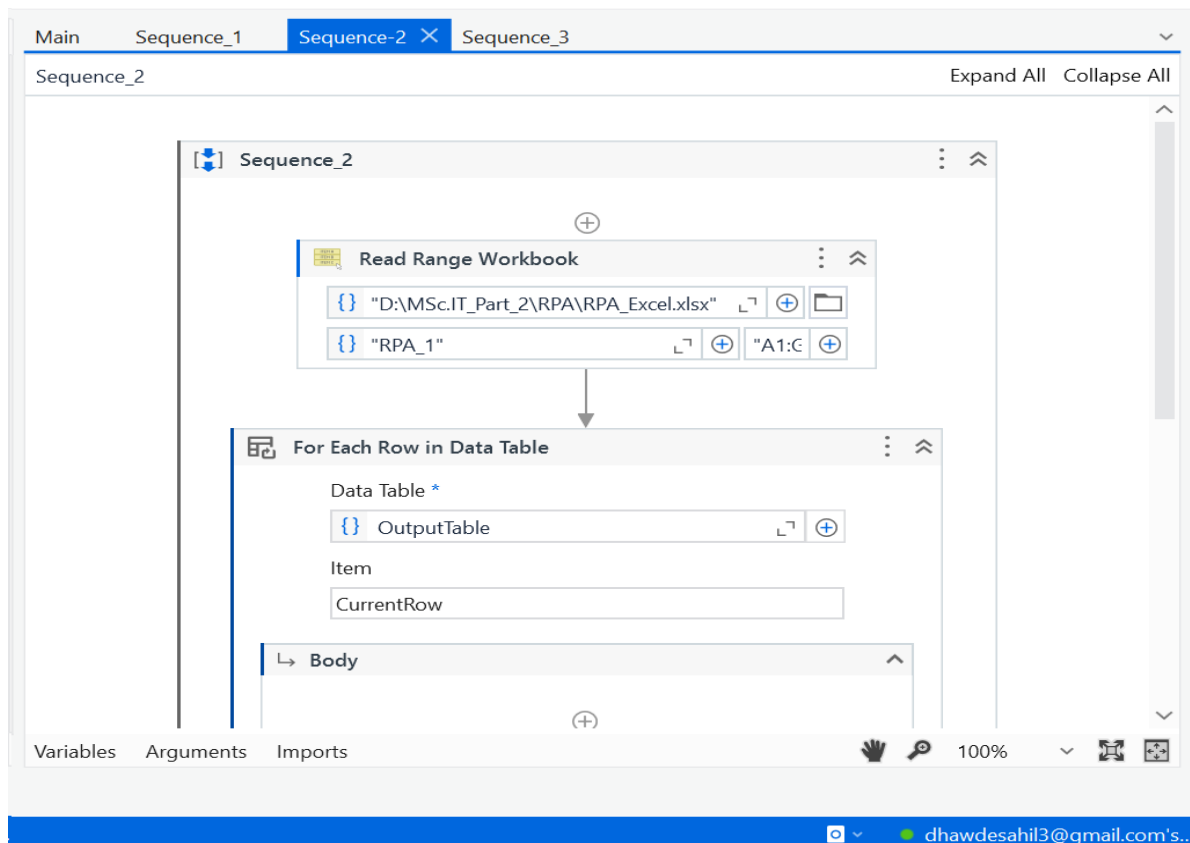
RPA_1 RPA_2

iii. Read Range

Steps:

1. Open Project.
2. Select Sequence activity.
3. Select from activities panel Read Range Workbook activity, and give the excel file name, sheet name and cell value as "A1:G7" to read whole table.
4. Create OutputTable variable as DataTable.
5. Select For Each in Data Table activity.
6. Set values of List of items as "OutputTable".
7. Set values of Item as "CurrentRow".
8. Inside the Body of For Each in DataTable.
9. Select the Write line activity & enter text "CurrentRow.Item(0).ToString + " CustomerID " + CurrentRow.Item(1).ToString + " CustomerName " + CurrentRow.Item(2).ToString + " ContactName " + CurrentRow.Item(3).ToString + " Address " + CurrentRow.Item(4).ToString + " City " + CurrentRow.Item(5).ToString + " PostalCode " + CurrentRow.Item(6).ToString + " Country "".
10. Select Message box and type "Reading of Excel Content has been completed.".
11. Run the project.

Process & Output:



Sequence_2

Expand All Collapse All

Body

Write Line

Text

{ } CurrentRow.Item(0).ToString + " Customer

Message Box

Text *

{ } "Reading of Excel Content has been compl

Name	Variable type	Scope	Default
OutputTable	DataTable	Sequence_2	Enter a VB expression

Variables Arguments Imports

100%

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Write Line

Text

{ }

Message Box

Reading of Excel Content has been completed.

OK

Message Box

Text *

{ } "Reading of Excel Content has been compl

Variable type	Scope	Default
DataTable	Sequence_2	Enter a VB expression

Arguments Imports

100%

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Output



Search

⌕ Execution started for file: Sequence-2

⌕ Excel_operations execution started

⌕ 1 CustomerID Alfreds Futterkiste CustomerName Maria Anders ContactName Obere Str. 57 Address Berlin City 12209 PostalCode Germany Country

⌕ 2 CustomerID Ana Trujillo Emparedados y helados CustomerName Ana Trujillo ContactName Avda. de la Constitución 2222 Address México D.F. City 05021 PostalCode Mexico Country

⌕ 3 CustomerID Antonio Moreno Taquería CustomerName Antonio Moreno ContactName Mataderos 2312 Address México D.F. City 05023 PostalCode Mexico Country

⌕ Excel_operations execution ended in: 00:00:16

Output Breakpoints Error List

 dhawdesahil3@gmail.com's...

iv. Write Range

Steps:

1. Open Project.
2. Select Sequence activity.
3. Select from activities panel Build Data Table activity, and create a Data Table.
4. Create OutputTable variable as DataTable.
5. Select from activities panel Write Range Workbook activity, and give the excel file name, sheet name and cell value as "A1" to write Data Table.
6. Give the content as "OutputTable".
7. Select Message box and type "Writing of excel file has been completed.".
8. Run the project.

Process & Output:

Sequence_3

Build Data Table

Write Range Workbook

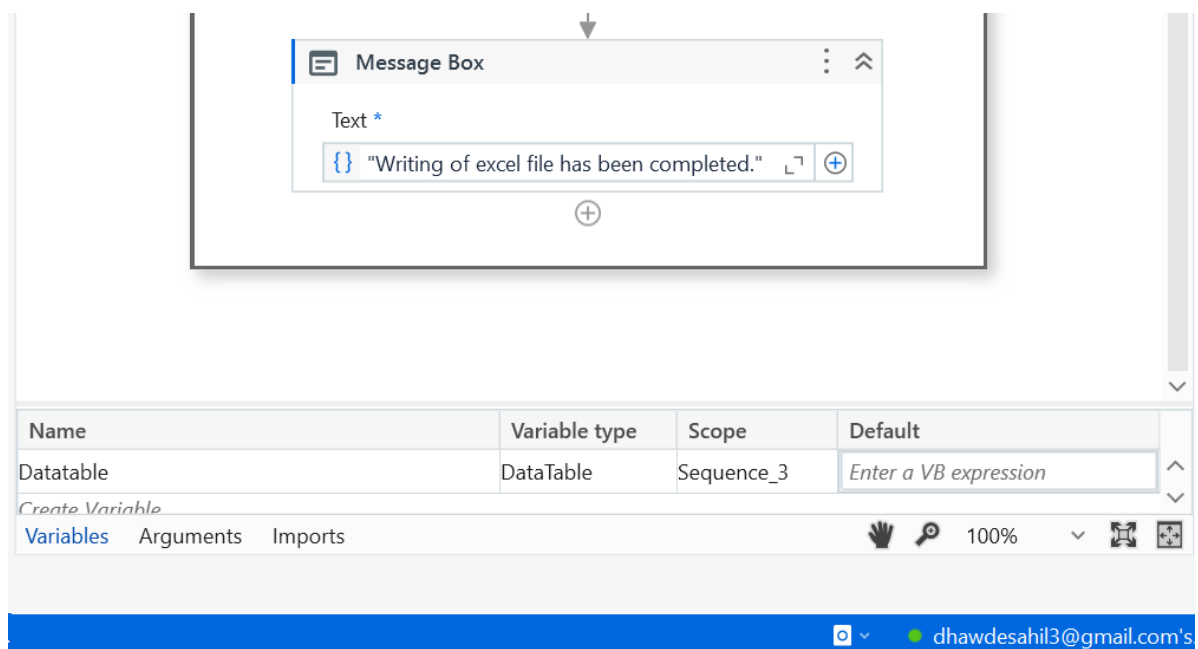
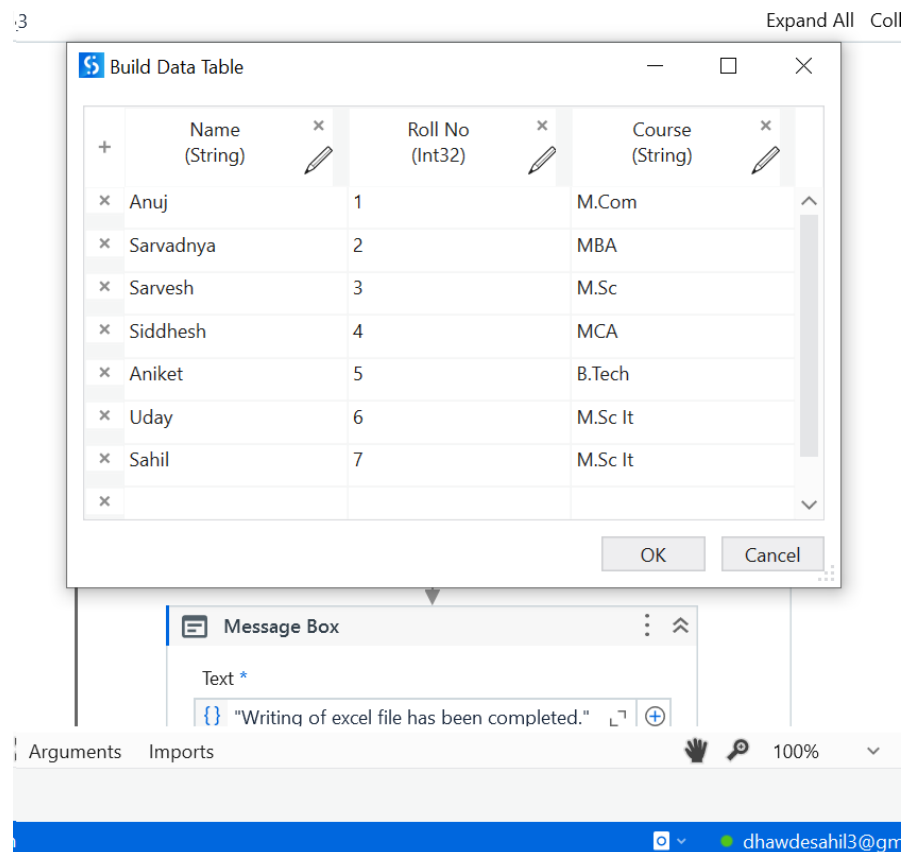
Message Box

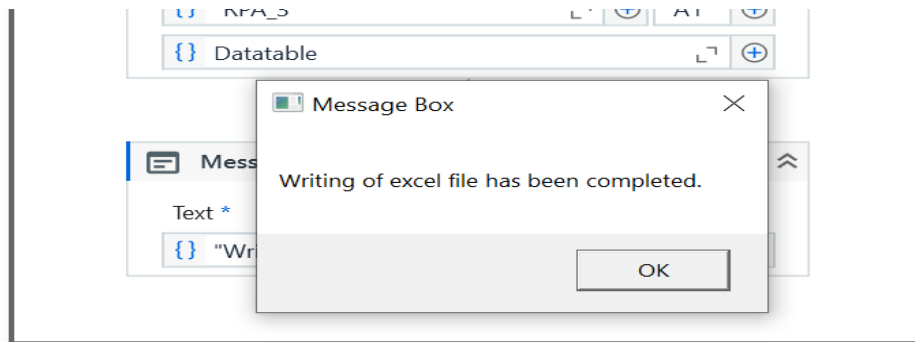
Name	Variable type	Scope	Default
Datatable	DataTable	Sequence_3	Enter a VB expression

Create Variable

Variables Arguments Imports

dhawdesahil3@gmail.com's..





	Variable type	Scope	Default
	DataTable	Sequence_3	Enter a VB expression

Documents Imports 100% dhawdesahil3@gmail.com

	A	B	C	D	E	F
1	Name	Roll No	Course			
2	Anuj	1	M.Com			
3	Sarvadnya	2	MBA			
4	Sarvesh	3	M.Sc			
5	Siddhesh	4	MCA			
6	Aniket	5	B.Tech			
7	Uday	6	M.Sc It			
8	Sahil	7	M.Sc It			
9						
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22						

RPA_1 RPA_2 RPA_3

v. Append Range

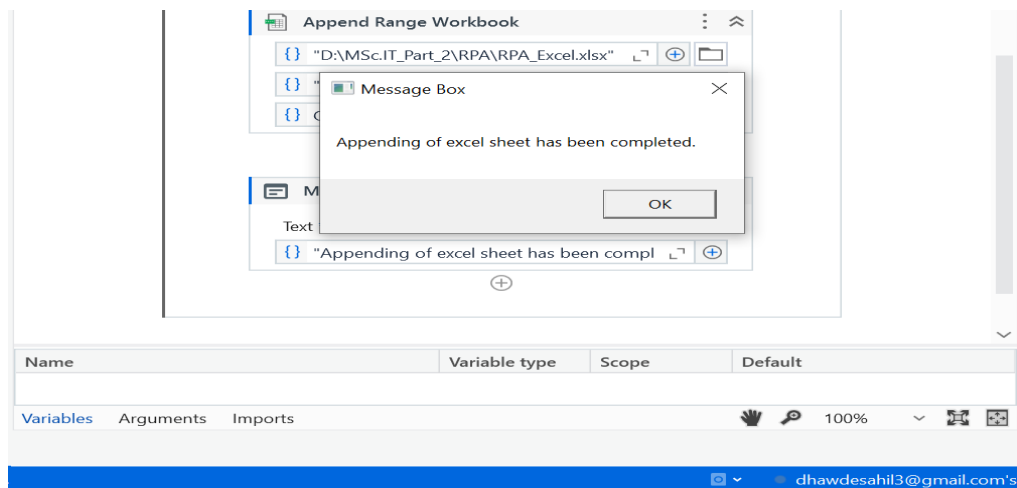
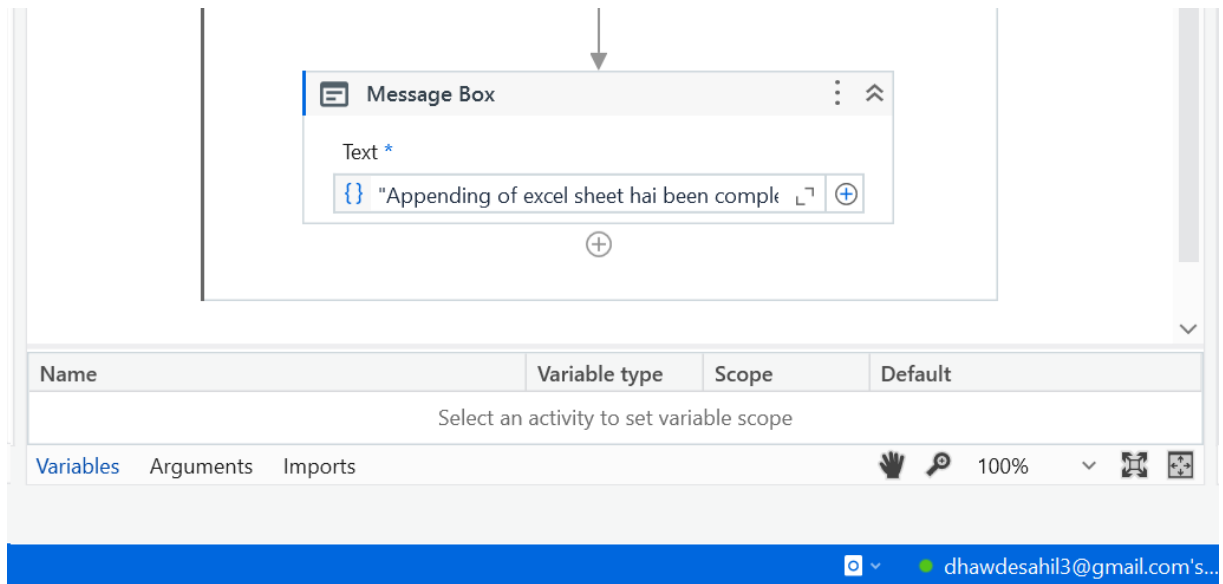
Steps:

1. Open Project.
2. Select Sequence activity.
3. Select from activities panel Read Range Workbook activity, and give the excel file name, sheet name and cell value as "A1:G7" to read whole table.
4. Create OutputTable variable as DataTable.
5. Select Append activity from activities panel.
6. Give the excel file name, sheet name and DataTable as "OutputTable" to append the Data Table.
7. Select Message box and type "Appending of excel sheet has been completed".
8. Run the project.

Process & Output:

The screenshot displays the UiPath Studio interface for a project named 'Main'. The 'Main Sequence' workflow is visible, containing two activities: 'Read Range Workbook' and 'Append Range Workbook'. The 'Read Range Workbook' activity is configured with the file path 'D:\MSc.IT_Part_2\RPA\RPA_Excel.xlsx', sheet name 'RPA_3', and cell range 'A1:C'. The 'Append Range Workbook' activity is configured with the same file path, sheet name 'Append', and the variable 'OutputDt'. Below the workflow, a table shows the variable 'OutputDt' with type 'DataTable' and scope 'Main Sequence'. The bottom status bar shows the user 'dhawdesahil3@gmail.com's...'.

Name	Variable type	Scope	Default
OutputDt	DataTable	Main Sequence	Enter a VB expression



RPA_Excel.xlsx									
File Home Insert Draw Page Layout Formulas Data Review View Help									
Calibri 11 A A									
B I U									
Wrap Text Merge & Center									
General									
E9									
A	B	C	D	E	F	G	H	I	J
1	Customer	Customer	ContactNa	Address	City	PostalCod	Country		
2	1	Alfreds Fu	Maria Anc	Obere Str.	Berlin	12209	Germany		
3	2	Ana Trujill	Ana Trujill	Avda. de l	México D.	05021	Mexico		
4	3	Antonio M	Antonio M	Matadero	México D.	05023	Mexico		
5	4	Around th	Thomas H	120 Hanov	London	WA1 1DP	UK		
6	5	Berglunds	Christina E	Berguvsvä	Luleå	S-958 22	Sweden		
7	Anuj	1	M.Com						
8	Sarvadnya	2	MBA						
9	Sarvesh	3	M.Sc						
10	Siddhesh	4	MCA						
11	Aniket	5	B.Tech						
12	Uday	6	M.Sc It						
13	Sahil	7	M.Sc It						
14									
15									
16									
17									
18									
19									
20									
21									
22									
RPA_1 RPA_2 RPA_3 Append									

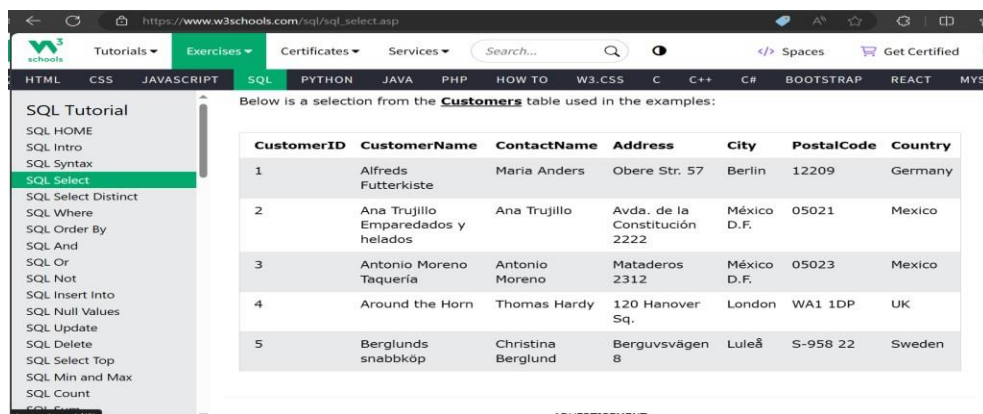
B. Automate the process to extract data from an excel file into a data table and vice versa

i. Extract Data Table from web to excel file

Steps:

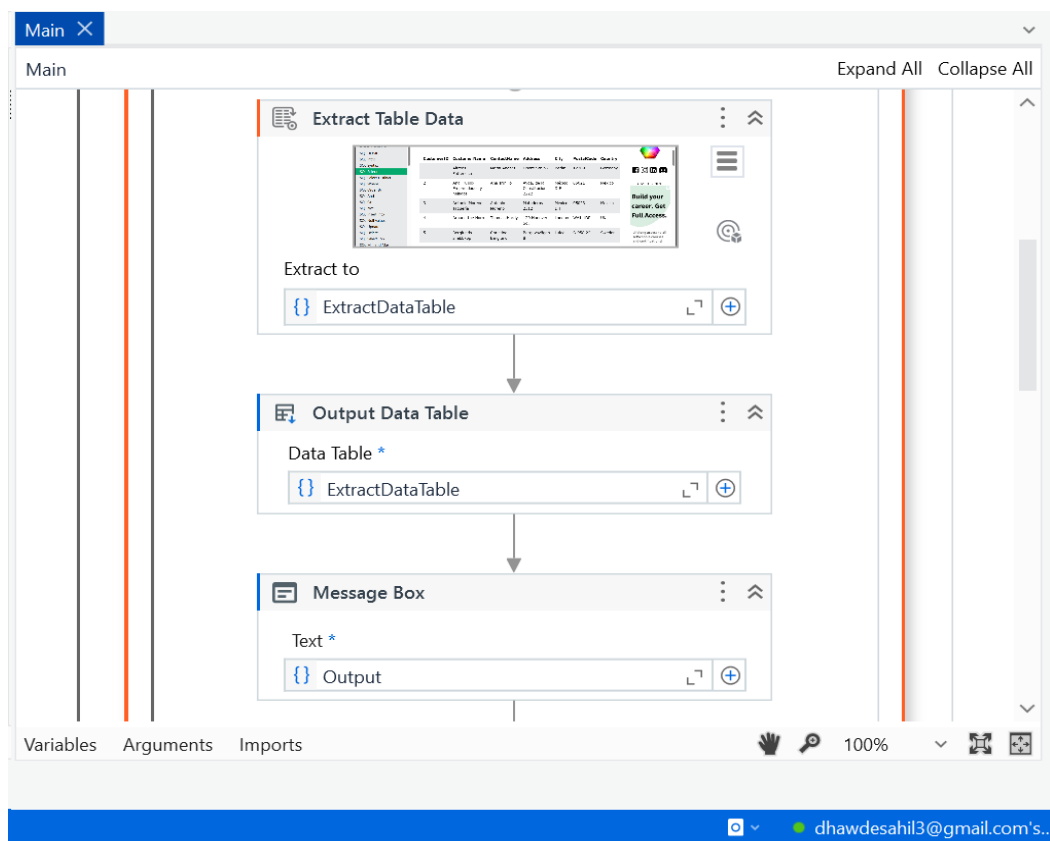
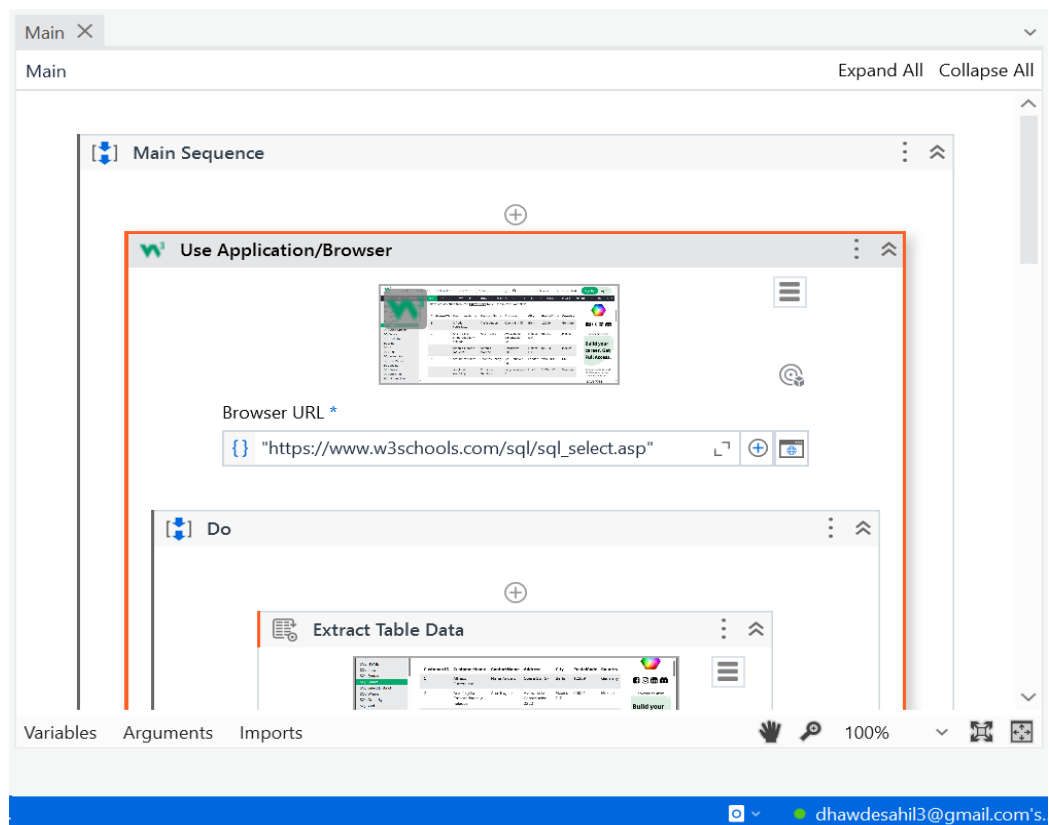
1. Open project.
2. Select Sequence activity.
3. Select Table extraction icon on Ribbon panel of Uipath studio.
4. Select the cell of table which do you want to extract.
5. Edit it if you want and save.
6. Select from activities panel Output Data Table activity.
7. Create Output variable as String.
8. Enter input as "ExtractDataTable" and output as "Output".
9. Select Message box and enter "Output".
10. Select For Each in Data Table activity.
11. Set values of List of items as "ExtractDataTable".
12. Set values of Item as "CurrentRow".
13. Inside the Body of For Each in DataTable.
14. Select the Write line activity & enter text "CurrentRow.Item(0).ToString + " CustomerID " + CurrentRow.Item(1).ToString + " CustomerName " + CurrentRow.Item(2).ToString + " ContactName " + CurrentRow.Item(3).ToString + " Address " + CurrentRow.Item(4).ToString + " City " + CurrentRow.Item(5).ToString + " PostalCode " + CurrentRow.Item(6).ToString + " Country "".
15. Select from activities panel Write Range Workbook activity, and give the excel file name, sheet name and cell value as "A1" to write Data Table.
16. Give the content as "ExtractDataTable".
17. Select Message box and type "Extraction of Data Table has been done".
18. Run the project.

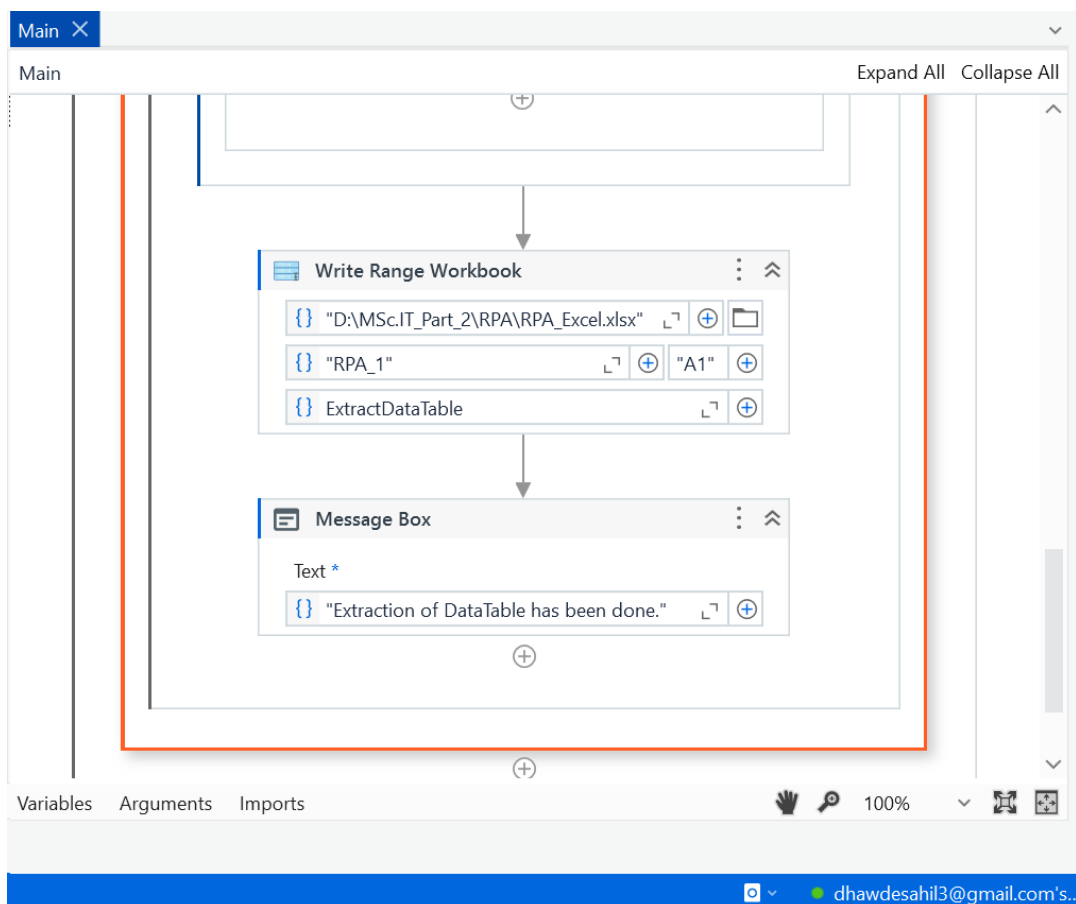
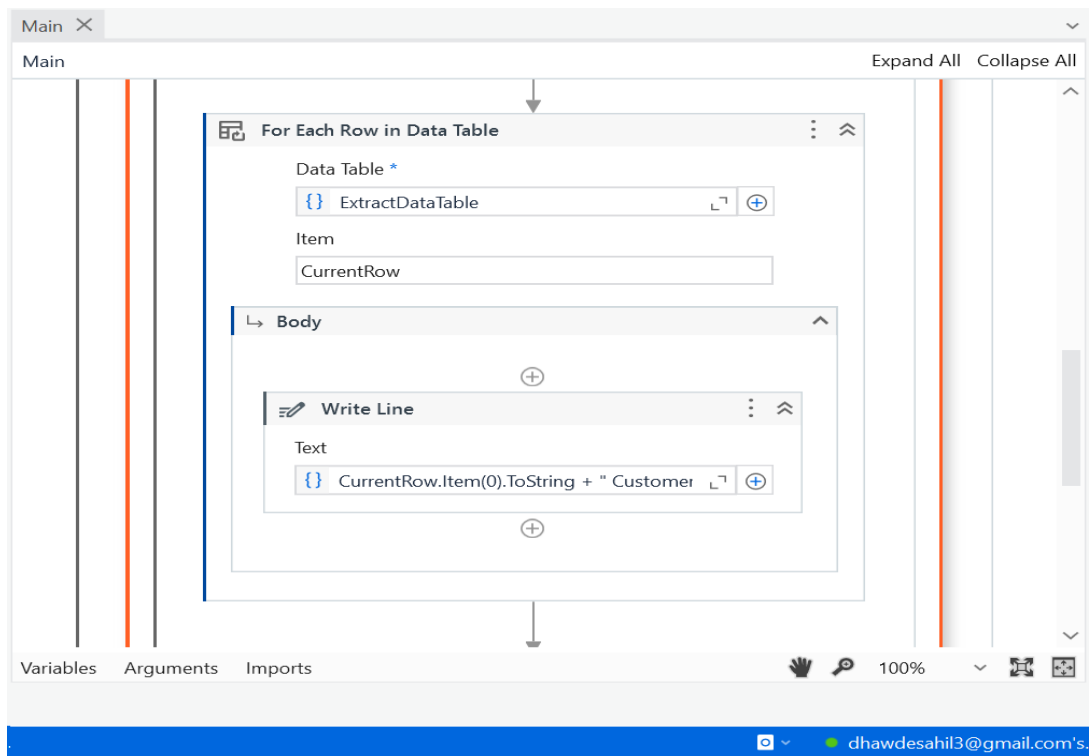
Process & Output:



Below is a selection from the **Customers** table used in the examples:

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
1	Alfreds Futterkiste	Maria Anders	Obere Str. 57	Berlin	12209	Germany
2	Ana Trujillo Emparedados y helados	Ana Trujillo	Avda. de la Constitución 2222	México D.F.	05021	Mexico
3	Antonio Moreno Taquería	Antonio Moreno	Mataderos 2312	México D.F.	05023	Mexico
4	Around the Horn	Thomas Hardy	120 Hanover Sq.	London	WA1 1DP	UK
5	Berglunds snabbköp	Christina Berglund	Berguvsvägen 8	Luleå	S-958 22	Sweden





The screenshot shows the UiPath Studio interface. A 'Message Box' dialog is displayed in the center, containing the following text:

```
CustomerID, CustomerName, ContactName, Address, City, PostalCode, Country
1, Alfreds Futterkiste, Maria Anders, Obere Str. 57, Berlin, 12209, Germany
2, Ana Trujillo Emparedados y helados, Ana Trujillo, Avda. de la Constitución 2222, México D.F., 05021, Mexico
3, Antonio Moreno Taquería, Antonio Moreno, Mataderos 2312, México D.F., 05023, Mexico
4, Around the Horn, Thomas Hardy, 120 Hanover Sq., London, WA1 1DP, UK
5, Berglunds snabbköp, Christina Berglund, Berguvsvägen 8, Luleå, S-958 22, Sweden
```

Below the dialog, the 'Write Range Workbook' activity is visible, with the file path 'D:\MSc.IT_Part 2\RPA\Excel.xlsx' and the range 'A1' selected. A smaller 'Message Box' dialog is also present, stating 'Extraction of DataTable has been done.' The bottom of the interface shows the 'Output' pane with a search bar and a list of audit logs.

Search

- Audit: Extract Data. From: <html app='msedge.exe' title='SQL SELECT Statement' />
- 1 CustomerID Alfreds Futterkiste CustomerName Maria Anders ContactName Obere Str. 57 Address Berlin City 12209 PostalCode Germany Country
- 2 CustomerID Ana Trujillo Emparedados y helados CustomerName Ana Trujillo ContactName Avda. de la Constitución 2222 Address México D.F. City 05021 PostalCode Mexico Country
- 3 CustomerID Antonio Moreno Taquería CustomerName Antonio Moreno ContactName Mataderos 2312 Address México D.F. City 05023 PostalCode Mexico Country
- 4 CustomerID Around the Horn CustomerName Thomas Hardy ContactName 120 Hanover Sq. Address London City WA1 1DP PostalCode UK Country
- 5 CustomerID Berglunds snabbköp CustomerName Christina Berglund ContactName Berguvsvägen 8 Address Luleå City S-958 22 PostalCode Sweden Country

Breakpoints Error List

The screenshot shows an Excel spreadsheet titled 'RPA_Excel.xlsx'. The data is organized into columns: CustomerID, CustomerName, ContactName, Address, City, PostalCode, and Country. The data is as follows:

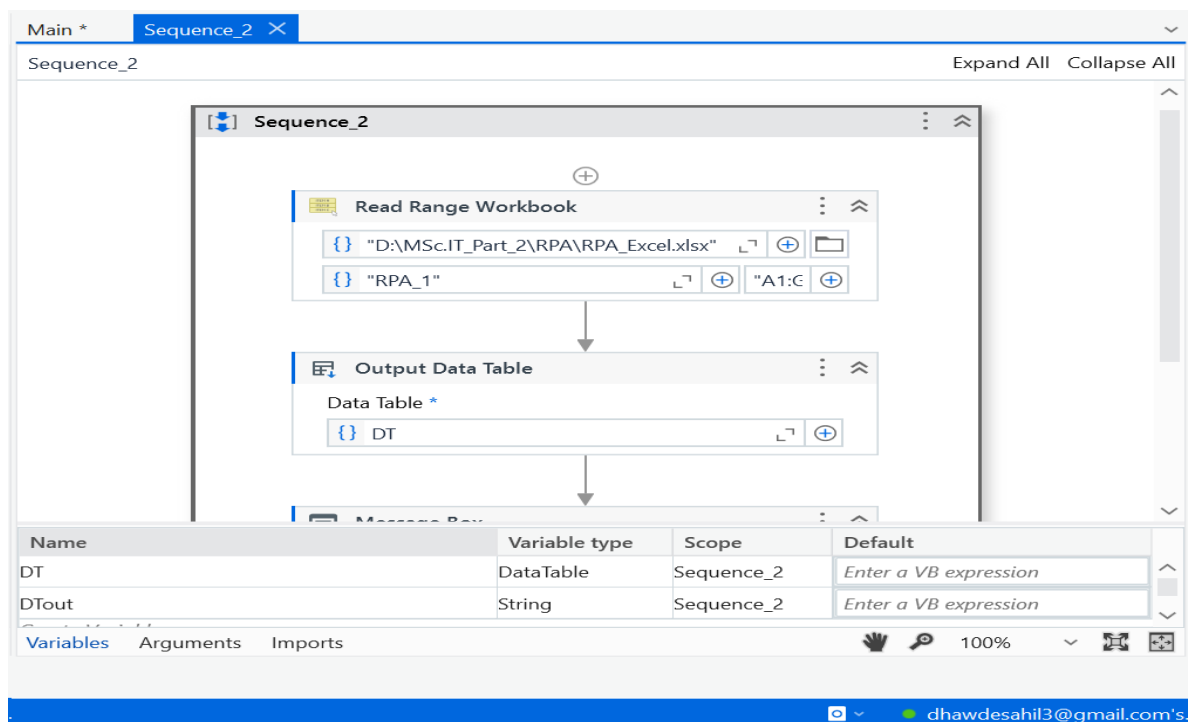
CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
1	Alfreds Futterkiste	Maria Anders	Obere Str. 57	Berlin	12209	Germany
2	Ana Trujillo Emparedados y helados	Ana Trujillo	Avda. de la Constitución 2222	México D.F.	05021	Mexico
3	Antonio Moreno Taquería	Antonio Moreno	Mataderos 2312	México D.F.	05023	Mexico
4	Around the Horn	Thomas Hardy	120 Hanover Sq.	London	WA1 1DP	UK
5	Berglunds snabbköp	Christina Berglund	Berguvsvägen 8	Luleå	S-958 22	Sweden

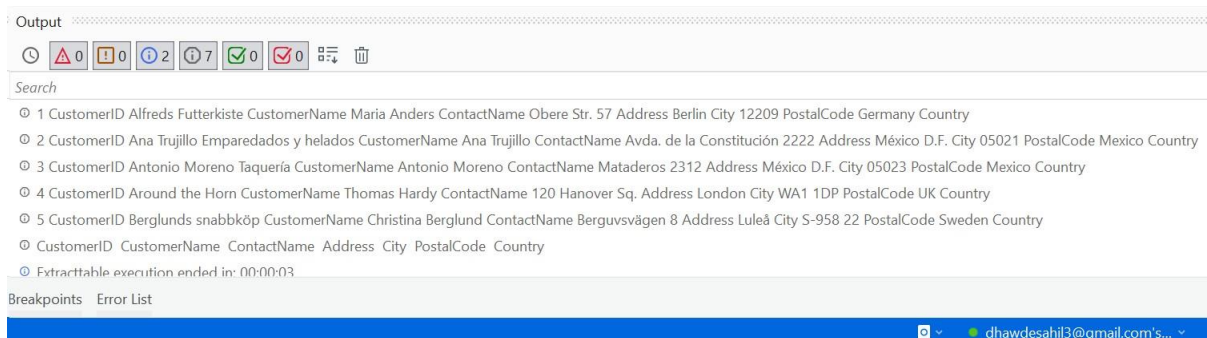
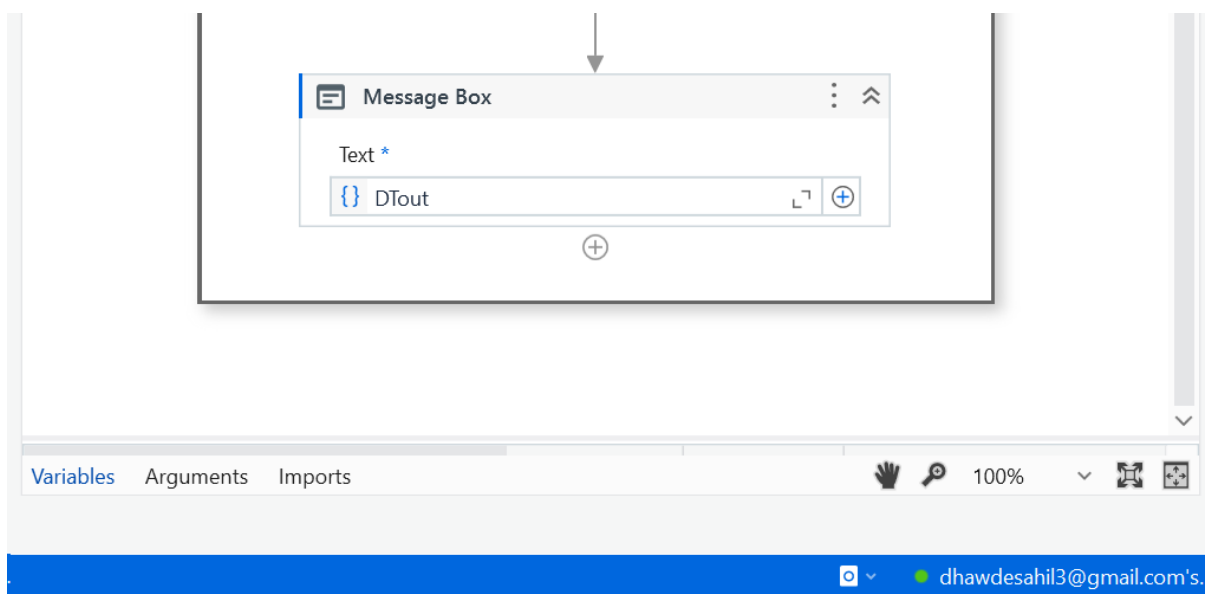
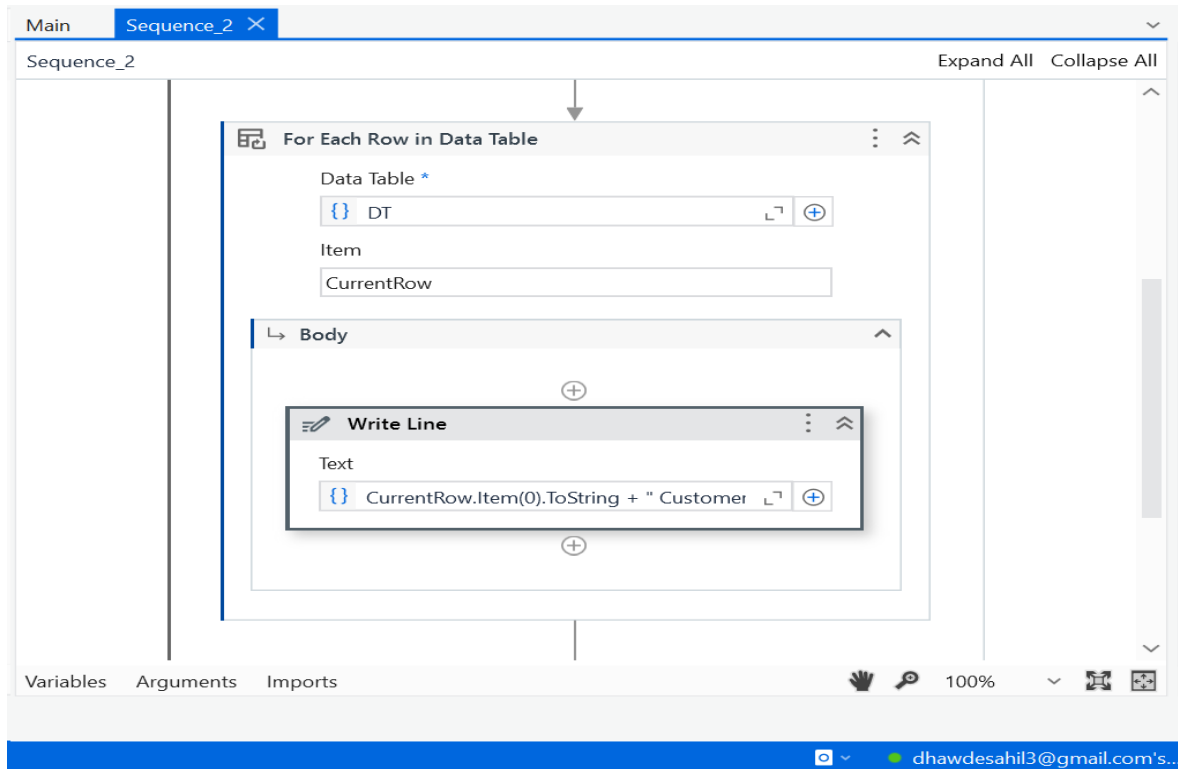
ii. Extract Data from an excel file into a Data Table

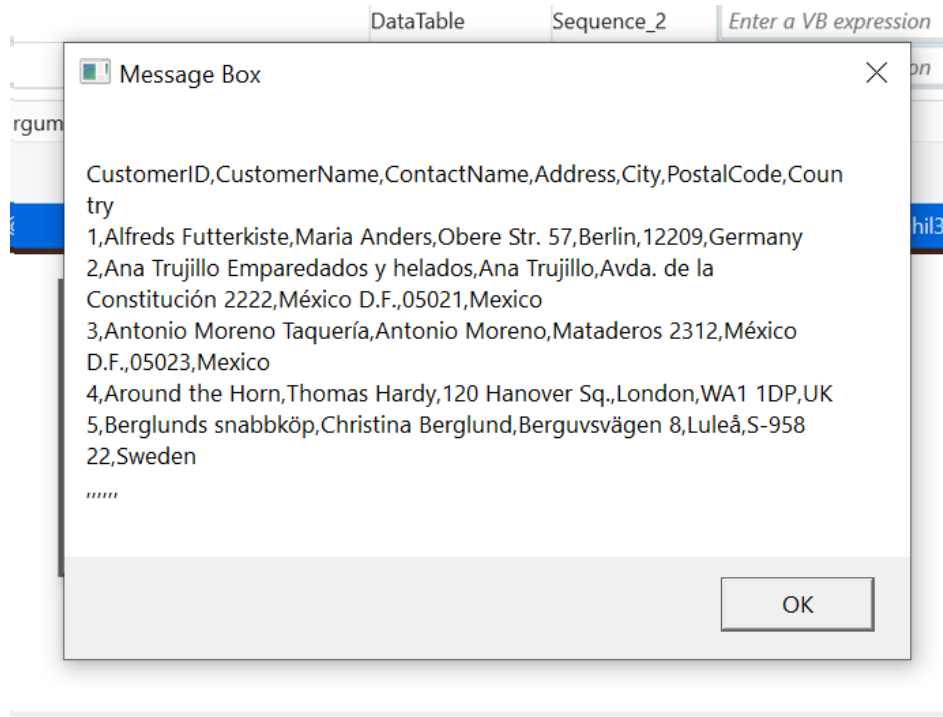
Steps:

1. Open Project.
2. Select Sequence activity.
3. Select from activities panel Read Range Workbook activity, and give the excel file name, sheet name and cell value as "A1:G7" to read whole table.
4. Create DT variable as DataTable.
5. Select from activities panel Output Data Table activity.
6. Create DTout variable as String.
7. Enter input as "DT" and output as "DTout".
8. Select For Each in Data Table activity.
9. Set values of List of items as "DT".
10. Set values of Item as "CurrentRow".
11. Inside the Body of For Each in DataTable.
12. Select the Write line activity & enter text "CurrentRow.Item(0).ToString + " CustomerID " + CurrentRow.Item(1).ToString + " CustomerName " + CurrentRow.Item(2).ToString + " ContactName " + CurrentRow.Item(3).ToString + " Address " + CurrentRow.Item(4).ToString + " City " + CurrentRow.Item(5).ToString + " PostalCode " + CurrentRow.Item(6).ToString + " Country "".
13. Select Message box and enter "DTout".
14. Run the project.

Process & Output:







Practical No: 7

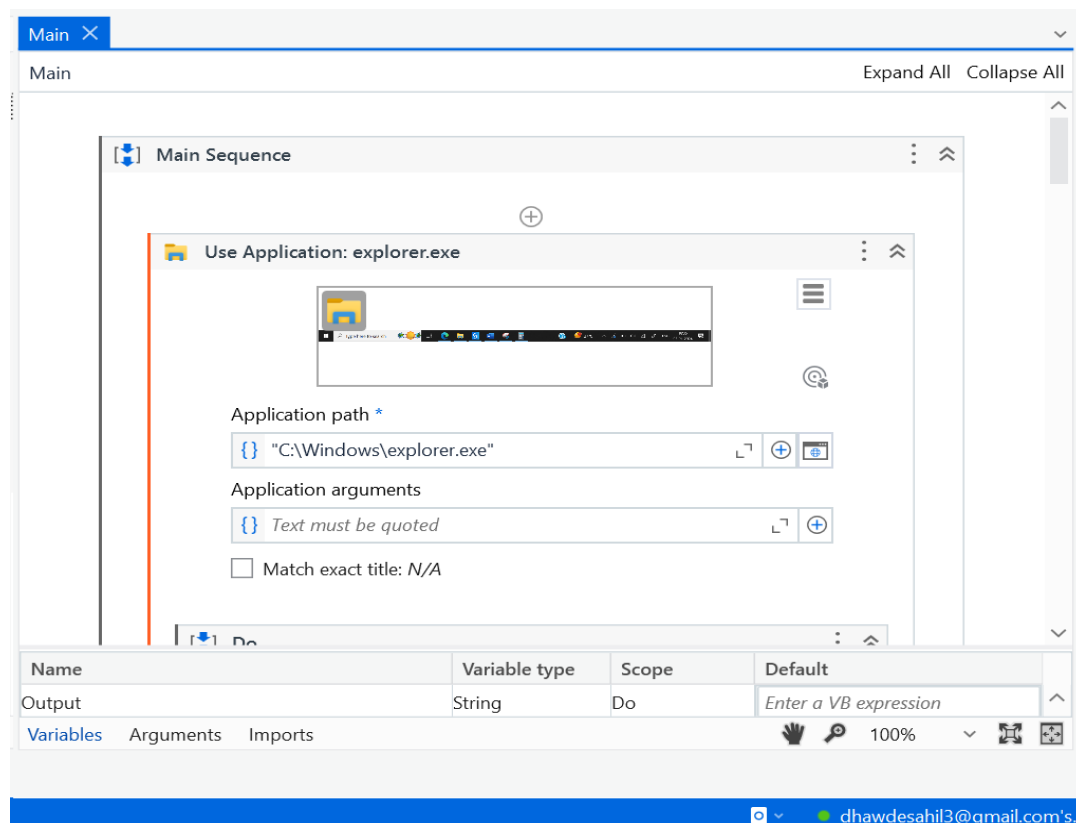
Aim: Implement the following:

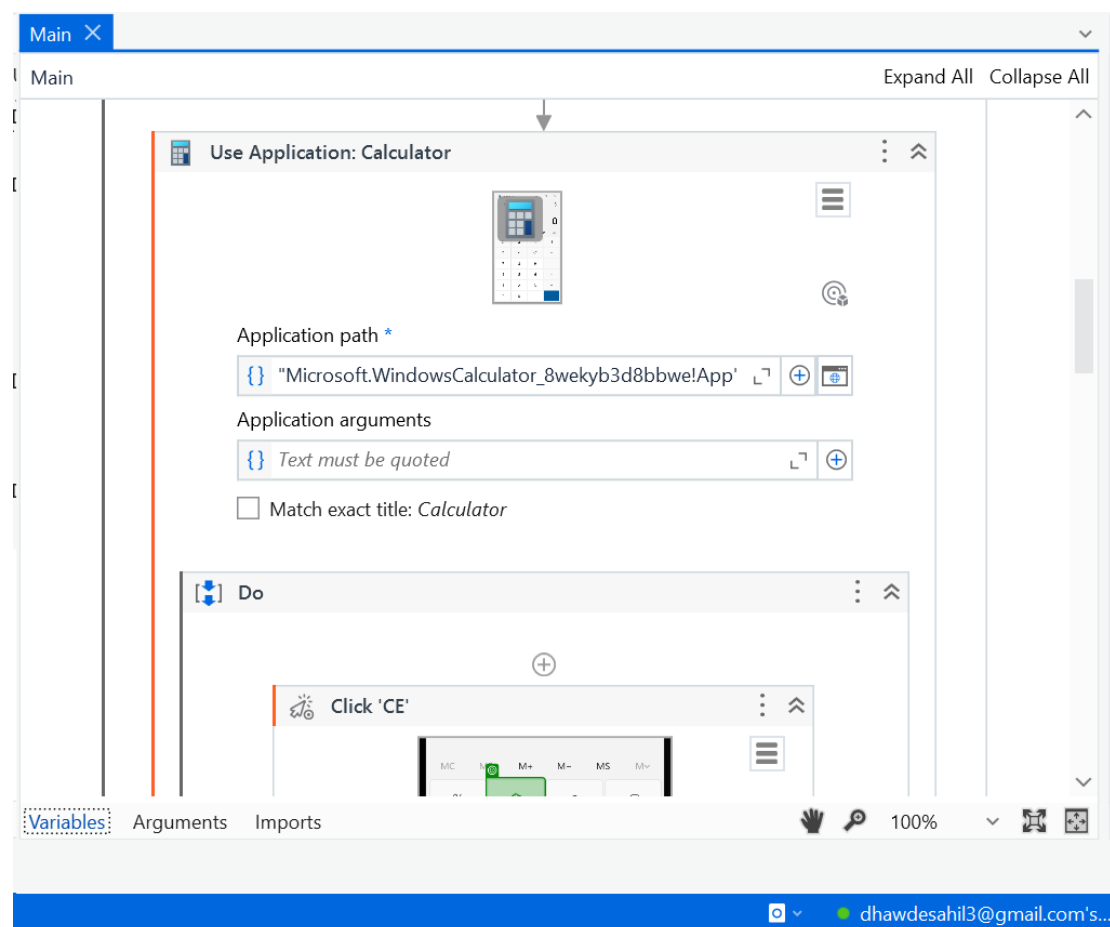
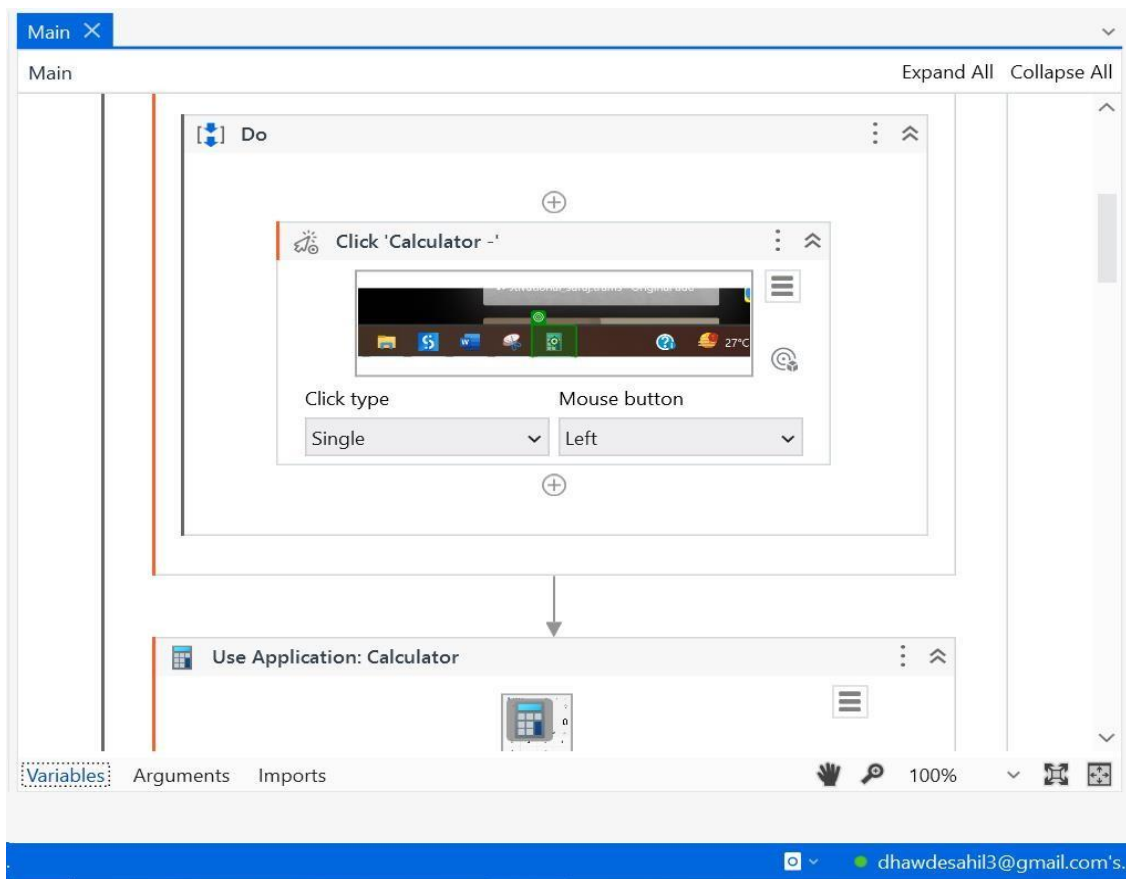
A. Implement the attach window activity.

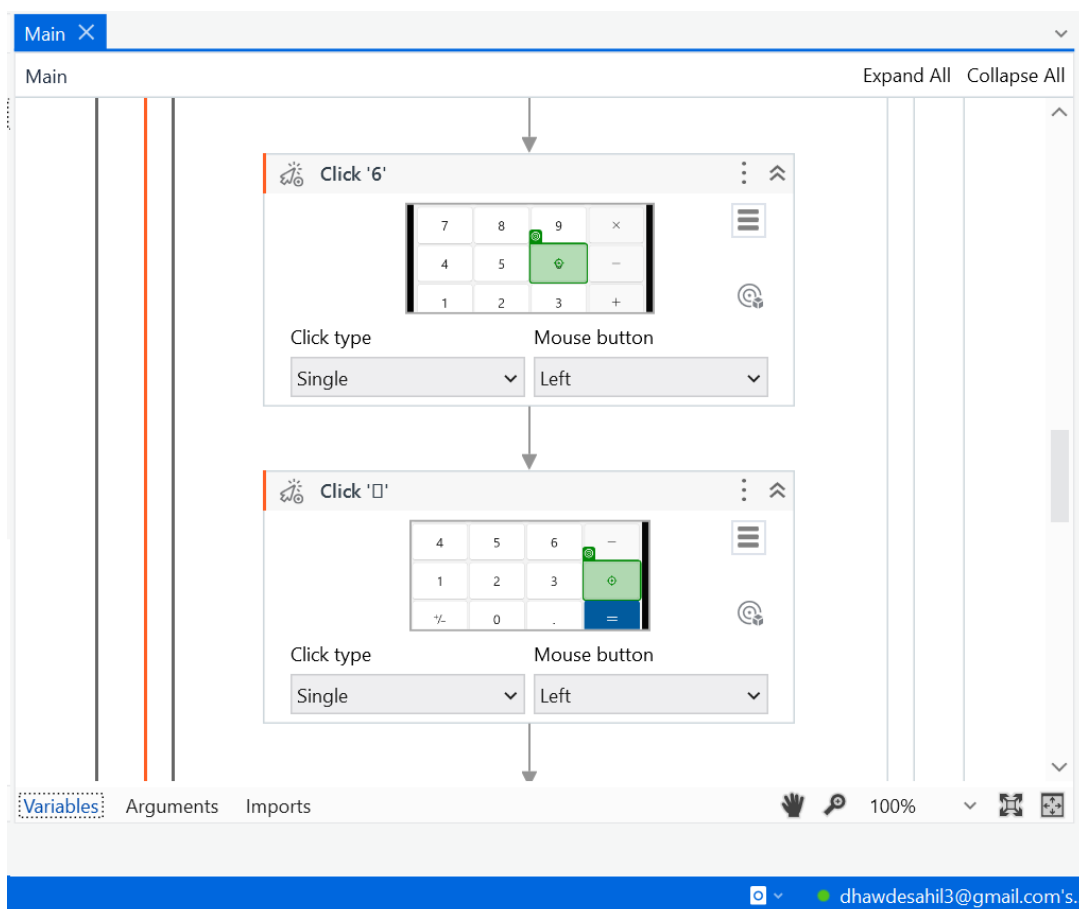
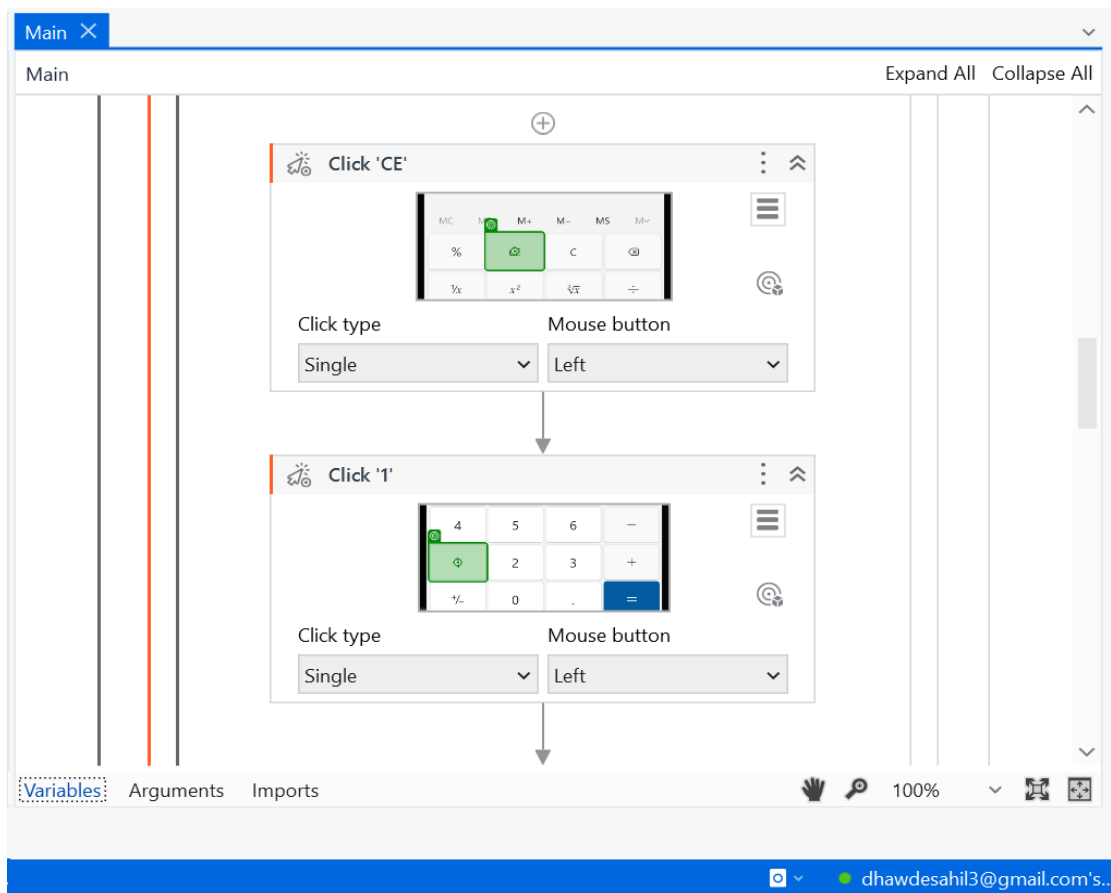
Steps:

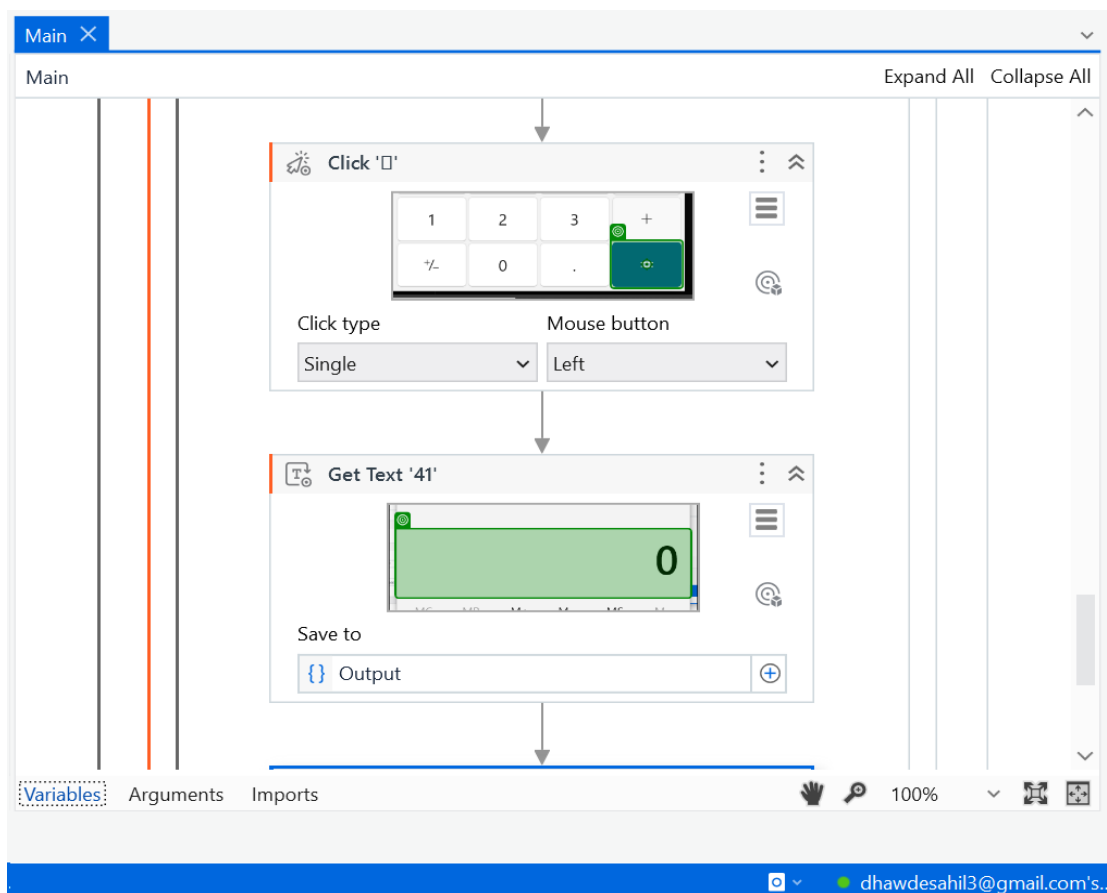
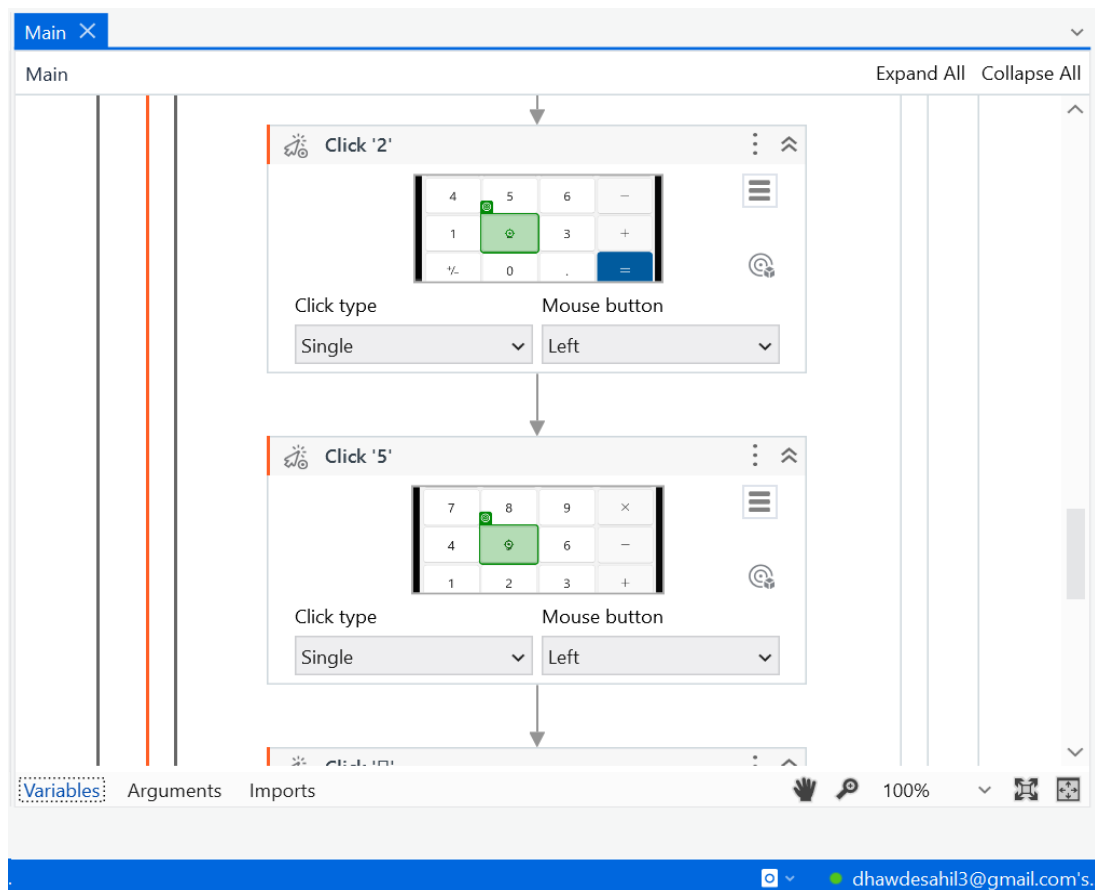
1. Open project.
2. Select Sequence activity.
3. Select App/Web Recorder icon on Ribbon panel of UiPath studio.
4. Select Calculator icon from task bar and select "Single Click" option.
5. In calculator, select number to calculate with "Single Click" option.
6. Select output display window of calculator and select "Get Text" option.
7. In Get Text activity, save to as "Output."
8. Select from activities panel Message box activity, type "Calculation has been completed and sum is " + Output."
9. Run the project.

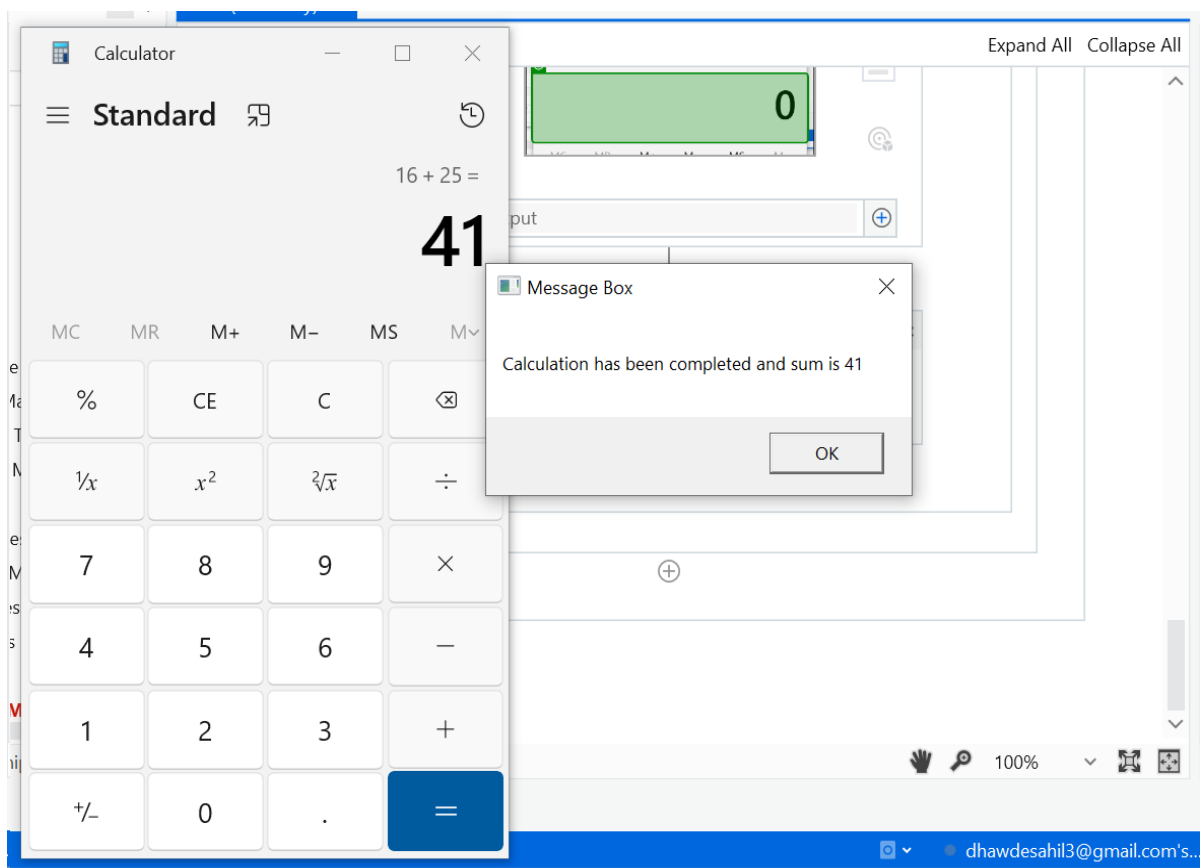
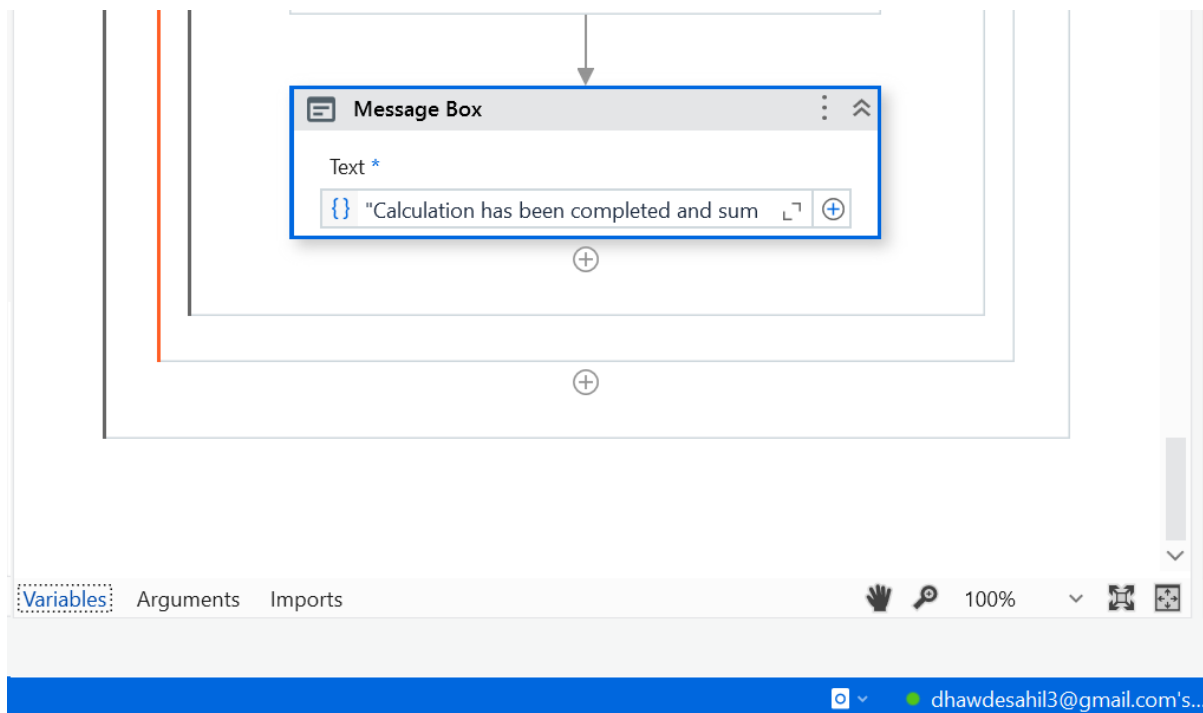
Process & Output:











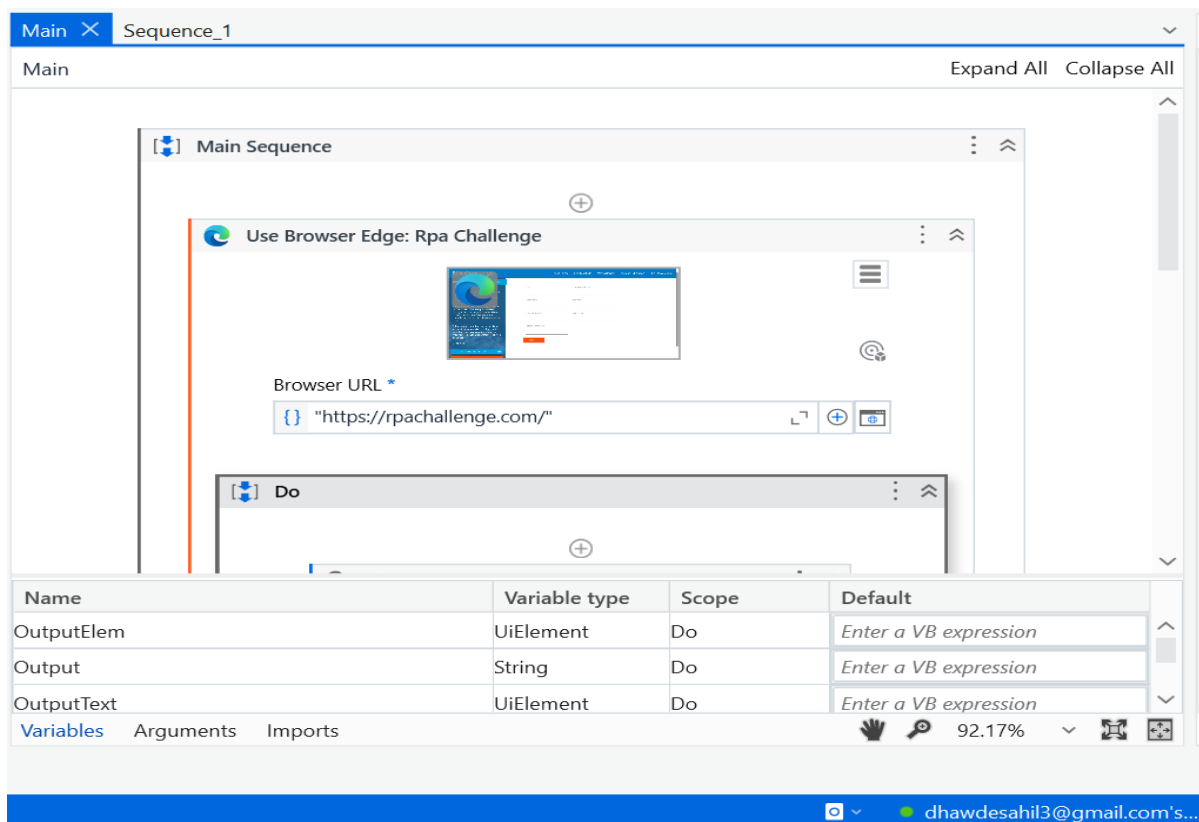
B. Find different controls using UiPath.

i. Find Element

Steps:

1. Open project.
2. Select Sequence activity.
3. Select from activities panel Use Application/Browser activity, and indicate the website screen.
4. Select from activities panel Find Element activity and indicate element on the screen to find.
5. Create OutputElem variable in FoundElement.
6. Select Get Attribute activity.
7. Set values of input element as "OutputElem".
8. Set values of Attribute as "aaname".
9. Create Output variable as String.
10. Set values of Save to as "Output".
11. Select Message box and enter "Output".
12. Run the project.

Process & Output:

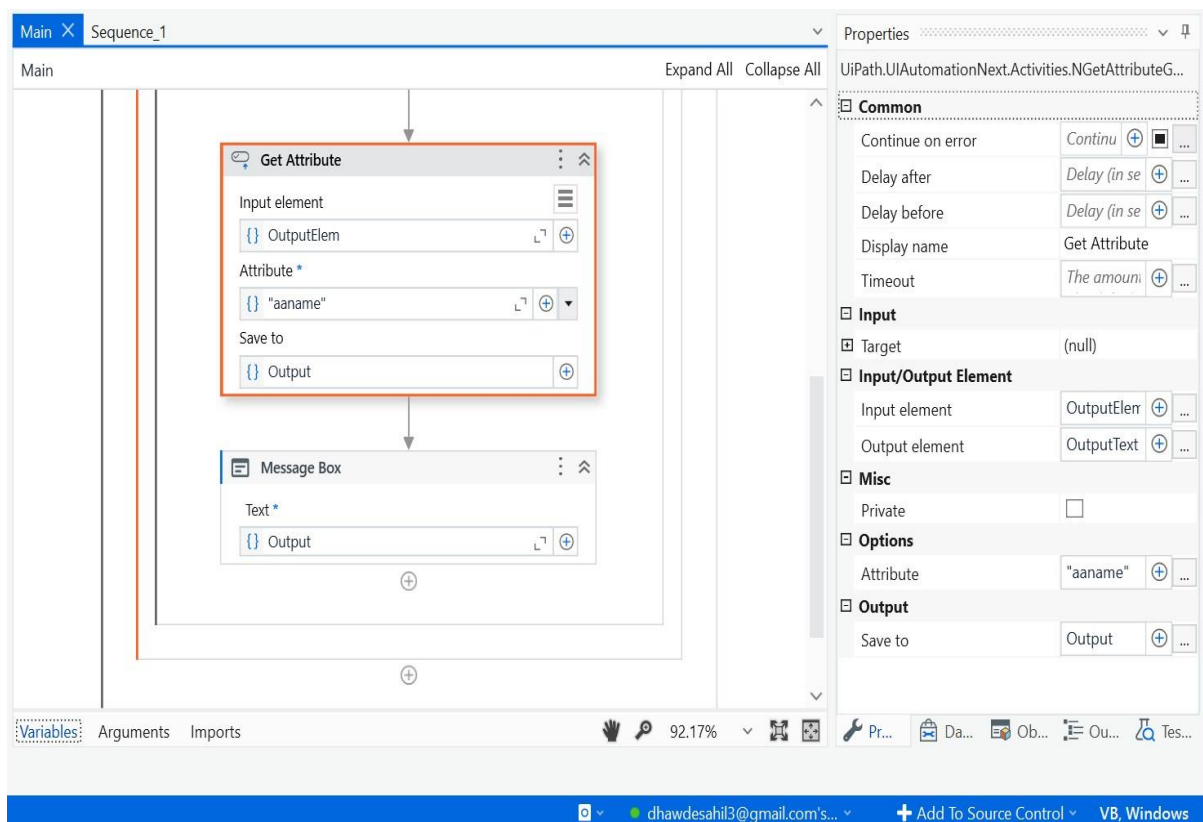
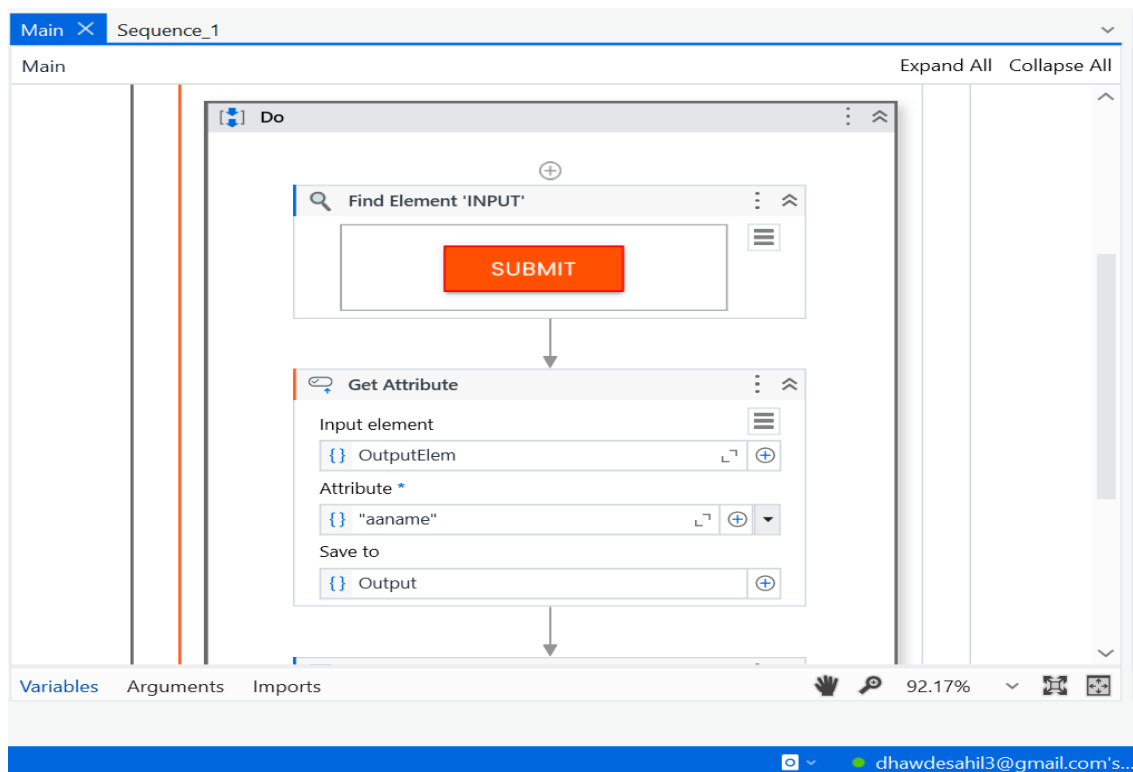


The screenshot displays the UiPath Studio interface. The main workspace shows a sequence activity named 'Main Sequence'. Inside this sequence, there is a 'Use Browser Edge: Rpa Challenge' activity, which is currently selected. Below it, there is a 'Do' activity, which is also selected. The 'Do' activity is expanded, showing a table with variables: 'OutputElem' (UiElement), 'Output' (String), and 'OutputText' (UiElement). The 'Browser URL' is set to 'https://rpachallenge.com/'.

Name	Variable type	Scope	Default
OutputElem	UiElement	Do	Enter a VB expression
Output	String	Do	Enter a VB expression
OutputText	UiElement	Do	Enter a VB expression

Variables Arguments Imports

92.17%



RPA Challenge

Input FormsShortest PathMovie Search

Instructions

EN

1. The goal of this challenge is to create a workflow that will input data from a spreadsheet into the form fields on the screen.

2. Beware! The fields will change position on the screen after every submission throughout 10 rounds thus the workflow must correctly identify where each spreadsheet record must be typed every time.

3. The actual countdown of the challenge will begin once you click the Start button until then you may submit the form as many times as you wish without receiving penalties.

Good luck!

EmailCompany Name

First NameAddress

Phone NumberName

Role in Company

SUBMIT

Message ...

Submit

OK

ii. Element Exist

Steps:

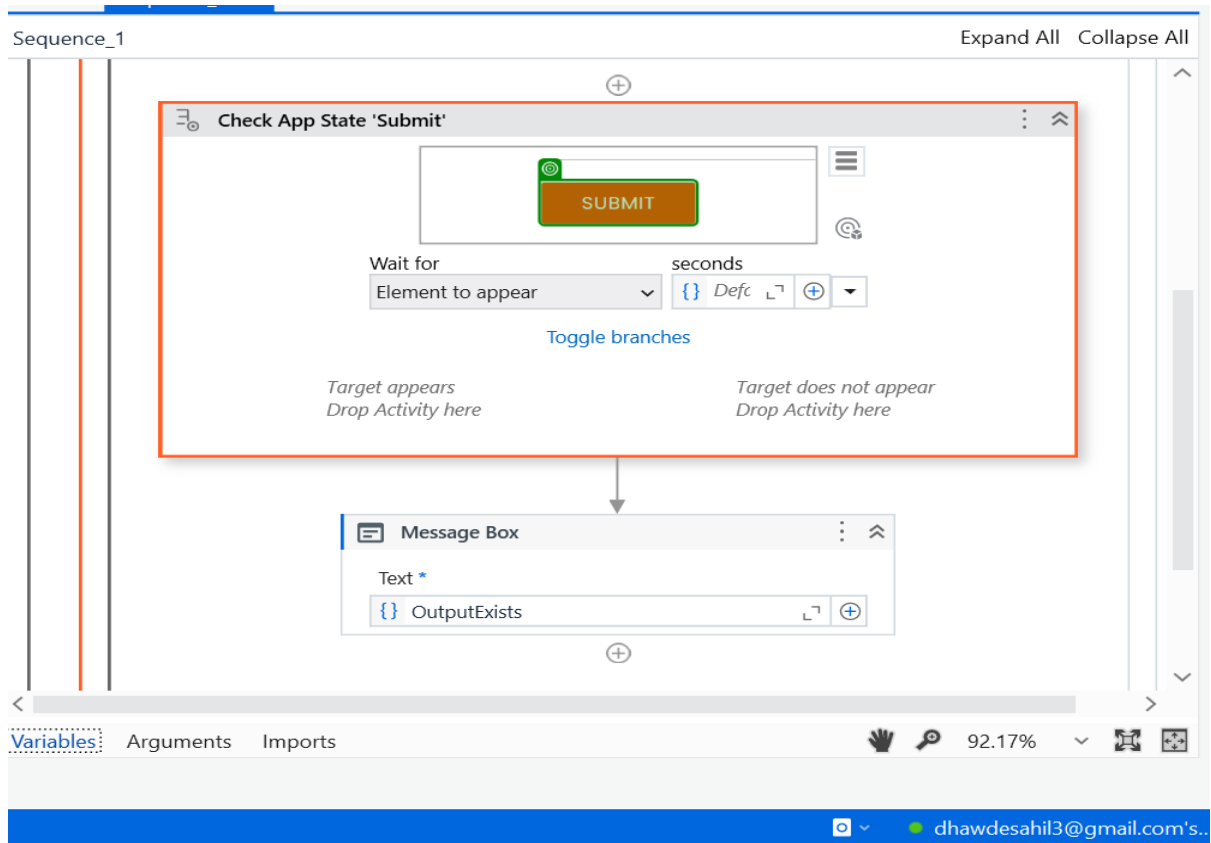
1. Open project.
2. Select Sequence activity.
3. Select from activities panel Use Application/Browser activity, and indicate the website screen.
4. Select from activities panel Check App State (new version) activity and indicate element on the screen to find.
5. Create OutputExist variable in result.
6. Select Message box and enter "OutputExist".
7. Run the project.

Process & Output:

The screenshot displays the RPA Studio interface with a sequence of activities. The first activity is 'Use Browser Edge: Rpa Challenge', which has a 'Browser URL' field set to 'https://rpachallenge.com/'. Below this is a 'Do' activity, which contains a 'Check App State \'Submit\'' activity. The 'OutputExists' variable is defined as Boolean, Scope Do, with a default value 'Enter a VB expression'.

Name	Variable type	Scope	Default
OutputExists	Boolean	Do	Enter a VB expression

Variables Arguments Imports 92.17%



RPA Challenge

Input Forms Shortest Path Mov

Instructions

EN

1. The goal of this challenge is to create a workflow that will input data from a spreadsheet into the form fields on the screen.
2. Beware! The fields will change position on the screen after every submission throughout 10 rounds thus the workflow must correctly identify where each spreadsheet record must be typed every time.
3. The actual countdown of the challenge will begin once you click the Start button until then you may submit the form as many times as you wish without receiving penalties.

Good luck!

Email _____ Company Name _____

First Name _____ Address _____

Phone Number _____ Name _____

Role in Company _____

SUBMIT

Message ... X

True

OK

C. Demonstrate the following activities in UiPath:

i. Type Secure text

Steps:

1. Create an Asset in UiPath Orchestrator of type “Credential.”
2. Open Project.
3. Select Sequence activity.
4. Select Get Credential activity from activities panel.
5. In Get Credential activity, select orchestrator folder place as your workspace and enter the Asset Name that created earlier.
6. Create two variables as Username and Passout.
7. Select from activities panel Use Application/Browser activity, and indicate the browser on the screen.
8. Select from activities panel Type into activity and indicate the input dialog of username of the website.
9. In Type this of Type into enter “Username”.
10. Select from activities panel Type Secure Text activity and indicate the input dialog of password of the website.
11. In Securetext of Type secure Text enter “Passout”.
12. Select Message box and type “Username and Password has been entered”.
13. Run the project.

Process & Output:

UiPath Orchestrator

My Workspace > Assets > Add Asset

General details

Asset name*
Gmail_Login

Type
Credential

Description
Gmail login username and password

Credential Store*
Orchestrator Database

Tags

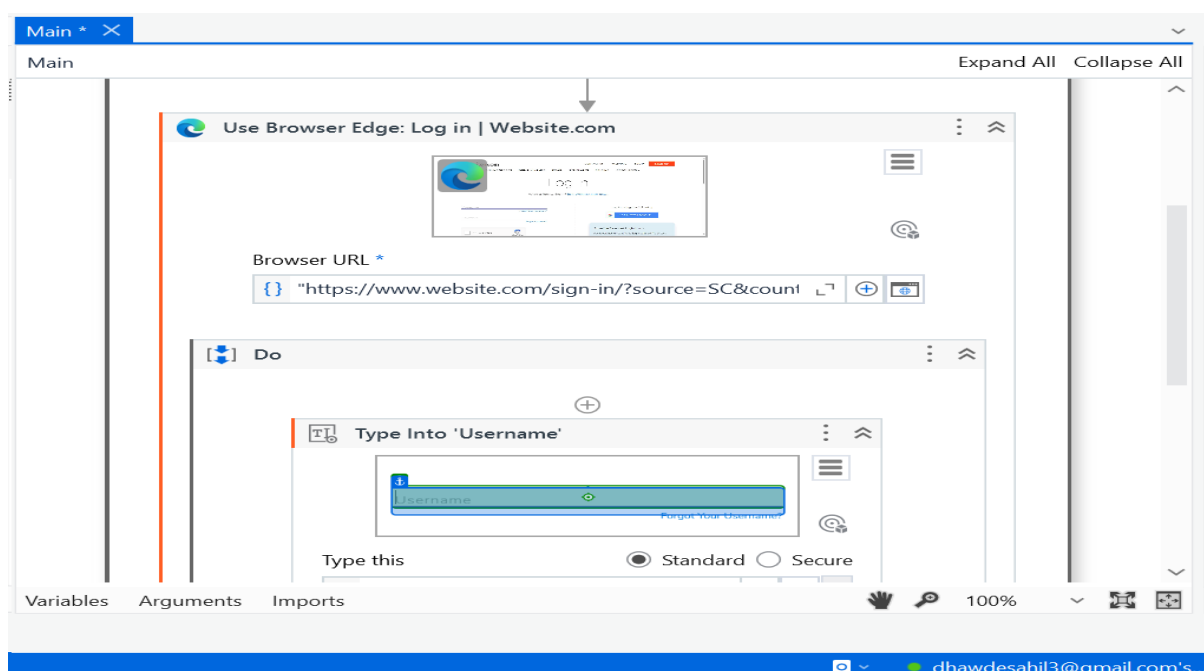
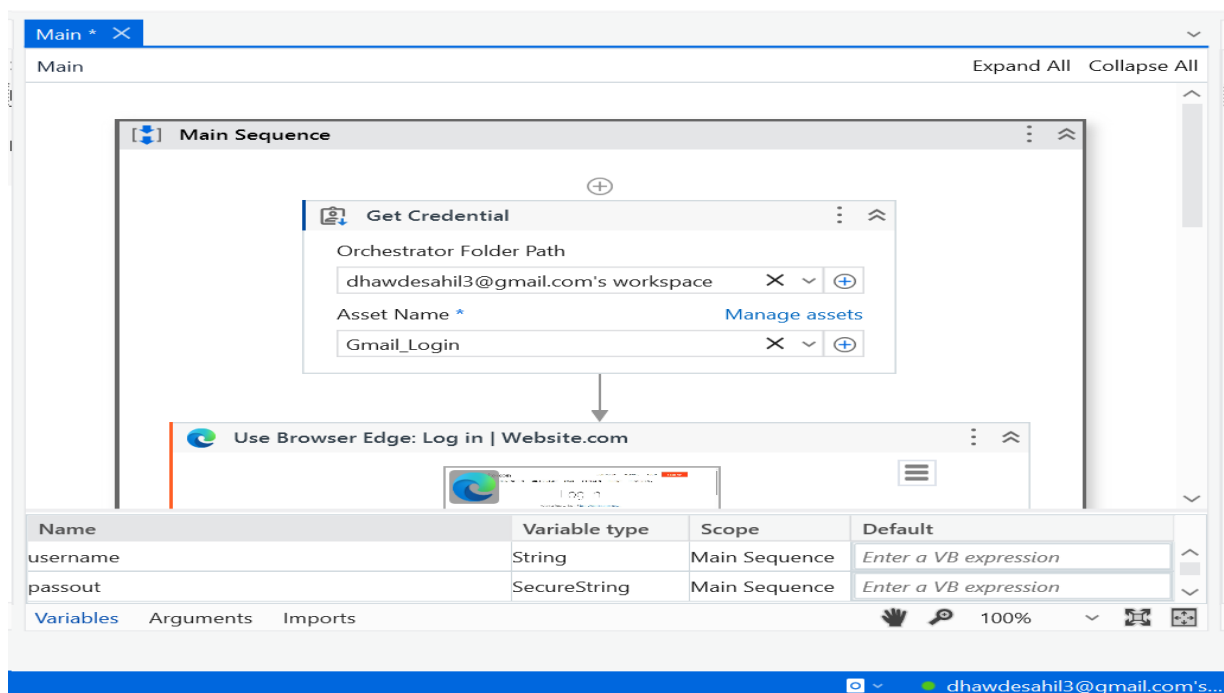
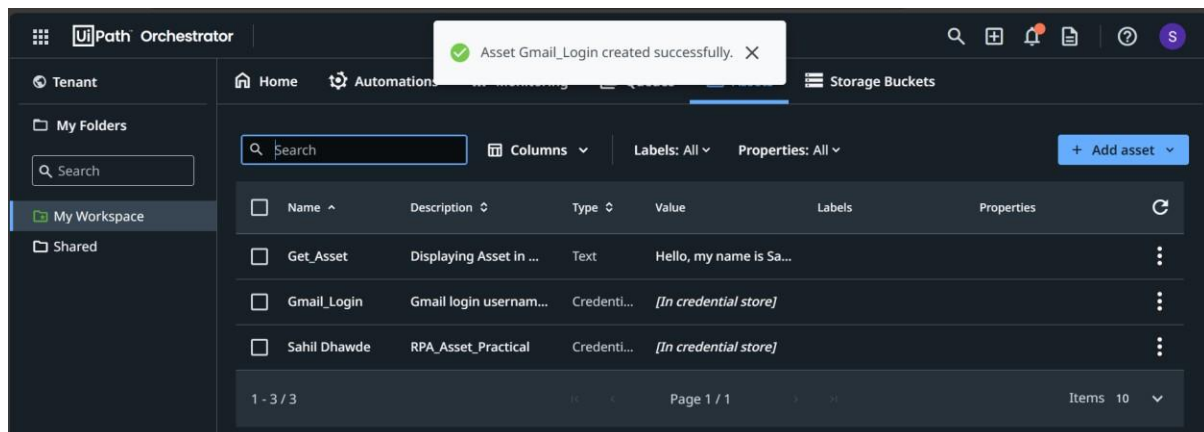
Asset value

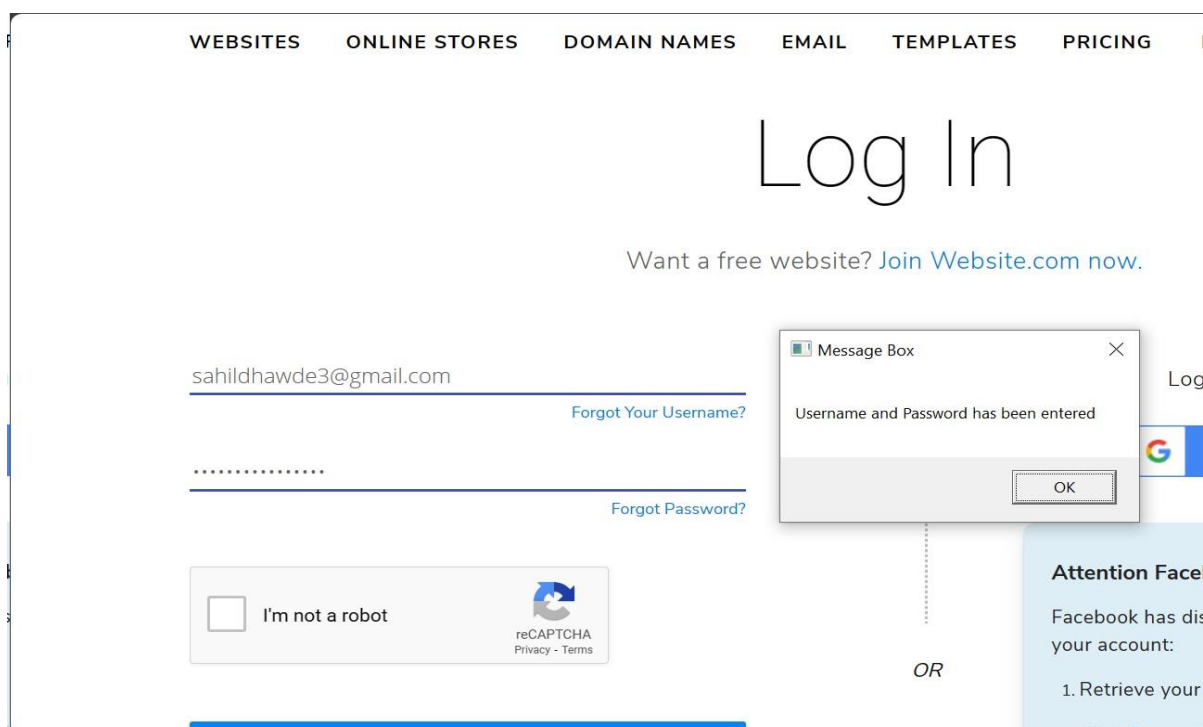
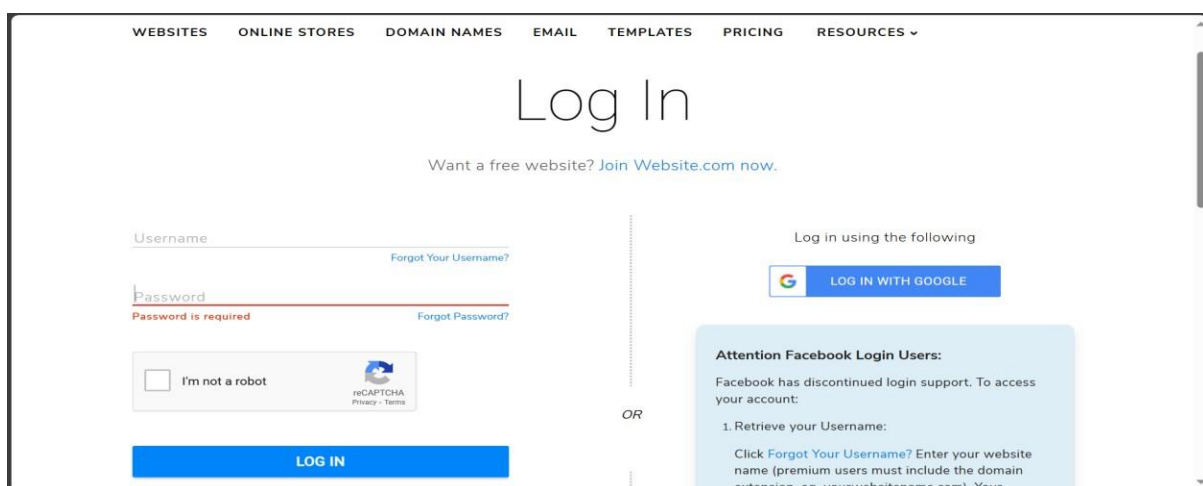
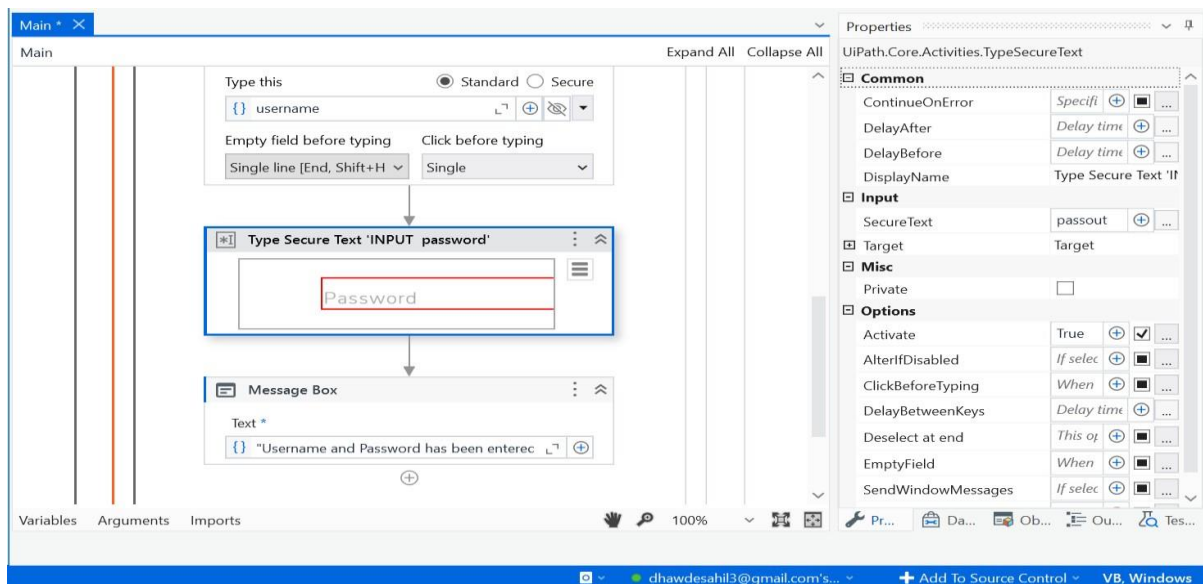
When the global value is enabled every user will receive it, unless specifically overridden in the table below.

Username*
sahildhawde3@gmail.com

Password*
.....

Cancel Create





Practical No: 8

Aim: Implement the following:

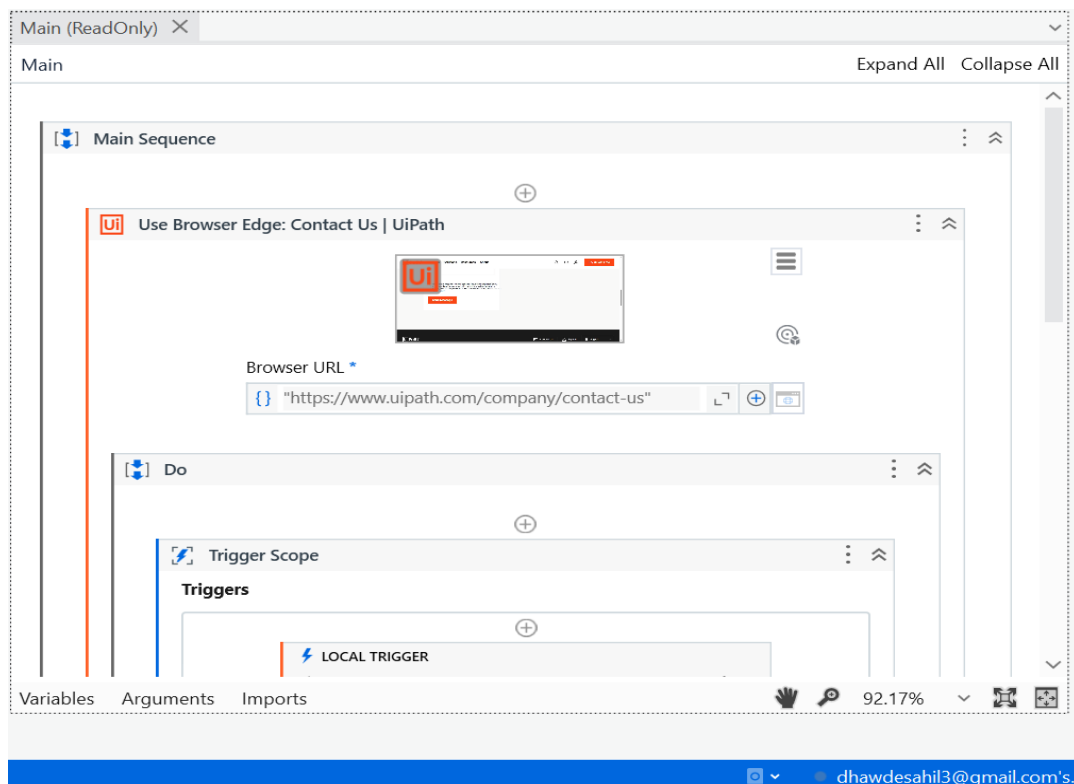
A. Demonstrate the following events in UiPath:

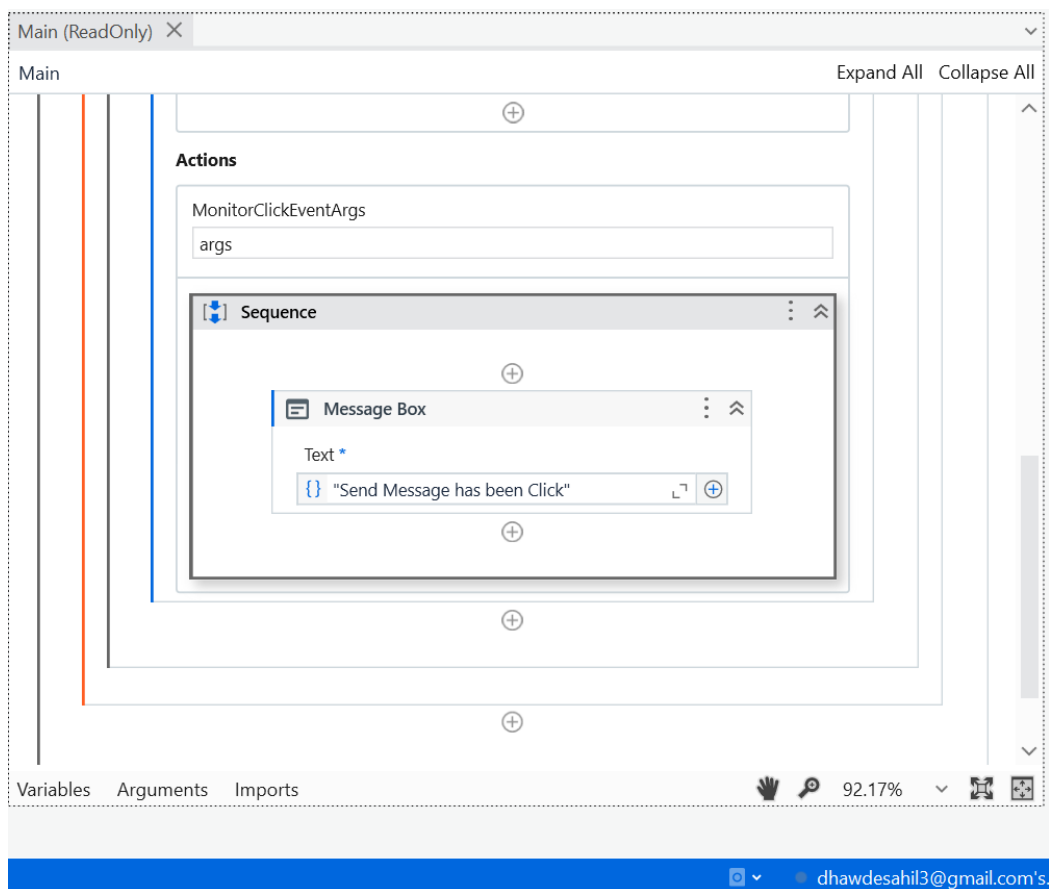
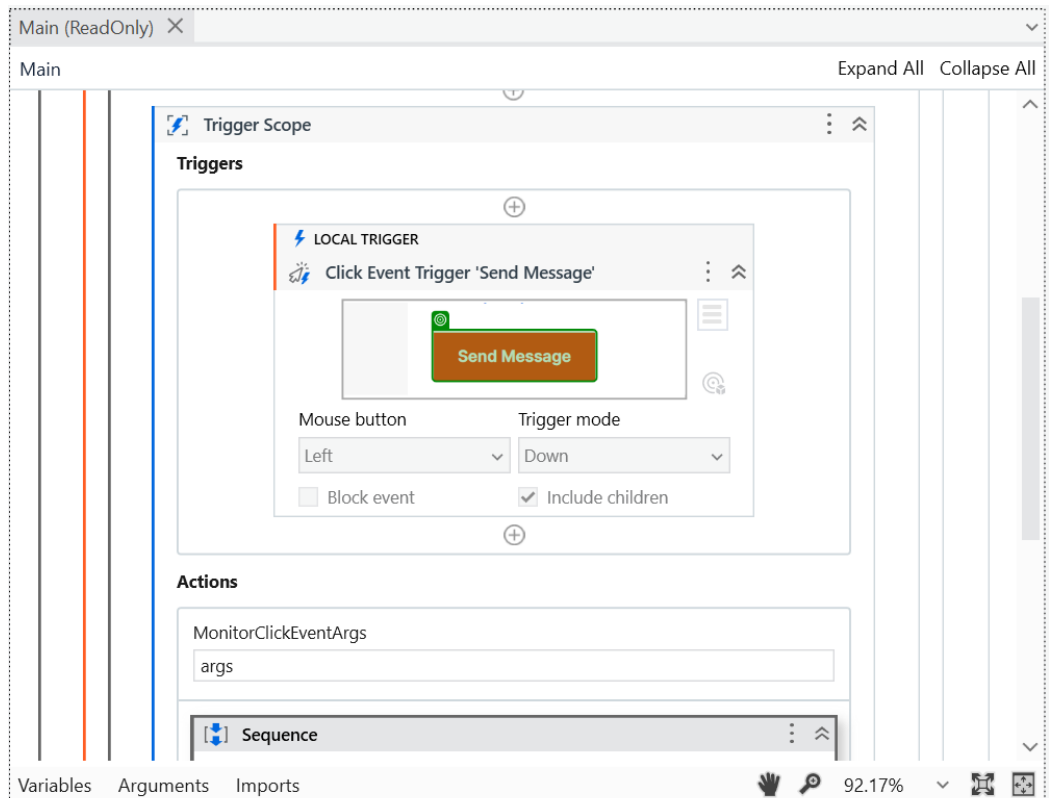
i. Element triggering event

Steps:

1. Open project.
2. Select Sequence activity.
3. Select from activities panel Use Application/Browser activity, and indicate the website screen.
4. Select from activities panel Trigger Scope activity.
5. In Trigger block, select from activities panel Click Event Trigger activity and indicate element on the screen to be triggered.
6. Enter Mouse Button as “LEFT” and Trigger mode as “Down”.
7. In Action block, select Message box and type “Send message has been clicked”.
8. Run the project.

Process & Output:





Platform Solutions Resources About

☐ I would like to receive communications about UiPath tailored to my interests and preferences, including latest news about products, services, events and promotions. For more information, please see [Privacy Policy](#)

Send Message

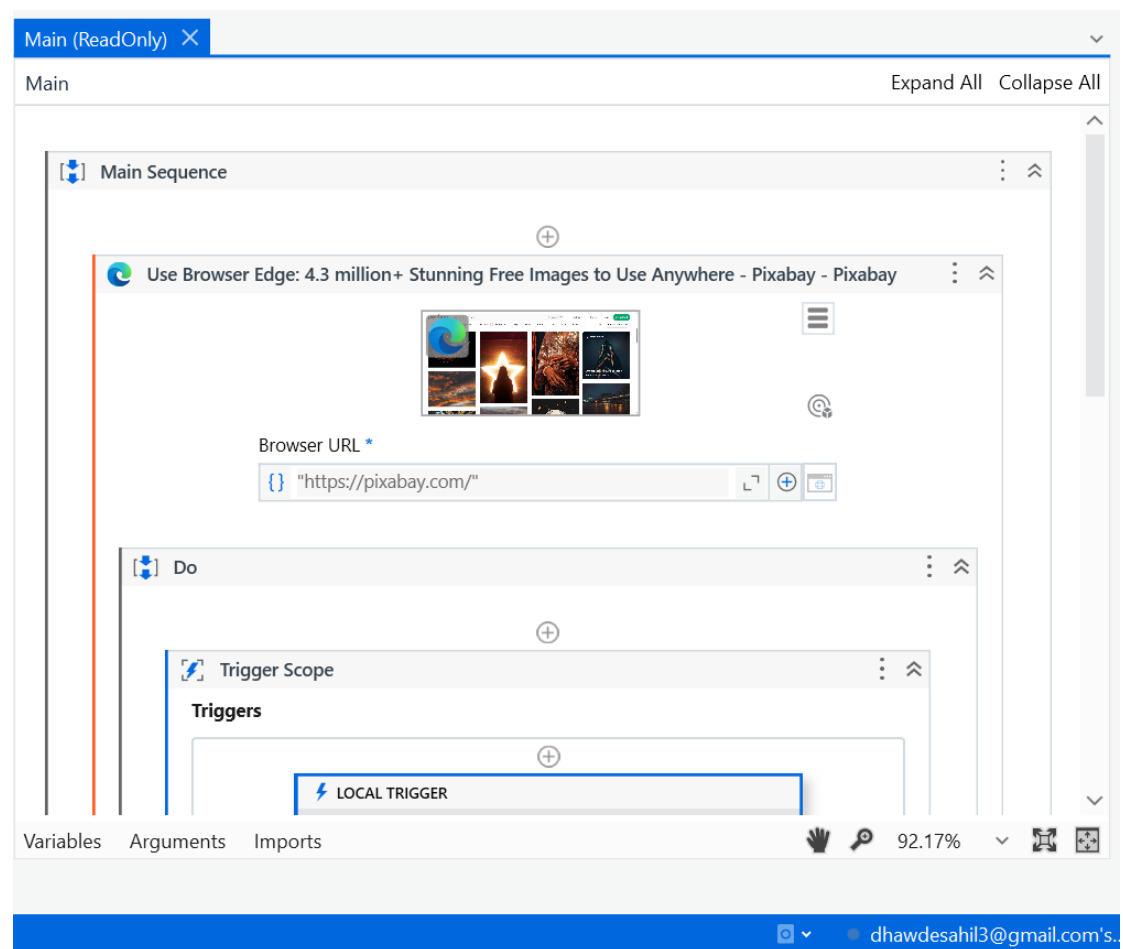
Message Box
Send Message has been Click
OK

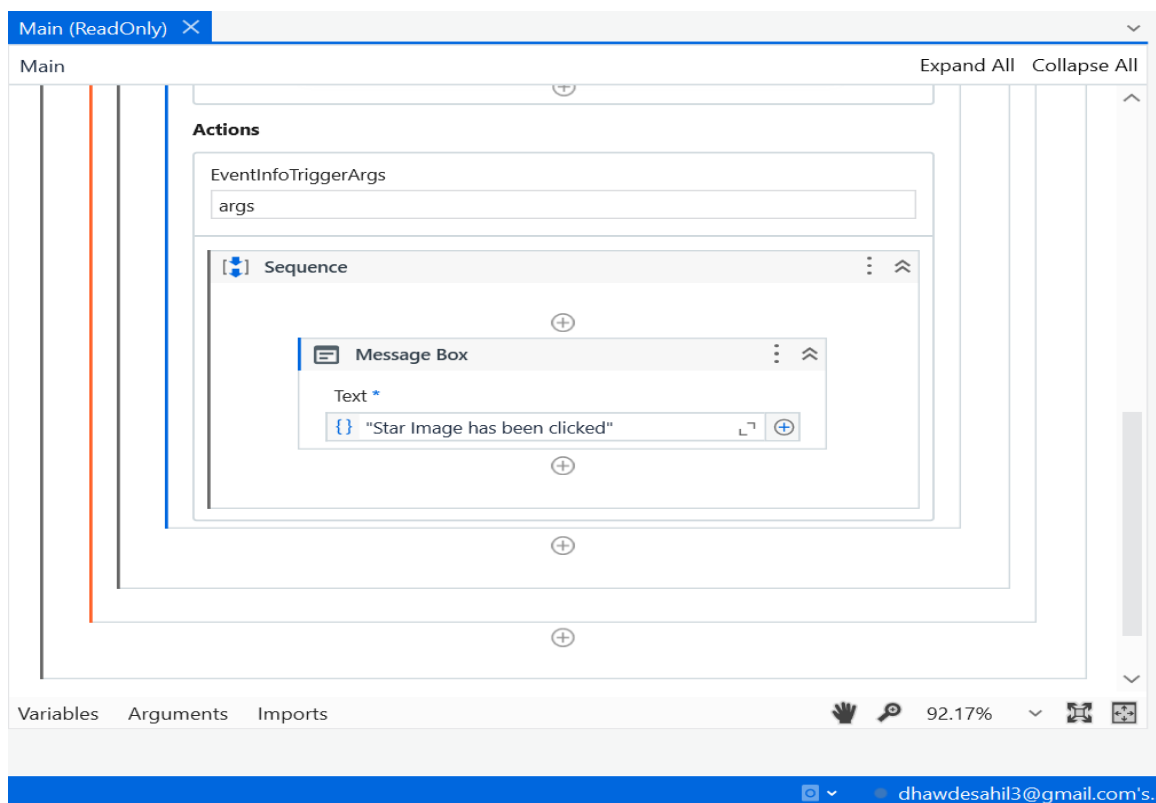
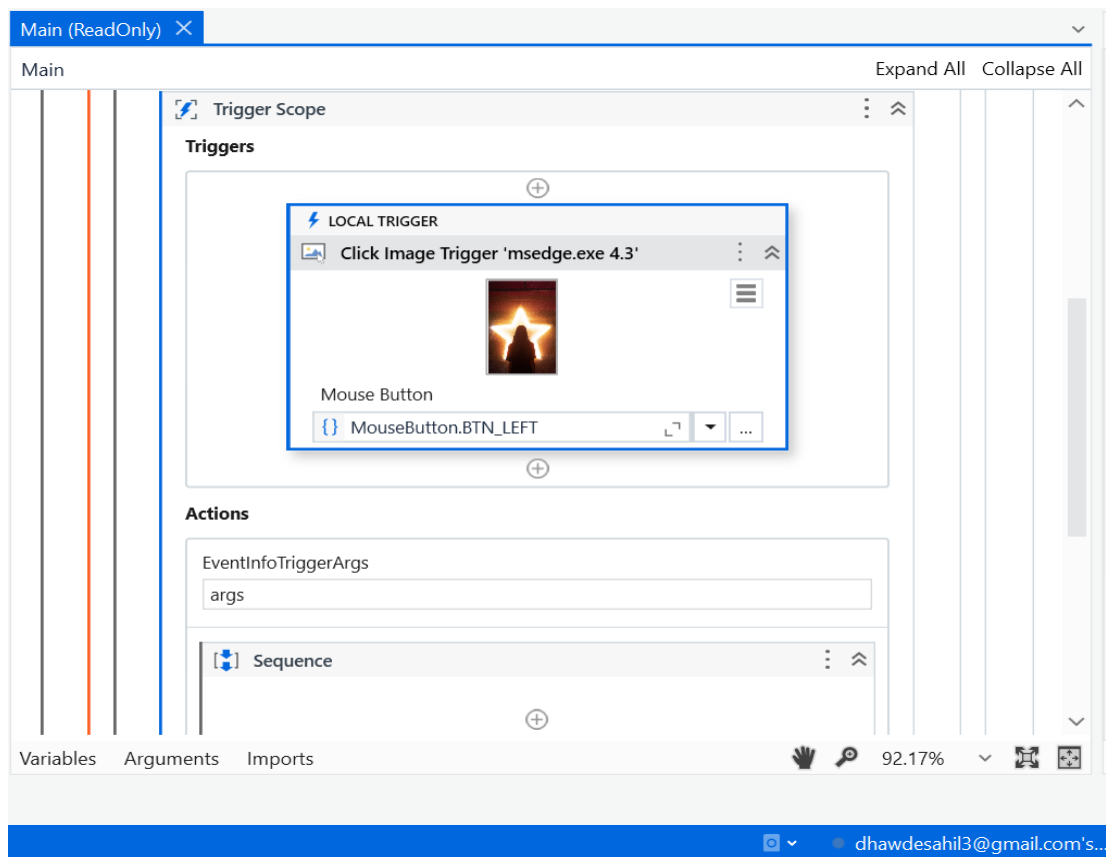
ii. Image triggering event

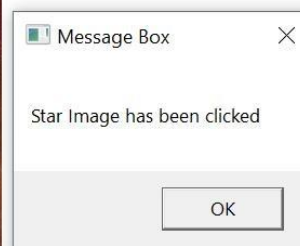
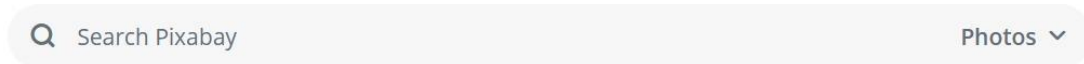
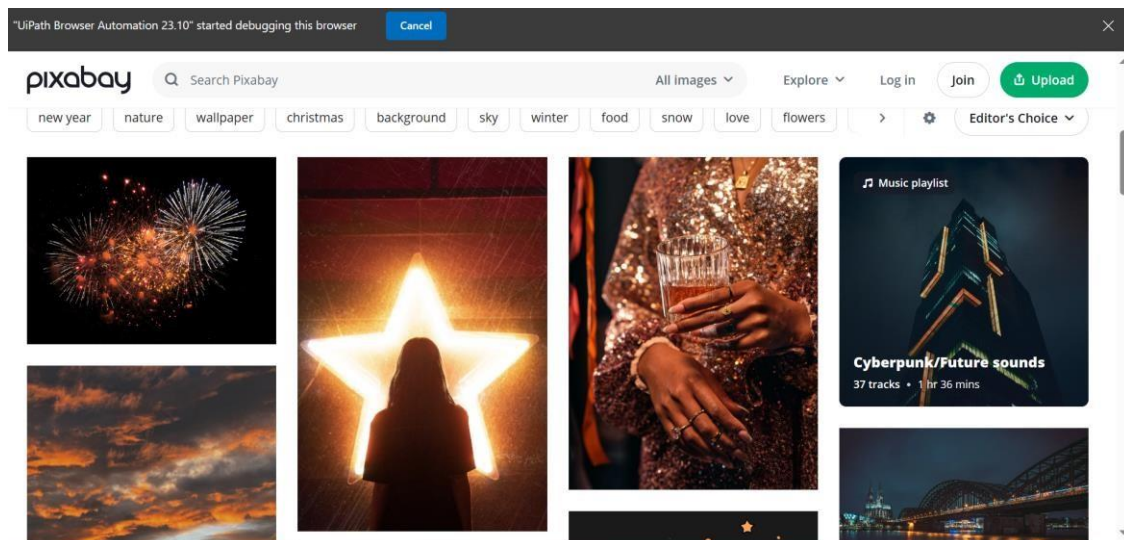
Steps:

1. Open project.
2. Select Sequence activity.
3. Select from activities panel Use Application/Browser activity, and indicate the website screen.
4. Select from activities panel Tigger Scope activity.
5. In Trigger block, select from activities panel Click Image Tigger activity and indicate image on the screen to be triggered.
6. Enter Mouse Button as "MouseButton.BTN_LEFT".
7. In Action block, select Message box and type "Star Image has been clicked".
8. Run the project.

Process & Output:







iii. System Triggering Event

M.L. Dahanukar College of Commerce

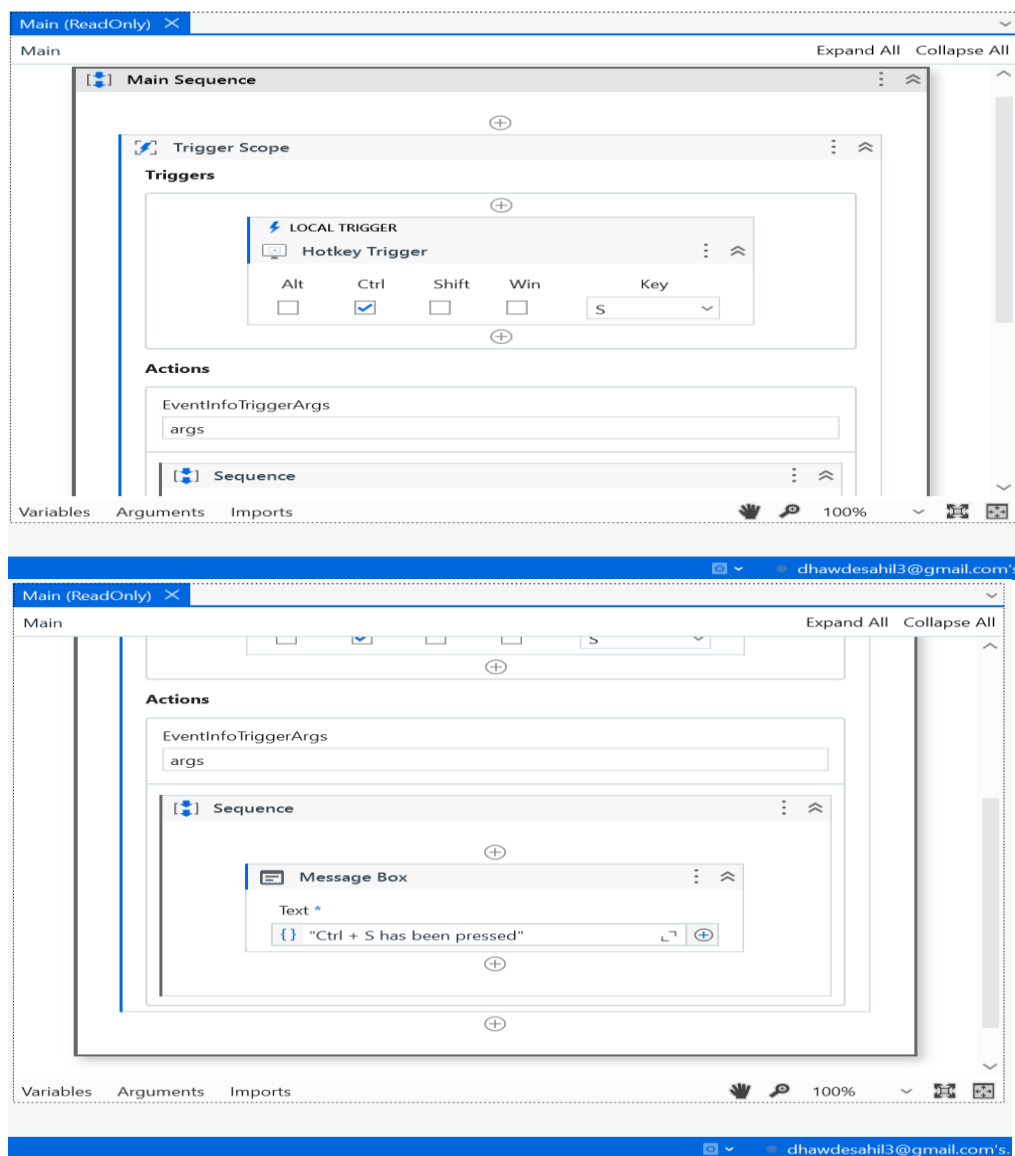
SHAIKH NAGHMA ARA

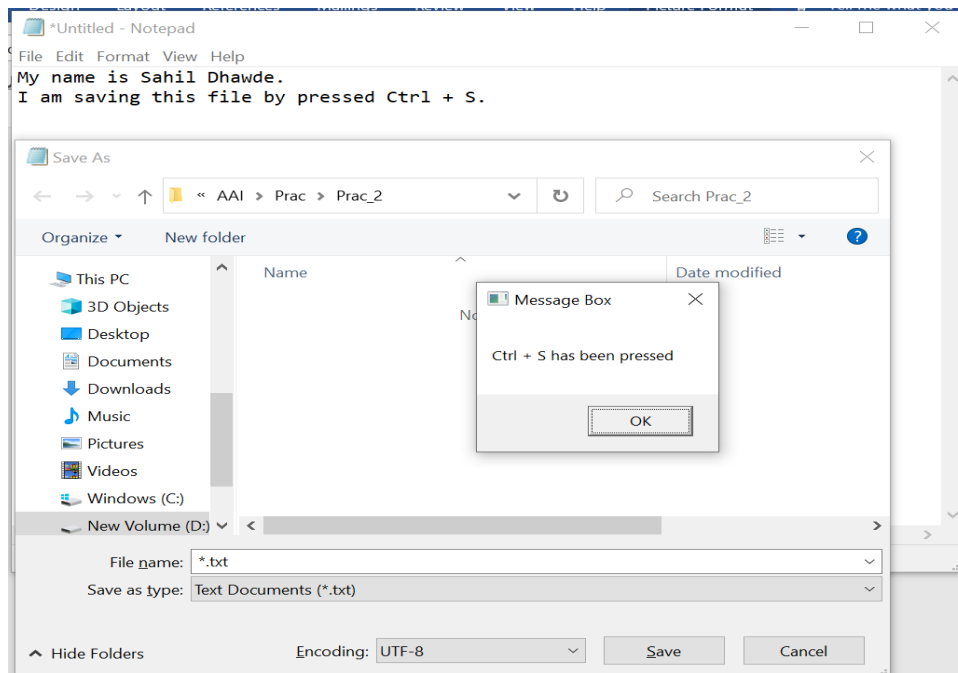
Hotkey Trigger

Steps:

1. Open project.
2. Select Sequence activity.
3. Select from activities panel Trigger Scope activity.
4. In Trigger block, select from activities panel Hotkey Trigger activity and check for “Ctrl” and enter Key as “S”.
5. In Action block, select Message box and type “Ctrl + S has been pressed”.
6. Run the project.

Process & Output:



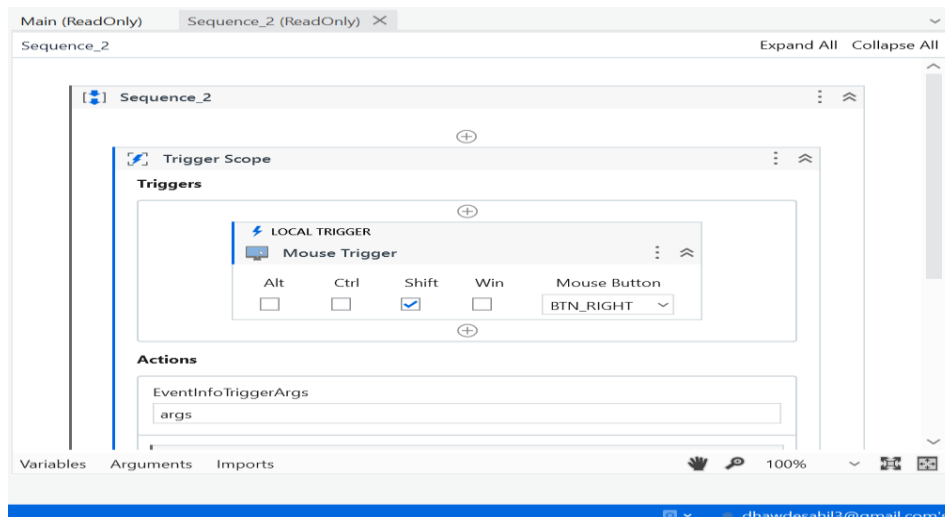


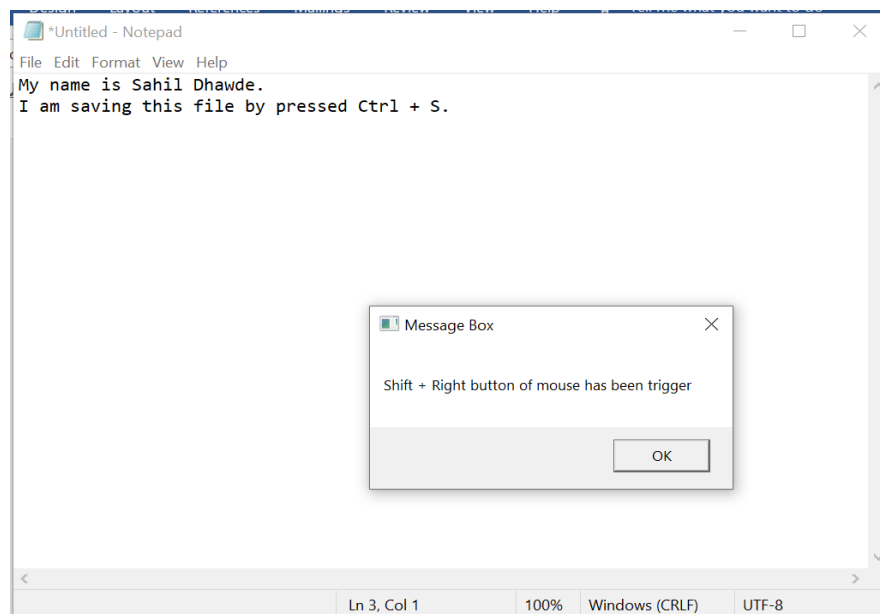
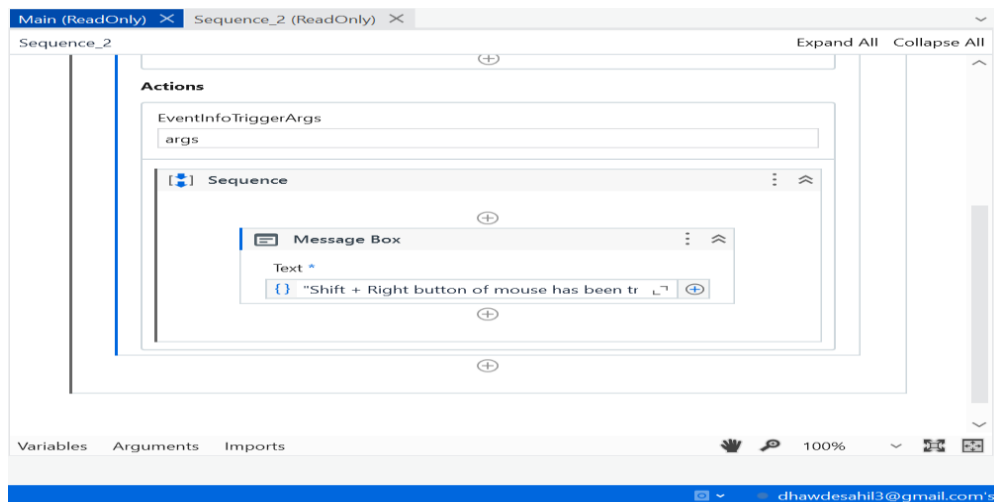
Mouse Trigger

Steps:

1. Open project.
2. Select Sequence activity.
3. Select from activities panel Tigger Scope activity.
4. In Trigger block, select from activities panel Mouse Tigger activity and check for “Shift” and enter Mouse Button as “Btn_Right”.
5. In Action block, select Message box and type “Shift + Right button of mouse has been trigger”.
6. Run the project.

Process & Output:



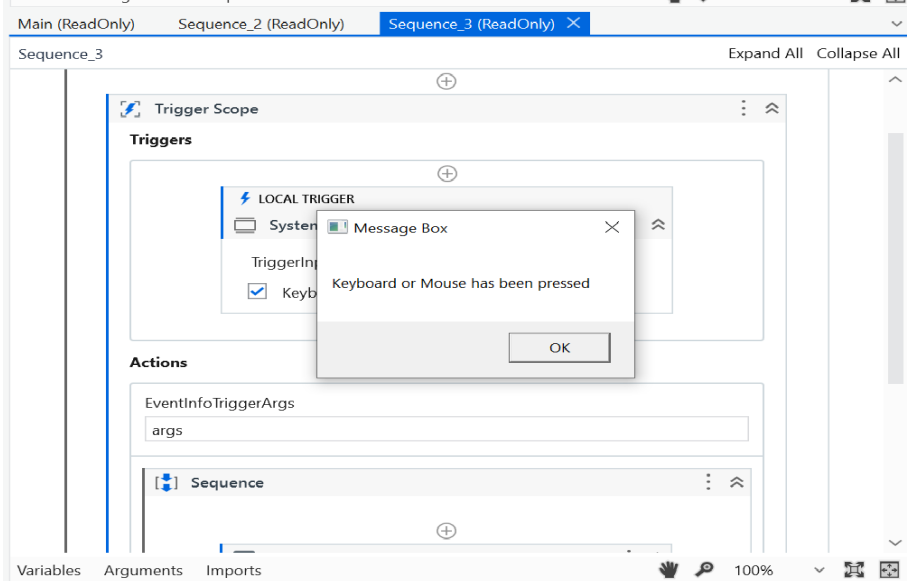
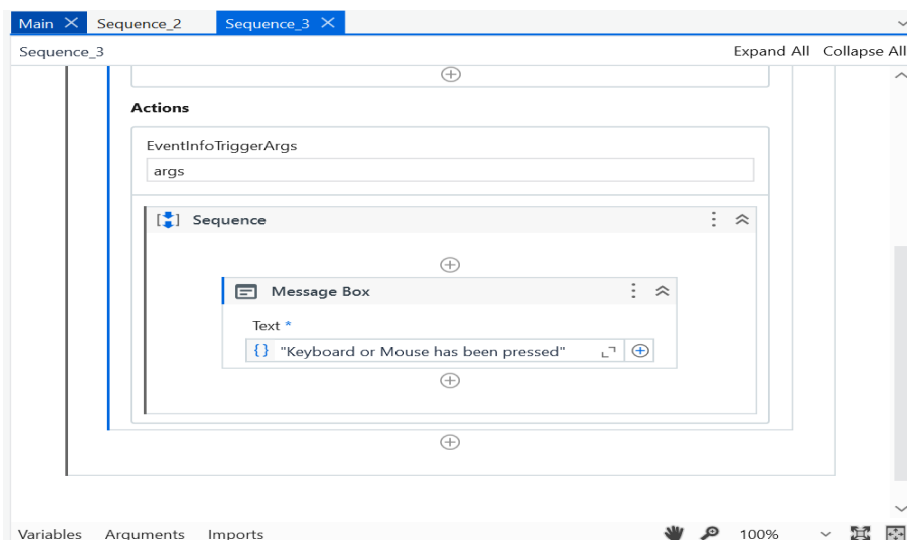
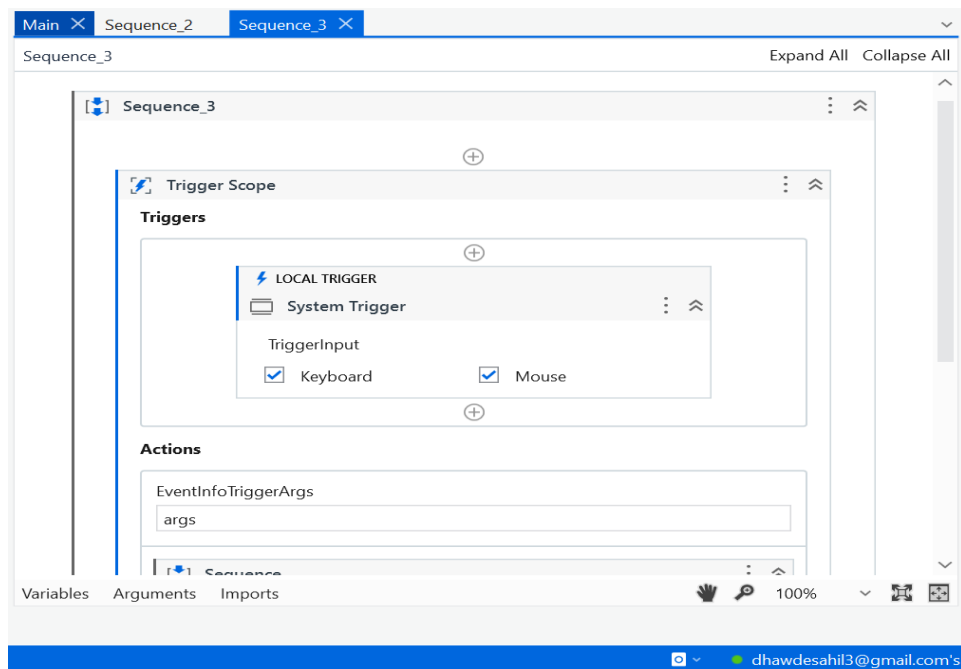


System Trigger

Steps:

1. Open project.
2. Select Sequence activity.
3. Select from activities panel Tigger Scope activity.
4. In Trigger block, select from activities panel System Tigger activity and check for both “Keyboard” and “Mouse”.
5. In Action block, select Message box and type “Keyboard or Mouse has been pressed”.
6. Run the project.

Process & Output:



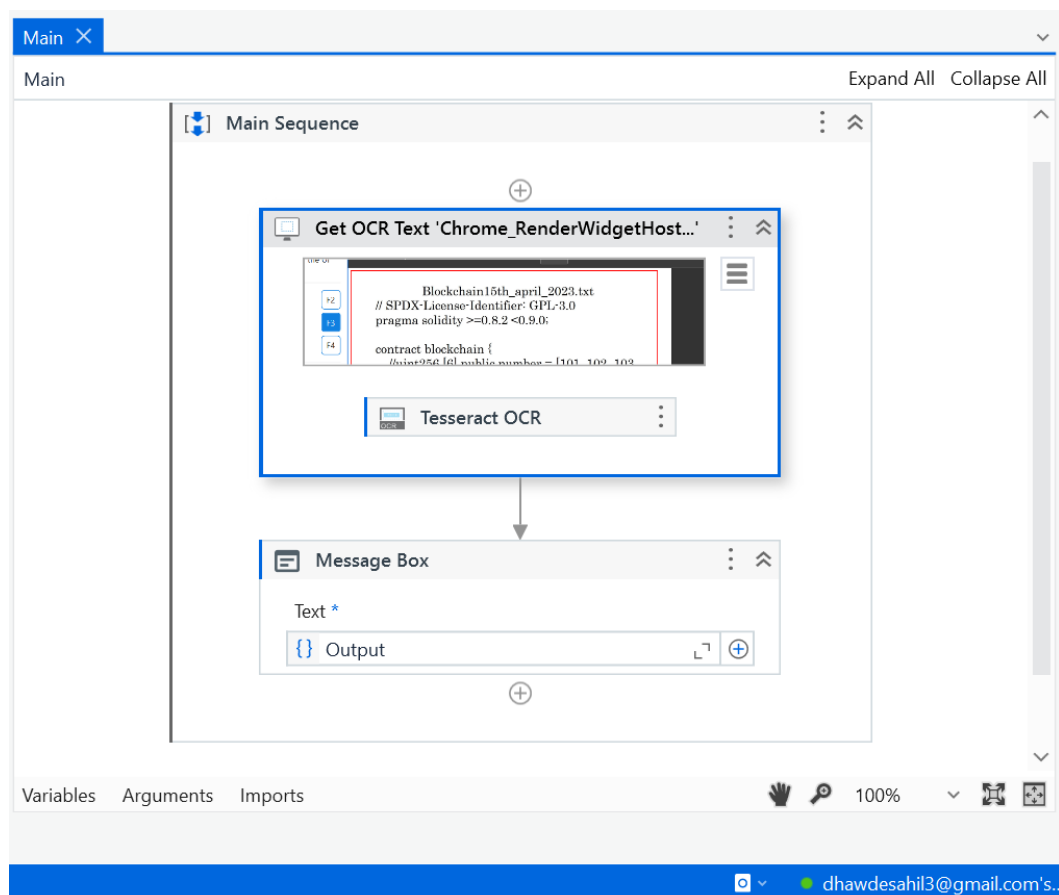
B. Automate the following screen scraping methods using UiPath

i. OCR

Steps:

1. Open project.
2. Select Sequence activity.
3. Select from activities panel Get OCR Text activity, and indicate to the file to get text.
4. Search and Drag n Drop Tesseract OCR from activities panel.
5. Create Output variable as String.
6. Select Message box and enter "Output".
7. Run the project.

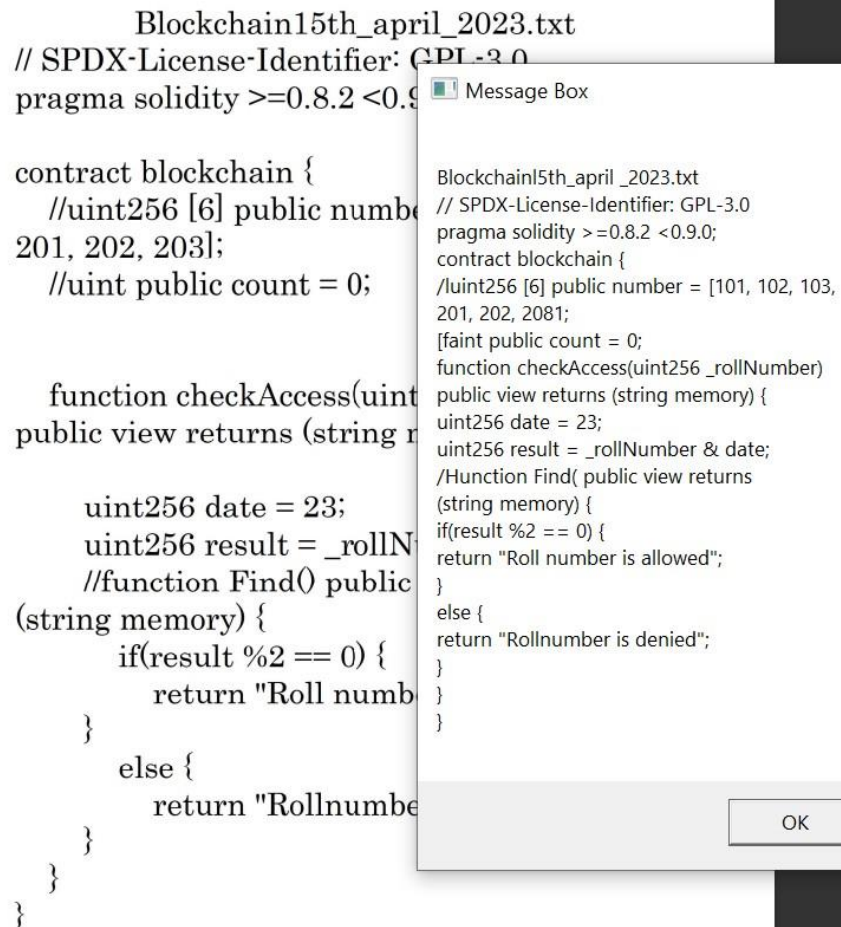
Process & Output:



```
Blockchain15th_april_2023.txt
// SPDX-License-Identifier: GPL-3.0
pragma solidity >=0.8.2 <0.9.0;

contract blockchain {
    //uint256 [6] public number = [101, 102, 103, 201, 202, 203];
    //uint public count = 0;

    function checkAccess(uint256 _rollNumber) public view returns (string memory) {
        uint256 date = 23;
        uint256 result = _rollNumber & date;
        //function Find() public view returns (string memory) {
        if(result %2 == 0) {
            return "Roll number is allowed";
        }
        else {
            return "Rollnumber is denied";
        }
    }
}
```

A screenshot of a code editor window titled 'Blockchain15th_april_2023.txt'. The code is a Solidity smart contract. It starts with a license identifier and a pragma statement for Solidity version 0.8.2 to 0.9.0. The contract is named 'blockchain'. It has a public array 'number' of type 'uint256' with 6 elements, and a public variable 'count' of type 'uint'. There is a function 'checkAccess' that takes a 'uint256' parameter '_rollNumber' and returns a 'string memory'. Inside this function, a 'uint256' variable 'date' is set to 23, and 'result' is calculated as '_rollNumber & date'. There is a commented-out function 'Find' that also returns 'string memory'. The 'checkAccess' function has an 'if' statement that checks if 'result % 2 == 0'. If true, it returns 'Roll number is allowed'; otherwise, it returns 'Rollnumber is denied'. A 'Message Box' dialog is overlaid on the code, showing the same code snippet. The dialog has an 'OK' button at the bottom right.

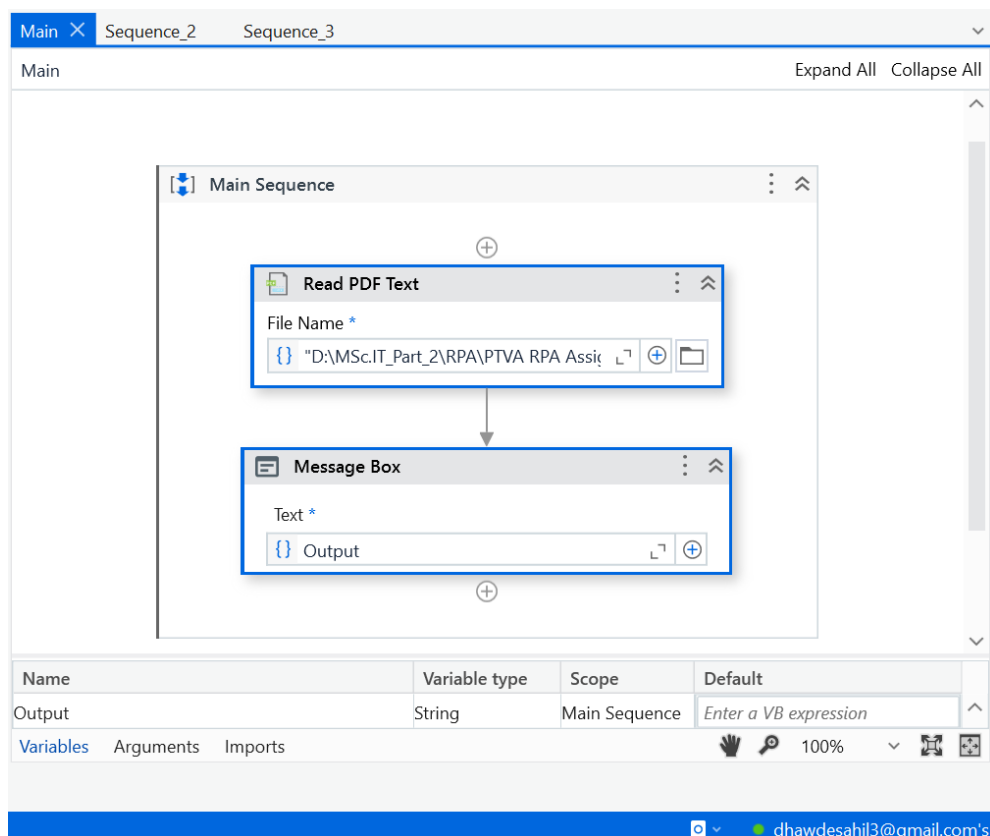
C. Install and automate any process using UiPath with the following plug-ins:

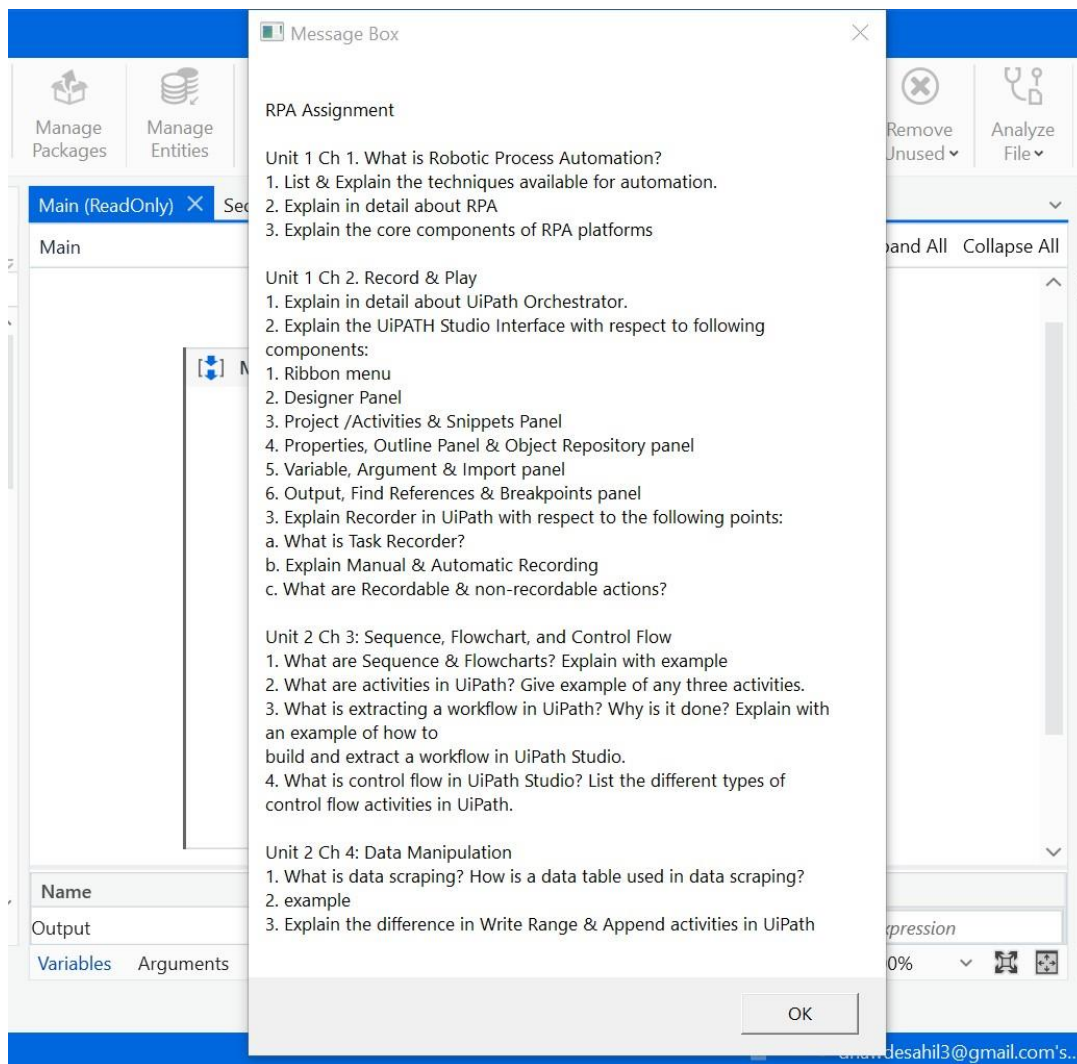
i. PDF Plugin

Steps:

1. Open project.
2. Select Sequence activity.
3. Download PDF plugin from Manage Packages icon on Ribbon panel of UiPath studio.
4. Select from activities panel Get PDF Text activity and give file name from which we have to get text.
5. Create Output variable as String.
6. Select Message box and enter "Output".
7. Run the project.

Process & Output:



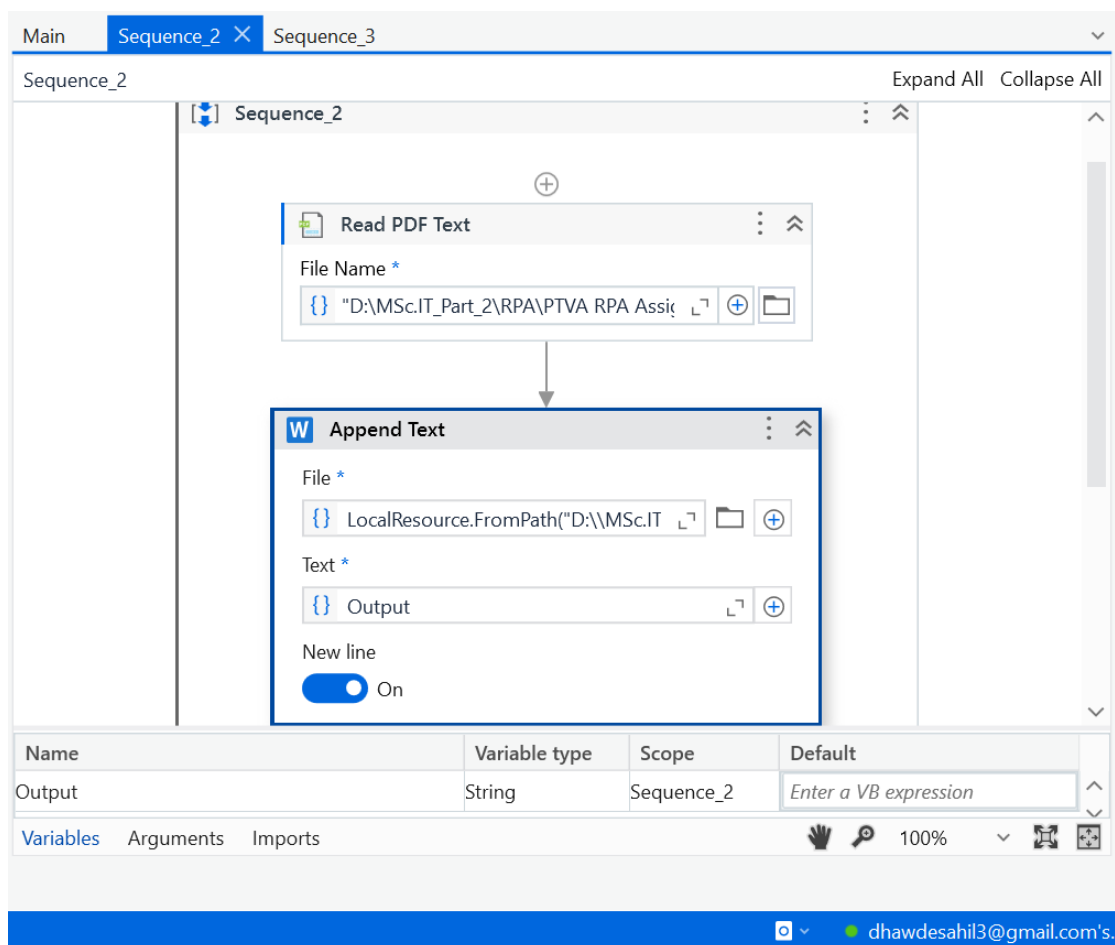


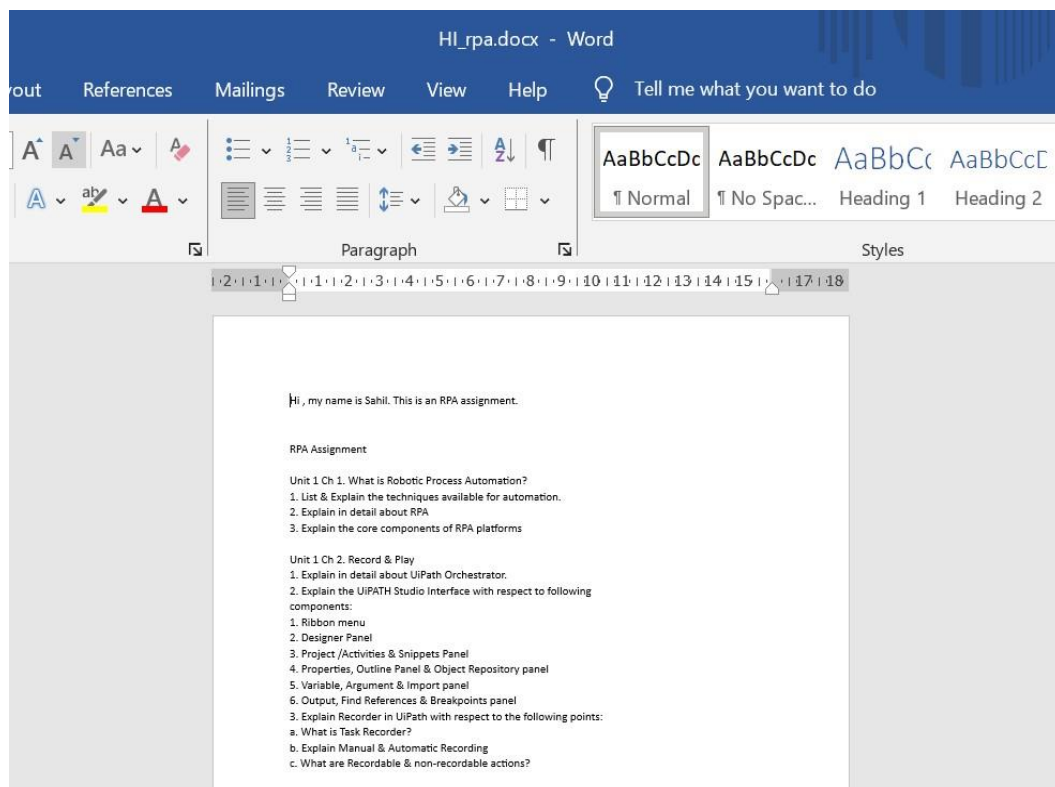
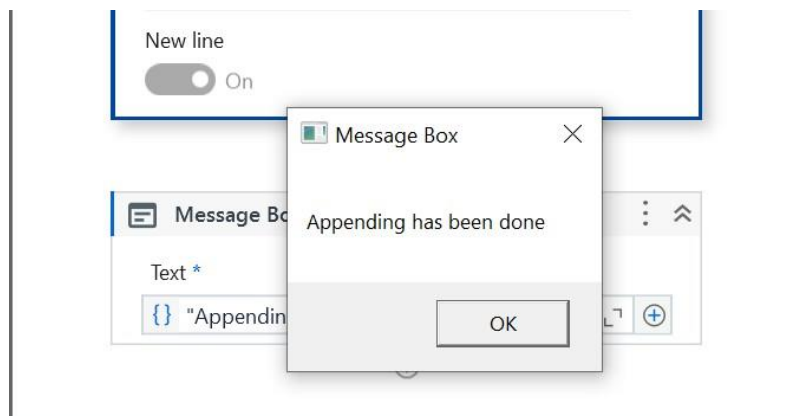
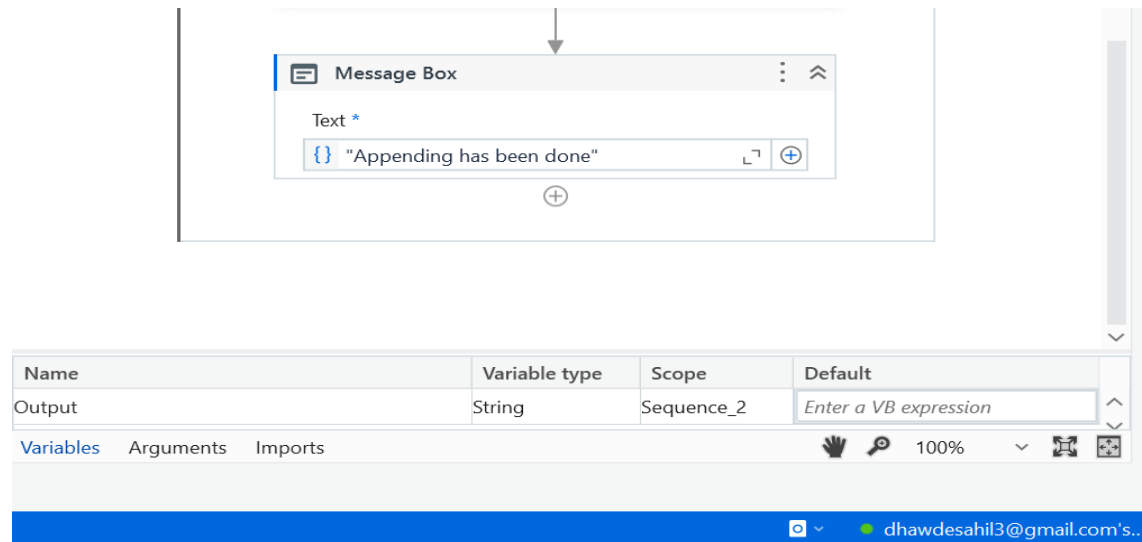
ii. Word Plugin

Steps:

1. Open project.
2. Select Sequence activity.
3. Download Word plugin from Manage Packages icon on Ribbon panel of UiPath studio.
4. Select from activities panel Get PDF Text activity and give file name from which we have to get text.
5. Create Output variable as String.
6. Select Word Append Text activity.
7. Browse the file in which we have to append and set Text as "Output".
8. Select Message box and type "Appending has been done".
9. Run the project.

Process & Output:





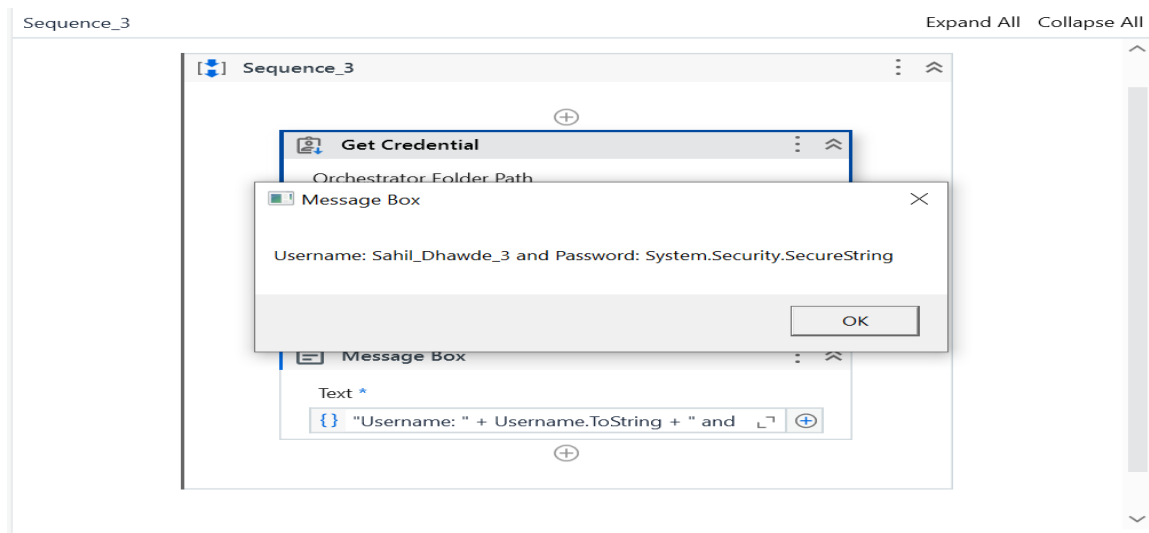
iii. Credential Management

Steps:

1. Create an Asset in UiPath Orchestrator of type "Credential."
2. Open Project.
3. Select Sequence activity.
4. Select Get Credential activity from activities panel.
5. In Get Credential activity, select orchestrator folder place as your workspace and enter the Asset Name that created earlier.
6. Create two variables as Username and Password.
7. Select Message box activity from activities panel and type "Username: " + Username.ToString + " and Password: " + Password.ToString".
8. Run the project.

Process & Output:

The image displays two screenshots from the UiPath ecosystem. The top screenshot shows the UiPath Orchestrator interface, specifically the 'Assets' tab. A table lists assets, with one entry for 'Sahil Dhawde' of type 'Credential' located in the 'RPA_Asset_Practical' folder. The bottom screenshot shows the UiPath Studio interface, specifically a 'Sequence' activity. The 'Get Credential' activity is configured with the 'Orchestrator Folder Path' set to 'dhawdesahil3@gmail.com's workspace' and the 'Asset Name' set to 'Sahil Dhawde'. Below it, a 'Message Box' activity is configured with the text 'Username: ' + Username.ToString + ' and Password: ' + Password.ToString'. The right-hand side of the Studio window shows the 'Properties' pane for the 'Get Credential' activity, with fields for 'Timeout (milliseconds)', 'Cache Strategy', and 'Output' (Username and Password).



Practical No: 9

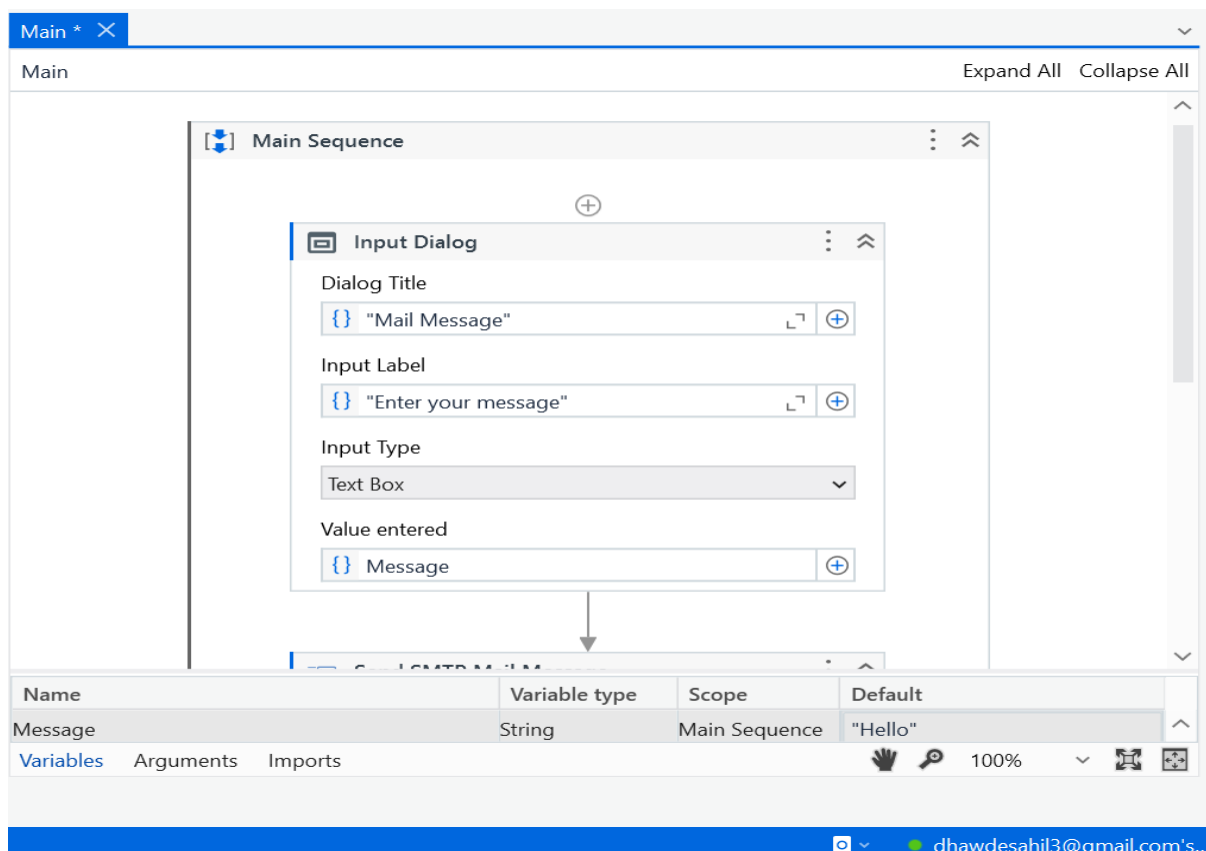
Aim: Implement the following:

A. Automate the process of send mail event (on any email).

Steps:

1. Open Project.
2. Select Sequence activity.
3. Create one string variables Message to store the message.
4. Select one Input dialog.
5. Configure the Input dialog boxes to take the values for Message to send.
6. Select Send SMTP mail message & configure the To, Subject & Message as per the variable created earlier.
7. In Properties panel. Set the Host values of Port to "587", Server Name to "smtp-mail.outlook.com".
8. And select Message box and enter text "Message has been sent".
9. Run the project.

Process & Output:



The screenshot shows the UiPath Studio interface. The main workspace displays a sequence of two activities: 'Send SMTP Mail Message' and 'Message Box'. The 'Send SMTP Mail Message' activity is configured with the following properties:

- To ***: {} "dhawdesahil3@gmail.com"
- Subject**: {} "Send testing SMTP mail message"
- Body**: {} Message

Below the 'Send SMTP Mail Message' activity is the 'Message Box' activity, configured with:

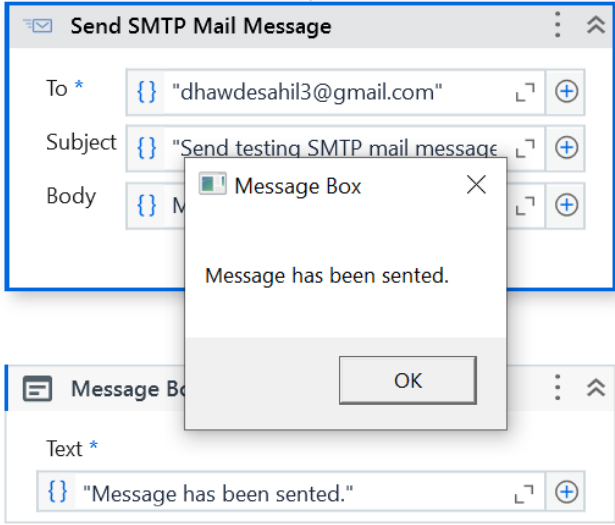
- Text ***: {} "Message has been sent."

The Properties window on the right shows the configuration for the 'Send SMTP Mail Message' activity, including fields for Body, Subject, Forward, Host (Port 587, Server smtp-ma), Logon (Email TangoCh, Password Overandl, SecurePassword The passw, UseOAuth Indica), Misc (Private checkbox), and Options (ContinueOnError, Ignore CRL, IsBodyHtml).

Name	Variable type	Scope	Default
Message	String	Main Sequence	"Hello"

The screenshot shows a 'Mail Message' dialog box overlaid on the UiPath Studio interface. The dialog has a title bar with a close button (X). Inside, it says 'Enter your message' and displays the text 'Hello Sahil, I am doing RPA Practical' in a text box. An 'Ok' button is located at the bottom center of the dialog.

Main Expand All Collapse All



Send SMTP Mail Message

To * {} "dhawdesahil3@gmail.com" L +

Subject {} "Send testing SMTP mail message" L +

Body {} M L +

Message Box

Message has been sent.

OK

Message Box

Text * {} "Message has been sent." L +

← 📁 ⚠️ 🗑️ ✉️ ⌚ ↺ 📁 📄 ⋮

Send testing SMTP mail message activity. Inbox x

 **TangoCharlie333@outlook.com**
Hello

 **TangoCharlie333@outlook.com**
to me ▼
Hello Sahil, I am doing RPA Practical.

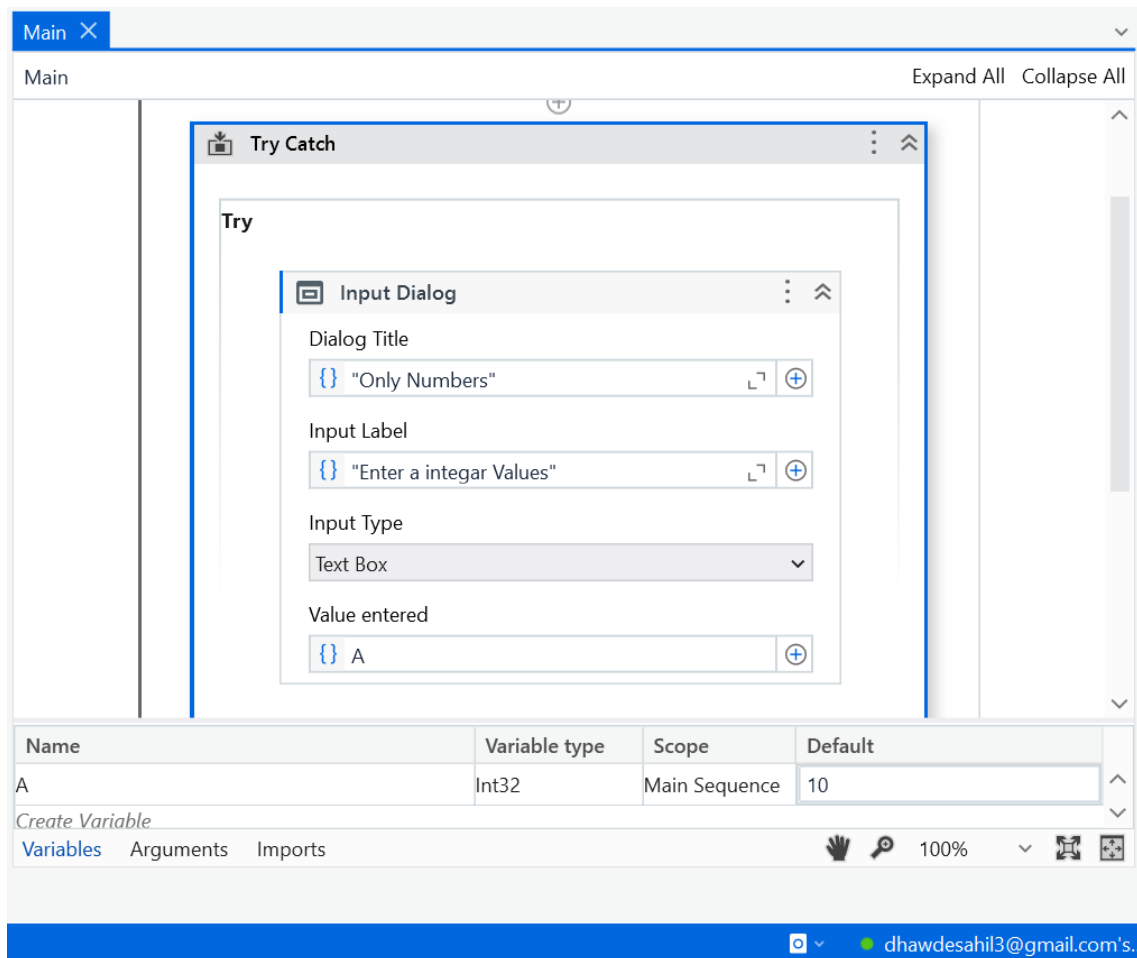
↩ Reply ↪ Forward 😊

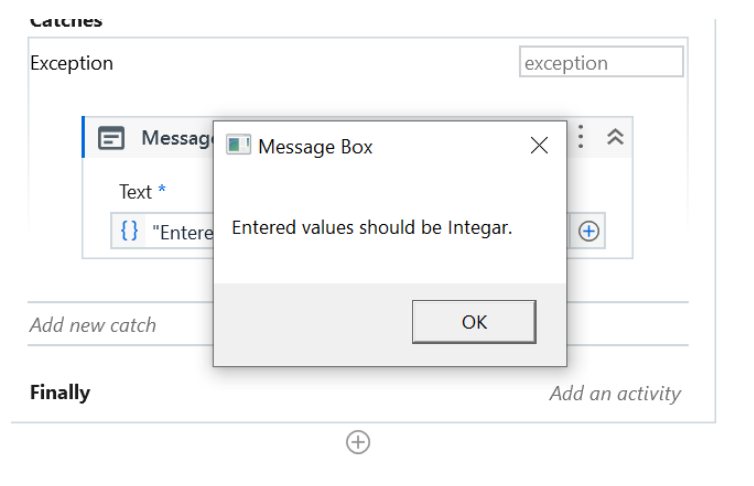
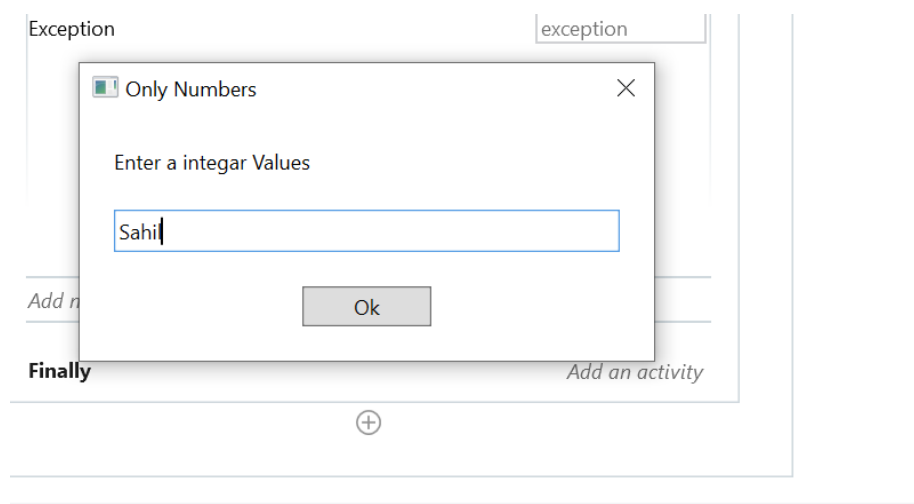
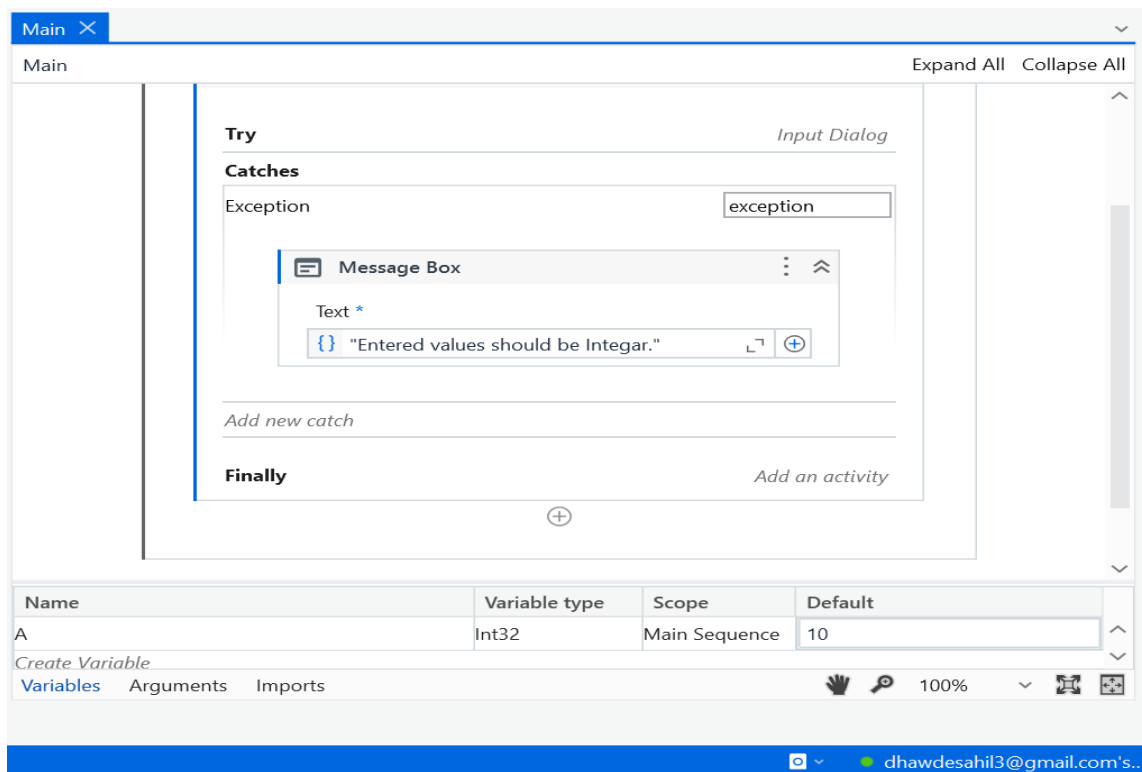
B. Demonstrate the Exception handling in UiPath.

Steps:

1. Open Project.
2. Select Sequence activity.
3. Create one variables A (int32).
4. Select Try Catch activity from activities panel.
5. In Try block, drag n drop Input dialog activity and configure it by entering A as a Value entered.
6. In Catch block, drag n drop Message box and type "Entered values should be integer".
7. And search for "System.Exception" for Exception.
8. Run the project.

Process & Output:





Practical No: 10

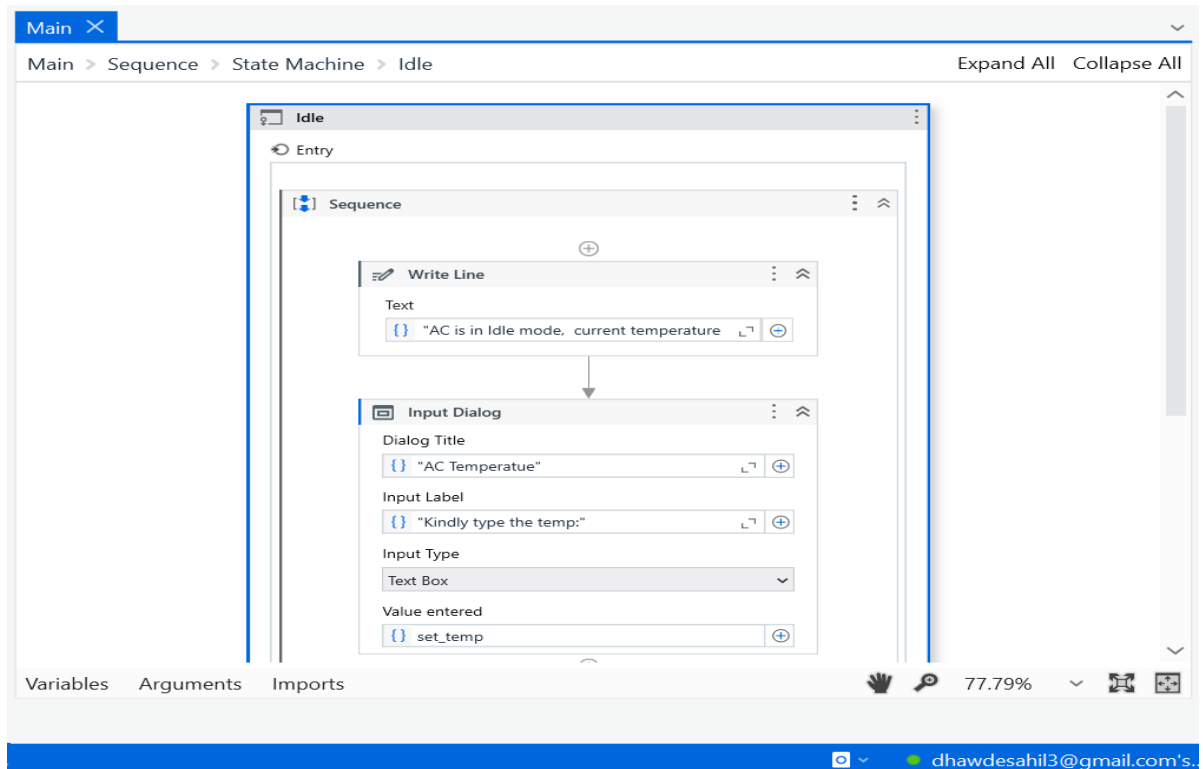
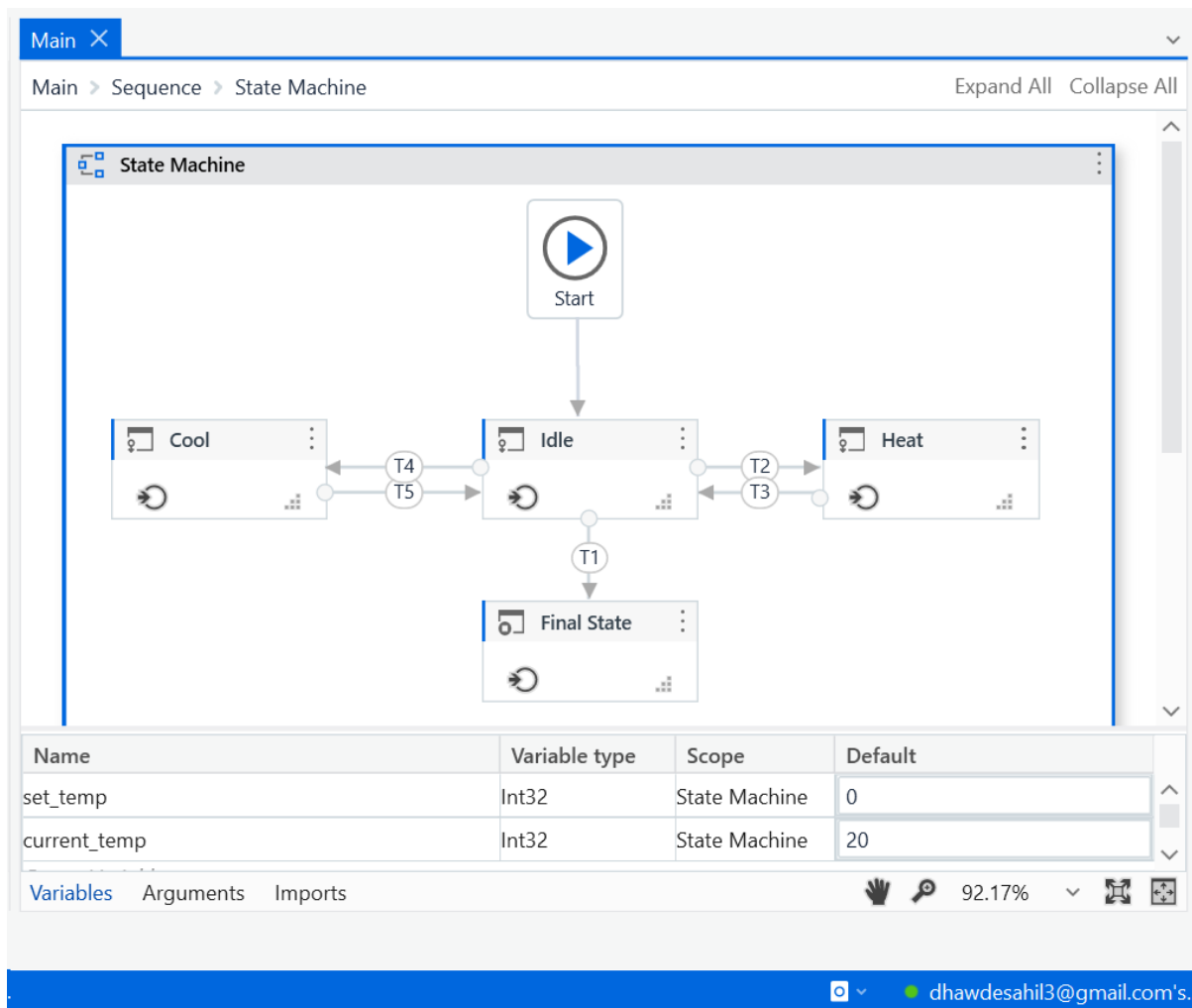
Aim: Implement the following:

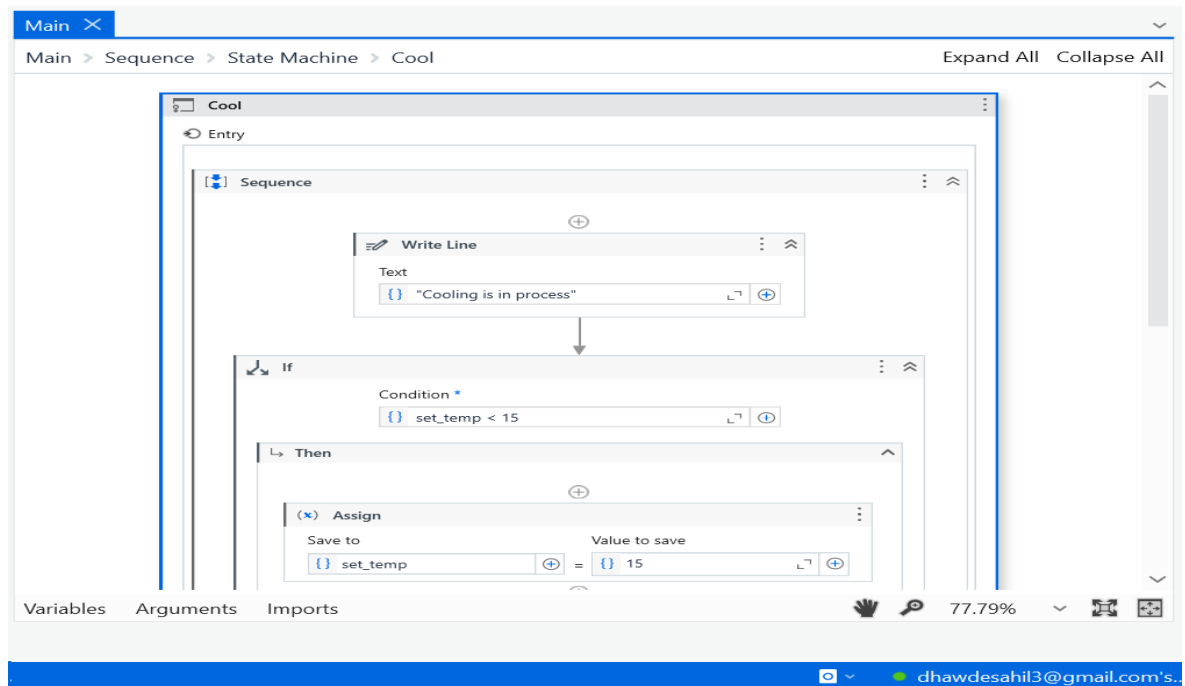
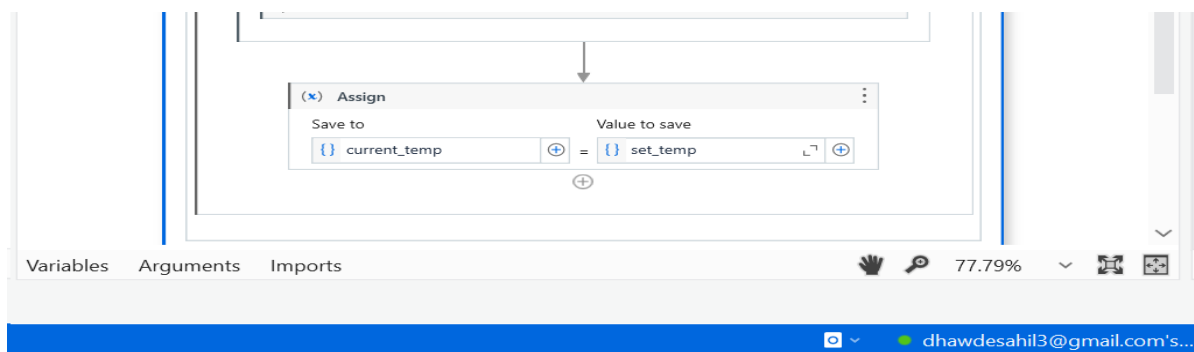
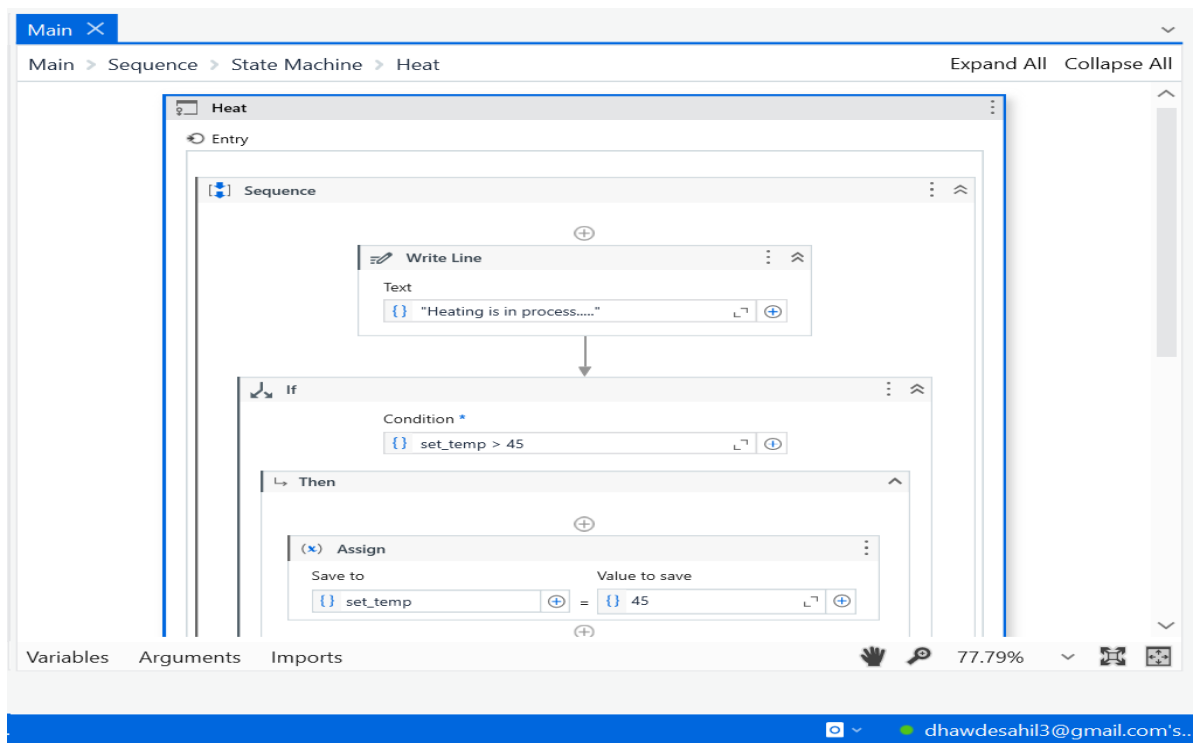
A. Automate any process using State Machine in UiPath.

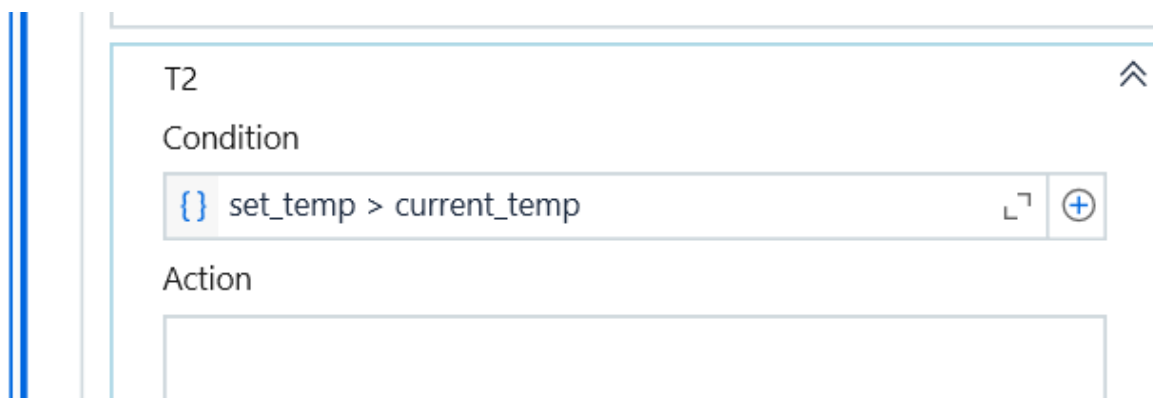
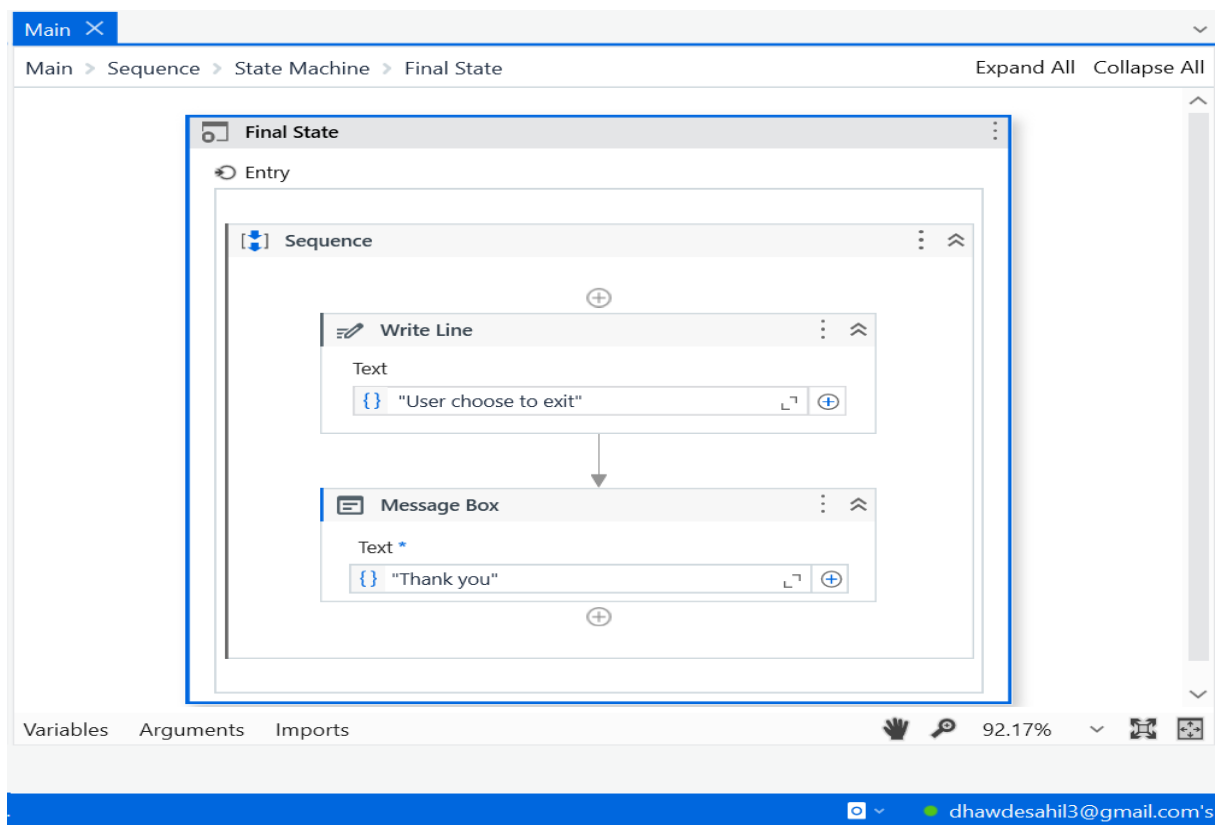
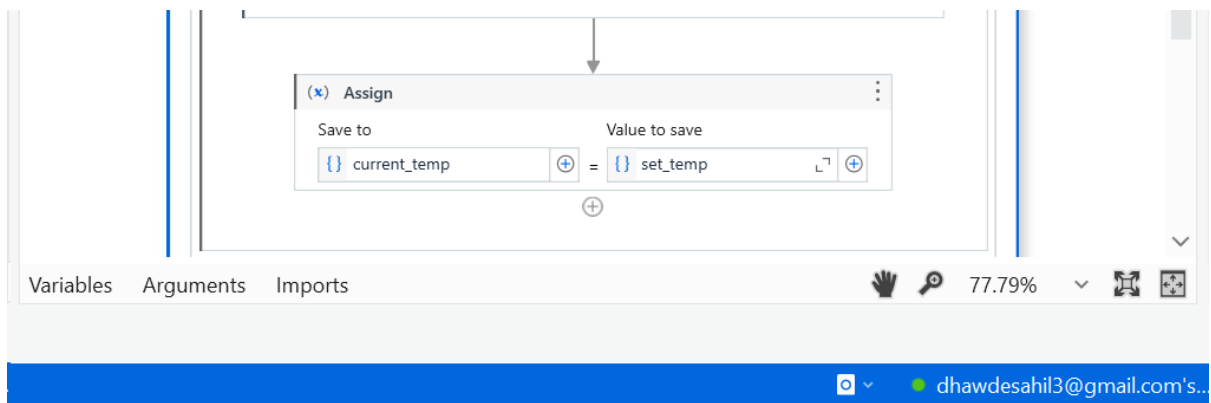
Steps:

1. Open Project.
2. Select State Machine activity, double click it.
3. Select from activities panel State activity and named them as Cool, Idle, Heat.
4. Place them beside near other in order as Cool, Idle, Heat.
5. Draw a two Transition line such as incoming (T3) and outgoing (T2) between from Idle to Heat.
6. Draw a two Transition line such as incoming (T5) and outgoing (T4) between from Idle to Heat.
7. Select from activities panel Final State activity.
8. Draw a one Transition line such as outgoing (T1) between from Idle to Final state.
9. Create two variables set_temp (int32) assigns to 0 and current_temp (int32) assigns to 20.
10. In Idle state, select one Write Line activity and type "AC is in Idle mode, current temperature = " + current_temp.ToString + " deg celsius".
11. And select Input dialog and configure it.
12. Set the value of Value Entered as "set_temp".
13. In Heat state, select one Write Line activity and type " Heating is in process. ...".
14. Select If activity and give condition as "set_temp>45".
15. In then block, select Assign activity to assign set_temp = 45.
16. Select Assign activity outside the If activity and assign current_temp = set_temp.
17. In Cool state, select one Write Line activity and type " Cooling is in process....".
18. Select If activity and give condition as "set_temp<15".
19. In then block, select Assign activity to assign set_temp = 15.
20. Select Assign activity outside the If activity and assign current_temp = set_temp.
21. Set the conditions of T2 as "set_temp > current_temp".
22. Set the conditions of T1 as "set_temp < current_temp".
23. Set the conditions of both T3, T5 transition as "current_temp.Equals(set_temp)".
24. Set the conditions of T1 as "set_temp.Equals(0)".
25. In Final state, select one Write Line activity and type "User choose to exit".
26. And Select Message box and type "Thank you".
27. Run the project.

Process & Output:







The screenshot displays a state machine editor with three states: T3, T4, and T5. Each state has a condition and an action field.

- T3:** Condition is `current_temp.Equals(set_temp)`. The action field is empty.
- T4:** Condition is `set_temp < current_temp`. The action field is empty.
- T5:** Condition is `current_temp.Equals(set_temp)`. The action field is empty.

Below the states is an 'Output' window showing a list of execution events:

- Execution started for file: Main
- State_Machine execution started
- AC is in Idle mode, current temperature = 20 deg celsius
- Heating is in process.....
- AC is in Idle mode, current temperature = 25 deg celsius

A modal dialog box titled 'AC Temperature' is open, prompting the user to 'Kindly type the temp:'. The input field contains the value '25', and there is an 'Ok' button.

At the bottom of the interface, there are tabs for 'Breakpoints' and 'Error List', and a status bar showing the user's email address: dhawdesahil3@gmail.com's...

Output

Search

- State_Machine execution started
- AC is in Idle mode, current temperature = 20 deg celsius
- Heating is in process.....
- AC is in Idle mode, current temperature = 25 deg celsius
- Cooling is in process
- AC is in Idle mode, current temperature = 18 deg celsius

Breakpoints Error List

AC Temperature

Kindly type the temp:

18

Ok

dhawdesahil3@gmail.com's

Output

Search

- Heating is in process.....
- AC is in Idle mode, current temperature = 25 deg celsius
- Cooling is in process
- AC is in Idle mode, current temperature = 18 deg celsius
- Heating is in process.....
- AC is in Idle mode, current temperature = 45 deg celsius

Breakpoints Error List

AC Temperature

Kindly type the temp:

50

Ok

dhawdesahil3@gmail.com's

Output

Search

- Cooling is in process
- AC is in Idle mode, current temperature = 18 deg celsius
- Heating is in process.....
- AC is in Idle mode, current temperature = 45 deg celsius
- Cooling is in process
- AC is in Idle mode, current temperature = 15 deg celsius

Breakpoints Error List

AC Temperature

Kindly type the temp:

12

Ok

dhawdesahil3@gmail.com's

Output

Search

- AC is in Idle mode, current temperature = 18 deg celsius
- Heating is in process.....
- AC is in Idle mode, current temperature = 45 deg celsius
- Cooling is in process
- AC is in Idle mode, current temperature = 15 deg celsius
- User choose to exit

Breakpoints Error List

Message ...

Thank you

OK

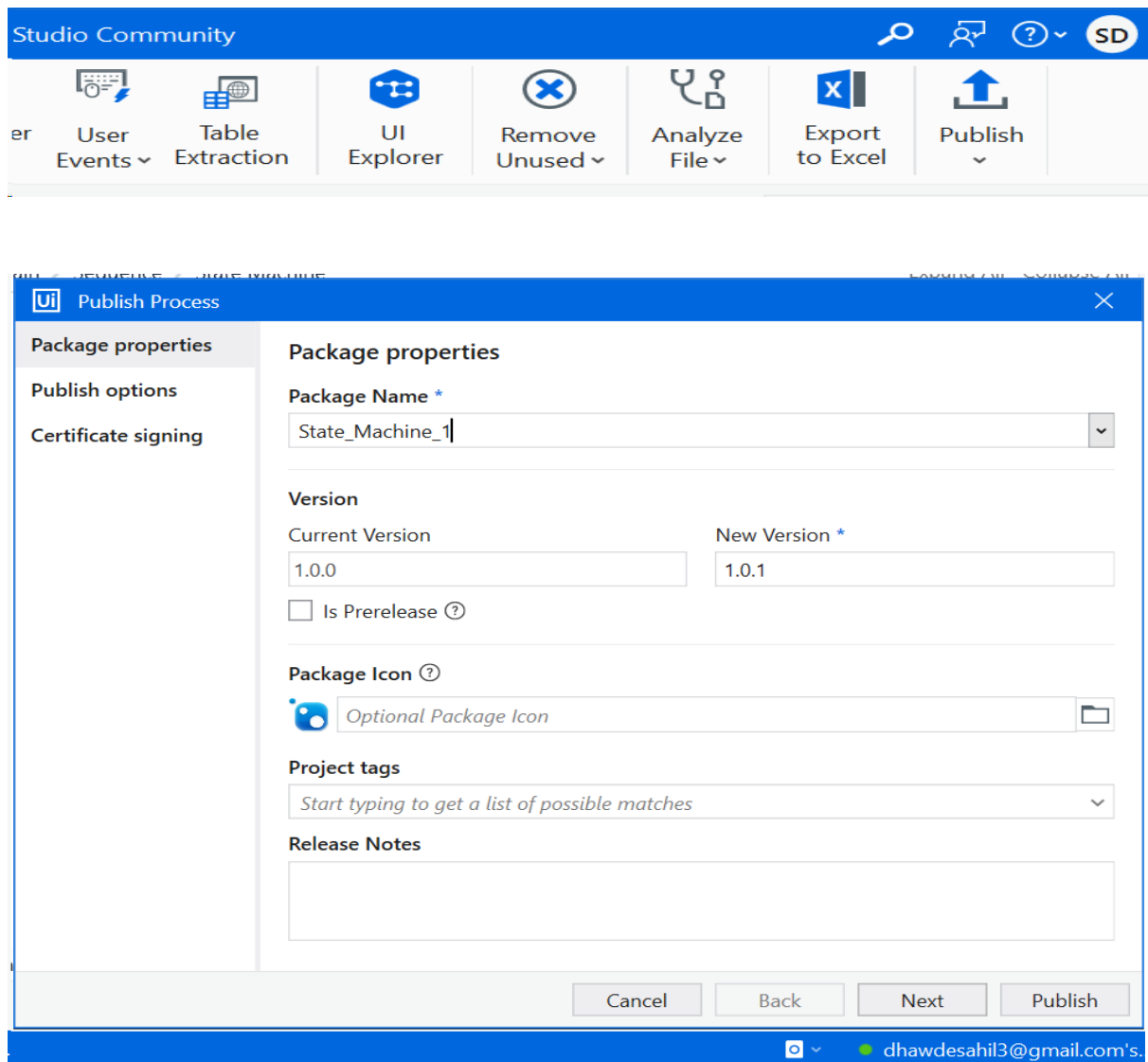
dhawdesahil3@gmail.com's

B. Demonstrate the use of publish utility.

Steps:

1. Open Saved project.
2. Select Publish icon on Ribbon panel of UiPath studio.
3. Publish process dialog box will be open.
4. In Package properties, give the Package Name and next.
5. In Publish options, select publish to “Orchestrator Personal Workspace Feed” and Next.
6. Then click on Publish button.
7. You will see your process publish in UiPath orchestrator and assistance.

Process & Output:



The screenshot shows the 'Publish Process' dialog box with the 'Publish options' tab selected. The left sidebar contains 'Package properties', 'Publish options', and 'Certificate signing'. The main area has the following fields:

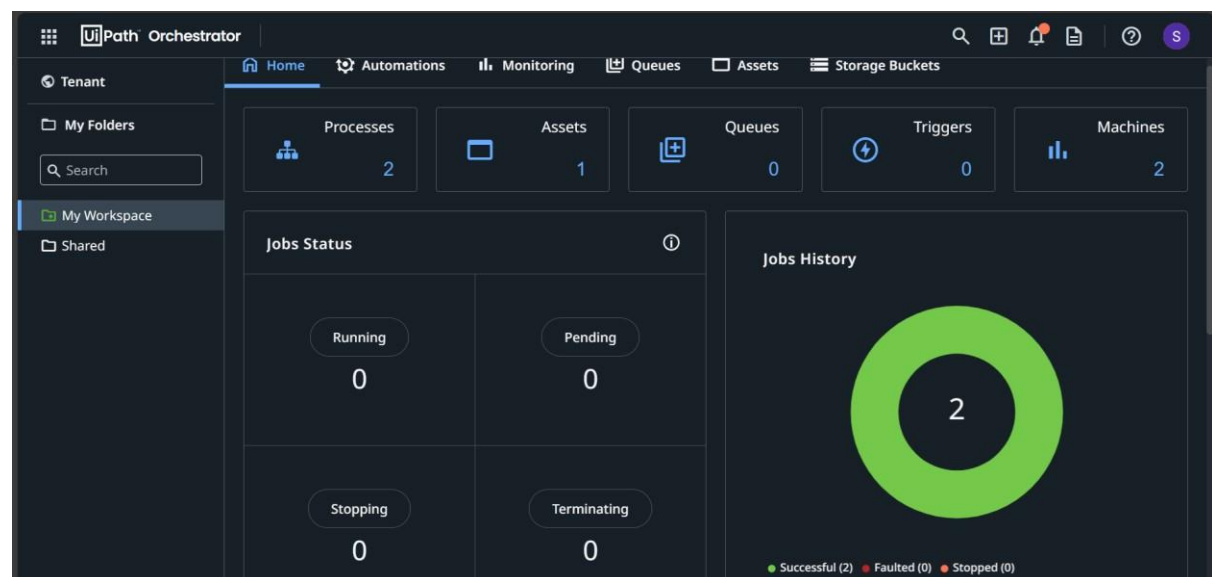
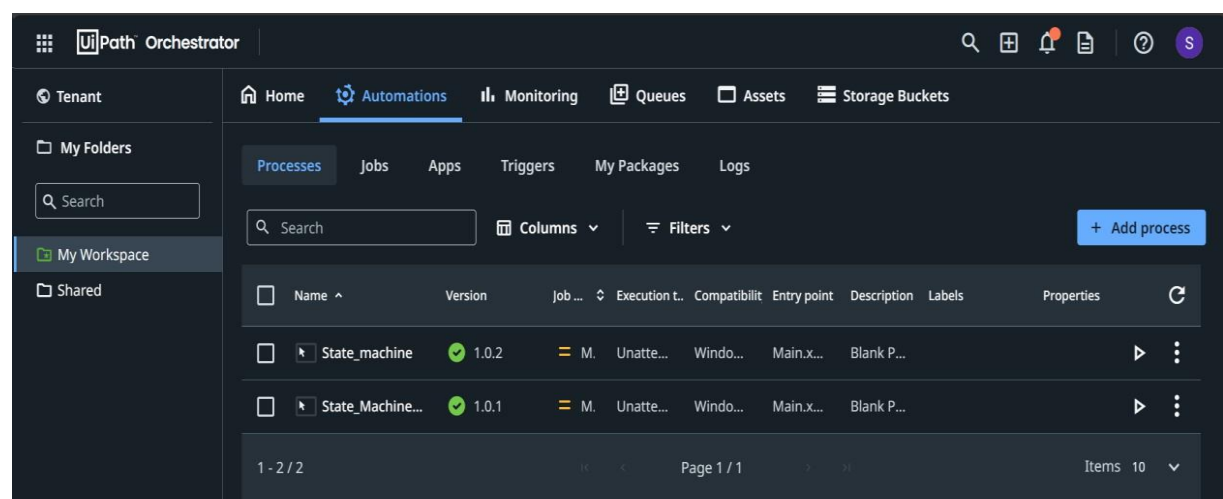
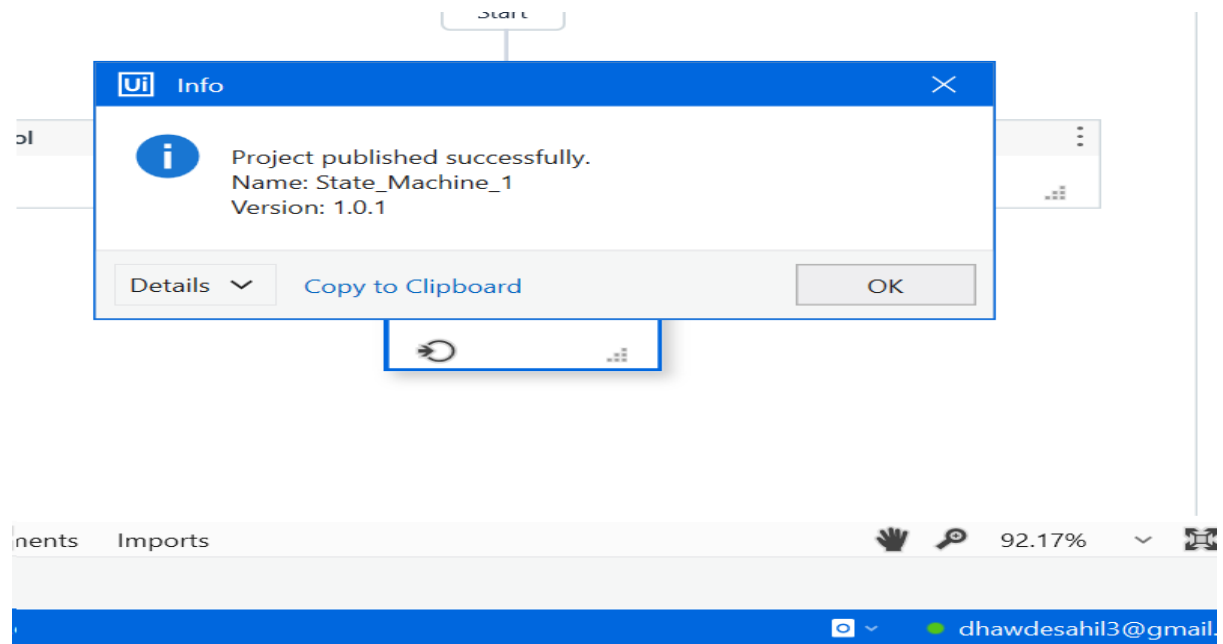
- Publish to:** A dropdown menu showing 'Orchestrator Personal Workspace Feed'.
- Custom URL:** A text field with the placeholder 'NuGet feed url or local folder' and a folder selection icon.
- API Key:** A text field with the placeholder 'Optional API Key'.
- Compilation Settings:** A checkbox labeled 'Remove Unused Dependencies' with a help icon.

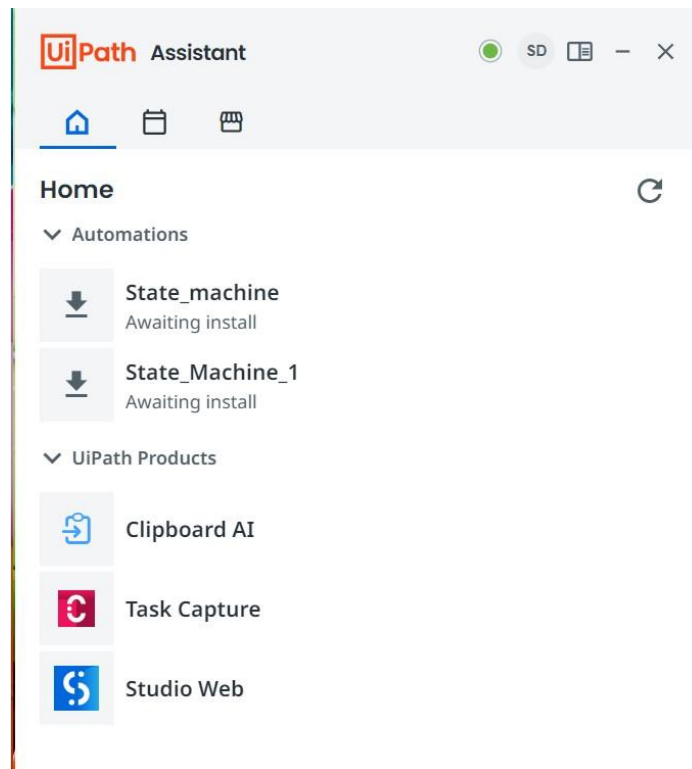
At the bottom are buttons for 'Cancel', 'Back', 'Next', and 'Publish'.

The screenshot shows the 'Publish Process' dialog box with the 'Certificate signing' tab selected. The left sidebar contains 'Package properties', 'Publish options', and 'Certificate signing'. The main area has the following fields:

- Certificate:** A text field with a folder selection icon.
- Certificate Password:** A text field.
- Timestamp:** A text field with the placeholder 'Optional Certificate Timestamp'.

At the bottom are buttons for 'Cancel', 'Back', 'Next', and 'Publish'.



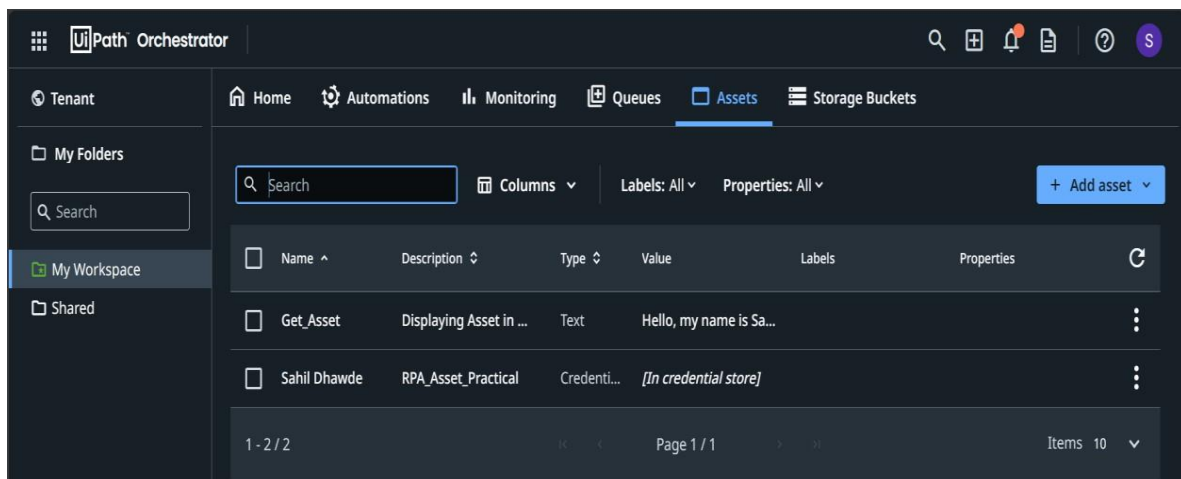
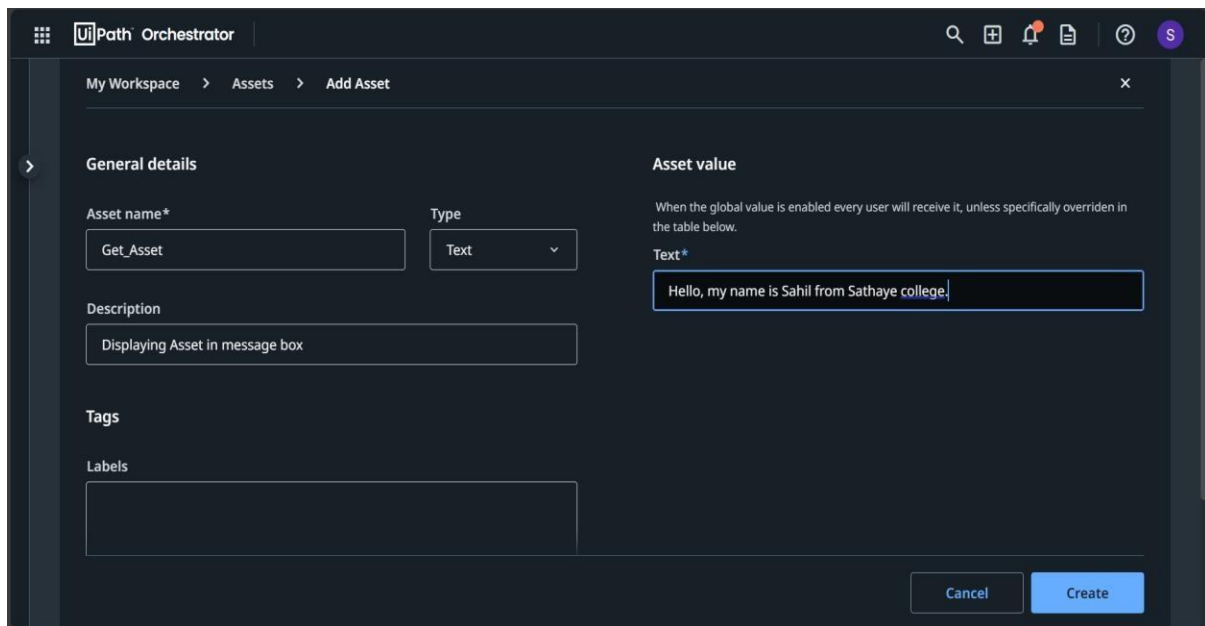


C. Demonstrate the use of asset using orchestrator.

Steps:

1. Create an Asset in UiPath Orchestrator of type "Text."
2. Open Project.
3. Select Sequence activity.
4. Create one variables Output (string).
5. Select Get Asset activity from activities panel.
6. In Get Asset activity, select orchestrator folder place as your workspace and enter the Asset Name that created earlier.
7. Select Message box activity from activities panel and enter "Output" variable.
8. Run the project.

Process & Output:



Name	Description	Type	Value	Labels	Properties
Get_Asset	Displaying Asset in ...	Text	Hello, my name is Sa...		
Sahil Dhawde	RPA_Asset_Practical	Credenti...	[In credential store]		

Main Sequence

Get Asset

Orchestrator Folder Path
dhawdesahil3@gmail.com's workspace

Asset Name *
Get_Asset

Message Box

Text *
Output

Name	Variable type	Scope	Default
Select an activity to set variable scope			

Variables Arguments Imports

Main Sequence

Get Asset

Orchestrator Folder Path
dhawdesahil3@gmail.com's workspace

Asset Name *
Get_Asset

Message Box

Text *
Output

Name	Variable type	Scope	Default
Output	String	Main Sequence	Enter a VB expression

Variables Arguments Imports

